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See-through computer display systems

Abstract

Aspects of the present invention relate to providing see-through computer display optics.

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D765076	12/2015	Rochat	N/A	N/A
9423612	12/2015	Border	N/A	N/A
9423842	12/2015	Osterhout	N/A	N/A
9436006	12/2015	Border	N/A	N/A
9448409	12/2015	Border	N/A	N/A
D768759	12/2015	Markovitz	N/A	N/A
D769873	12/2015	Cazalet	N/A	N/A
9482880	12/2015	Chandrasekhar	N/A	N/A
9523856	12/2015	Osterhout	N/A	N/A
9529195	12/2015	Osterhout	N/A	N/A
9529199	12/2015	Osterhout	N/A	N/A
9651787	12/2016	Haddick	N/A	N/A
9651788	12/2016	Osterhout	N/A	N/A
9651789	12/2016	Osterhout	N/A	N/A
9672210	12/2016	Osterhout	N/A	N/A
9684172	12/2016	Border	N/A	N/A
D792400	12/2016	Osterhout	N/A	N/A
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D793467	12/2016	Krause	N/A	N/A
D794022	12/2016	Limaye	N/A	N/A
D795865	12/2016	Porter	N/A	N/A
9720505	12/2016	Gribetz et al.	N/A	N/A
9746676	12/2016	Osterhout	N/A	N/A
D796504	12/2016	Natsume	N/A	N/A
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D800118	12/2016	Xing	N/A	N/A
D803832	12/2016	Lin	N/A	N/A
9846308	12/2016	Osterhout	N/A	N/A
9897822	12/2017	Osterhout	N/A	N/A
10013053	12/2017	Cederlund et al.	N/A	N/A
10025379	12/2017	Drake et al.	N/A	N/A
10379365	12/2018	Osterhout	N/A	N/A
11126003	12/2020	Osterhout et al.	N/A	N/A
11892644	12/2023	Osterhout	N/A	G06F 3/03542
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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. Non-Provisional application Ser. No. 17/389,174, filed on Jul. 29, 2021, which is a continuation of U.S. Non-Provisional application Ser. No. 16/538,720, filed on Aug. 12, 2019, now U.S. Pat. No. 11,126,003, issued on Sep. 21, 2021, which is a continuation of U.S. Non-Provisional application Ser. No. 15/488,916, filed on Apr. 17, 2017, now U.S. Pat. No. 10,379,365, issued on Aug. 13, 2019, which is a continuation of U.S. Non-Provisional application Ser. No. 14/742,249, filed on Jun. 17, 2015, now U.S. Pat. No. 9,746,676, issued on Aug. 29, 2017. U.S. Non-Provisional application Ser. No. 14/742,249 is a continuation of U.S. Non-Provisional application Ser. No. 14/589,713, filed on Jan. 5, 2015, now U.S. Pat. No. 9,529,195, issued on Dec. 27, 2016.

U.S. Non-Provisional application Ser. No. 14/589,713 is a continuation-in-part of the following: U.S. Non-Provisional application Ser. No. 14/160,377, filed Jan. 21, 2014, now abandoned. U.S. Non-Provisional application Ser. No. 14/163,646, filed Jan. 24, 2014, now U.S. Pat. No. 9,400,390, issued on Jul. 26, 2016. U.S. Non-Provisional application Ser. No. 14/172,901, filed Feb. 4, 2014, now abandoned. U.S. Non-Provisional application Ser. No. 14/181,459, filed Feb. 14, 2014, now U.S. Pat. No. 9,715,112, issued on Jul. 25, 2017, which is a continuation-in-part of U.S. Non-Provisional application Ser. No. 14/178,047, filed Feb. 11, 2014, now U.S. Pat. No. 9,229,233, issued on Jan. 5, 2016. U.S. Non-Provisional application Ser. No. 14/589,713 is also a continuation-in-part of the following: U.S. Non-Provisional application Ser. No. 14/216,175, filed on Mar. 17, 2014, now U.S. Pat. No. 9,298,007, issued on Mar. 29, 2016. U.S. Non-Provisional application Ser. No. 14/296,699, filed Jun. 5, 2014, now U.S. Pat. No. 9,841,599, issued on Dec. 12, 2017. U.S. Non-Provisional application Ser. No. 14/325,991, filed Jul. 8, 2014, now U.S. Pat. No. 9,366,867, issued on Jun. 14, 2016. U.S. Non-Provisional application Ser. No. 14/489,706, filed Sep. 18, 2014, now abandoned. U.S. Non-Provisional application Ser. No. 14/498,765, filed Sep. 26, 2014, now U.S. Pat. No. 9,366,868, issued on Jun. 14, 2016. U.S. Non-Provisional application Ser. No. 14/504,723, filed Oct. 2, 2014, now U.S. Pat. No. 9,671,613, issued on Jun. 6, 2017. U.S. Non-Provisional application Ser. No. 14/559,126, filed Dec. 3, 2014, now U.S. Pat. No. 10,684,687, issued on Jun. 16, 2020. U.S. Non-Provisional application Ser. No. 14/254,253, filed Apr. 16, 2014, now U.S. Pat. No. 9,952,664, issued on Apr. 24, 2018. U.S. Non-Provisional application Ser. No. 14/561,146, filed Dec. 4, 2014, now U.S. Pat. No. 9,594,246, issued on Mar. 14, 2017. U.S. Non-Provisional application Ser. No. 14/554,044, filed Nov. 26, 2014, now U.S. Pat. No. 9,448,409, issued on Sep. 20, 2016. (2) All of the above applications are incorporated herein by reference in their entirety.

BACKGROUND

Field of the Invention

(1) This invention relates to see-through computer display systems.

Description of Related Art

(2) Head mounted displays (HMD) and particularly HMDs that provide a see-through view of the environment are sensitive to the effects of stray light. Where, stray light includes light that is not intended to be included in the displayed image, including light that is scattered or inadvertently reflected from surfaces within the optics of the HMD. This stray light reduces the sharpness and contrast of the displayed image in the HMD. In addition, the stray light causes the black areas of the displayed image to be gray and this effect adversely affects the see-through view of the environment because the see-through view is produced by the combination of the light from the environment and the light from the displayed image, which is in the best case the black portions of displayed image. As a result, it is important to reduce stray light within the optics so that very dark black areas and high contrast can be provided in the images displayed by the HMDs.

SUMMARY

(3) Aspects of the present invention relate to methods and systems for the see-through computer display systems.

(4) These and other systems, methods, objects, features, and advantages of the present invention will be apparent to those skilled in the art from the following detailed description of the preferred embodiment and the drawings. All documents mentioned herein are hereby incorporated in their entirety by reference.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Embodiments are described with reference to the following Figures. The same numbers may be

used throughout to reference like features and components that are shown in the Figures:

- (2) FIG. 1 illustrates a head worn computing system in accordance with the principles of the present invention.
- (3) FIG. 2 illustrates a head worn computing system with optical system in accordance with the principles of the present invention.
- (4) FIG. 3a illustrates a large prior art optical arrangement.
- (5) FIG. 3b illustrates an upper optical module in accordance with the principles of the present invention.
- (6) FIG. 4 illustrates an upper optical module in accordance with the principles of the present invention.
- (7) FIG. 4a illustrates an upper optical module in accordance with the principles of the present invention.
- (8) FIG. 4b illustrates an upper optical module in accordance with the principles of the present invention.
- (9) FIG. 5 illustrates an upper optical module in accordance with the principles of the present invention.
- (10) FIG. 5a illustrates an upper optical module in accordance with the principles of the present invention.
- (11) FIG. 5b illustrates an upper optical module and dark light trap according to the principles of the present invention.
- (12) FIG. 5c illustrates an upper optical module and dark light trap according to the principles of the present invention.
- (13) FIG. 5d illustrates an upper optical module and dark light trap according to the principles of the present invention.
- (14) FIG. 5e illustrates an upper optical module and dark light trap according to the principles of the present invention.
- (15) FIG. 6 illustrates upper and lower optical modules in accordance with the principles of the present invention.
- (16) FIG. 7 illustrates angles of combiner elements in accordance with the principles of the present invention.
- (17) FIG. 8 illustrates upper and lower optical modules in accordance with the principles of the present invention.
- (18) FIG. 8a illustrates upper and lower optical modules in accordance with the principles of the present invention.
- (19) FIG. 8b illustrates upper and lower optical modules in accordance with the principles of the present invention.
- (20) FIG. 8c illustrates upper and lower optical modules in accordance with the principles of the present invention.
- (21) FIG. 9 illustrates an eye imaging system in accordance with the principles of the present invention.
- (22) FIG. 10 illustrates a light source in accordance with the principles of the present invention.
- (23) FIG. 10a illustrates a back lighting system in accordance with the principles of the present invention.
- (24) FIG. 10b illustrates a back lighting system in accordance with the principles of the present invention.
- (25) FIGS. 11a to 11d illustrate light source and filters in accordance with the principles of the present invention.
- (26) FIGS. 12a to 12c illustrate light source and quantum dot systems in accordance with the principles of the present invention.
- (27) FIGS. 13a to 13c illustrate peripheral lighting systems in accordance with the principles of the

present invention.

(28) FIGS. **14a** to **14c** illustrate a light suppression systems in accordance with the principles of the present invention.

(29) FIG. **15** illustrates an external user interface in accordance with the principles of the present invention.

(30) FIG. **16a** to **16c** illustrate distance control systems in accordance with the principles of the present invention.

(31) FIG. **17a** to **17c** illustrate force interpretation systems in accordance with the principles of the present invention.

(32) FIG. **18a** to **18c** illustrate user interface mode selection systems in accordance with the principles of the present invention.

(33) FIG. **19** illustrates interaction systems in accordance with the principles of the present invention.

(34) FIG. **20** illustrates external user interfaces in accordance with the principles of the present invention.

(35) FIG. **21** illustrates mD trace representations presented in accordance with the principles of the present invention.

(36) FIG. **22** illustrates mD trace representations presented in accordance with the principles of the present invention.

(37) FIG. **23** illustrates an mD scanned environment in accordance with the principles of the present invention.

(38) FIG. **23a** illustrates mD trace representations presented in accordance with the principles of the present invention.

(39) FIG. **24** illustrates a stray light suppression technology in accordance with the principles of the present invention.

(40) FIG. **25** illustrates a stray light suppression technology in accordance with the principles of the present invention.

(41) FIG. **26** illustrates a stray light suppression technology in accordance with the principles of the present invention.

(42) FIG. **27** illustrates a stray light suppression technology in accordance with the principles of the present invention.

(43) FIGS. **28a** to **28c** illustrate DLP mirror angles.

(44) FIGS. **29** to **33** illustrate eye imaging systems according to the principles of the present invention.

(45) FIGS. **34** and **34a** illustrate structured eye lighting systems according to the principles of the present invention.

(46) FIG. **35** illustrates eye glint in the prediction of eye direction analysis in accordance with the principles of the present invention.

(47) FIG. **36a** illustrates eye characteristics that may be used in personal identification through analysis of a system according to the principles of the present invention.

(48) FIG. **36b** illustrates a digital content presentation reflection off of the wearer's eye that may be analyzed in accordance with the principles of the present invention.

(49) FIG. **37** illustrates eye imaging along various virtual target lines and various focal planes in accordance with the principles of the present invention.

(50) FIG. **38** illustrates content control with respect to eye movement based on eye imaging in accordance with the principles of the present invention.

(51) FIG. **39** illustrates eye imaging and eye convergence in accordance with the principles of the present invention.

(52) FIG. **40** illustrates content position dependent on sensor feedback in accordance with the principles of the present invention.

(53) FIG. **41** illustrates content position dependent on sensor feedback in accordance with the principles of the present invention.

(54) FIG. **42** illustrates content position dependent on sensor feedback in accordance with the principles of the present invention.

(55) FIG. **43** illustrates content position dependent on sensor feedback in accordance with the principles of the present invention.

(56) FIG. **44** illustrates content position dependent on sensor feedback in accordance with the principles of the present invention.

(57) FIG. **45** illustrates various headings over time in an example.

(58) FIG. **46** illustrates content position dependent on sensor feedback in accordance with the principles of the present invention.

(59) FIG. **47** illustrates content position dependent on sensor feedback in accordance with the principles of the present invention.

(60) FIG. **48** illustrates content position dependent on sensor feedback in accordance with the principles of the present invention.

(61) FIG. **49** illustrates content position dependent on sensor feedback in accordance with the principles of the present invention.

(62) FIG. **50** illustrates light impinging an eye in accordance with the principles of the present invention.

(63) FIG. **51** illustrates a view of an eye in accordance with the principles of the present invention.

(64) FIGS. **52a** and **52b** illustrate views of an eye with a structured light pattern in accordance with the principles of the present invention.

(65) FIG. **53** illustrates an optics module in accordance with the principles of the present invention.

(66) FIG. **54** illustrates an optics module in accordance with the principles of the present invention.

(67) FIG. **55** shows a series of example spectrum for a variety of controlled substances as measured using a form of infrared spectroscopy.

(68) FIG. **56** shows an infrared absorbance spectrum for glucose.

(69) FIG. **57** illustrates a scene where a person is walking with a HWC mounted on his head.

(70) FIG. **58** illustrates a system for receiving, developing and using movement heading, sight heading, eye heading and/or persistence information from HWC(s).

(71) FIG. **59** illustrates a presentation technology in accordance with the principles of the present invention.

(72) FIG. **60** illustrates a presentation technology in accordance with the principles of the present invention.

(73) FIG. **61** illustrates a presentation technology in accordance with the principles of the present invention.

(74) FIG. **62** illustrates a presentation technology in accordance with the principles of the present invention.

(75) FIG. **63** illustrates a presentation technology in accordance with the principles of the present invention.

(76) FIG. **64** illustrates a presentation technology in accordance with the principles of the present invention.

(77) FIG. **65** illustrates a presentation technology in accordance with the principles of the present invention.

(78) FIG. **66** illustrates a presentation technology in accordance with the principles of the present invention.

(79) FIG. **67** illustrates an optical configuration in accordance with the principles of the present invention.

(80) FIG. **68** illustrates an optical configuration in accordance with the principles of the present invention.

(81) FIG. **69** illustrates an optical configuration in accordance with the principles of the present invention.

(82) FIG. **70** illustrates an optical configuration in accordance with the principles of the present invention.

(83) FIG. **71** illustrates an optical configuration in accordance with the principles of the present invention.

(84) FIG. **72** illustrates an optical element in accordance with the principles of the present invention.

(85) FIG. **73** illustrates an optical element in accordance with the principles of the present invention.

(86) FIG. **74** illustrates an optical element in accordance with the principles of the present invention.

(87) FIG. **75** illustrates an optical element in accordance with the principles of the present invention.

(88) FIG. **76** illustrates an optical element in a see-through computer display in accordance with the principles of the present invention.

(89) FIG. **77** illustrates an optical element in accordance with the principles of the present invention.

(90) FIG. **78** illustrates an optical element in accordance with the principles of the present invention.

(91) FIG. **79a** illustrates a schematic of an upper optic in accordance with the principles of the present invention.

(92) FIG. **79** illustrates a schematic of an upper optic in accordance with the principles of the present invention.

(93) FIG. **80** illustrates a stray light control technology in accordance with the principles of the present invention.

(94) FIGS. **81a** and **81b** illustrate a display with a gap and masked technologies in accordance with the principles of the present invention.

(95) FIG. **82** illustrates an upper module with a trim polarizer in accordance with the principles of the present invention.

(96) FIG. **83** illustrates an optical system with a laminated multiple polarizer film in accordance with the principles of the present invention.

(97) FIGS. **84a** and **84b** illustrate partially reflective layers in accordance with the principles of the present invention.

(98) FIG. **84c** illustrates a laminated multiple polarizer with a complex curve in accordance with the principles of the present invention.

(99) FIG. **84d** illustrates a laminated multiple polarizer with a curve in accordance with the principles of the present invention.

(100) FIG. **85** illustrates an optical system adapted for a head-mounted display in accordance with the principles of the present invention.

(101) FIG. **86** illustrates an optical system adapted for a head-mounted display in accordance with the principles of the present invention.

(102) FIG. **87** illustrates an optical system adapted for a head-mounted display in accordance with the principles of the present invention.

(103) FIG. **88** illustrates an optical system adapted for a head-mounted display in accordance with the principles of the present invention.

(104) FIG. **89** illustrates an optical system adapted for a head-mounted display in accordance with the principles of the present invention.

(105) FIG. **90** illustrates an optical system adapted for a head-mounted display in accordance with the principles of the present invention.

(106) FIG. **91** illustrates an optical system in accordance with the principles of the present invention.

(107) FIG. **92** illustrates an optical system in accordance with the principles of the present invention.

(108) FIG. **93** illustrates an optical system in accordance with the principles of the present invention.

(109) FIG. **94** illustrates an optical system in accordance with the principles of the present invention.

(110) FIG. **95** illustrates an optical system in accordance with the principles of the present invention.

(111) FIG. **96** illustrates an optical system in accordance with the principles of the present invention.

(112) FIG. **97** illustrates an optical system in accordance with the principles of the present invention.

(113) FIG. **98** illustrates an optical system in accordance with the principles of the present invention.

(114) FIG. **99** illustrates an optical system in accordance with the principles of the present invention.

(115) FIG. **100** illustrates an optical system in accordance with the principles of the present invention.

(116) FIG. **101** illustrates an optical system in accordance with the principles of the present invention.

(117) FIG. **102** illustrates an optical system in accordance with the principles of the present invention.

(118) FIGS. **103**, **103a** and **103b** illustrate optical systems in accordance with the principles of the present invention.

(119) FIG. **104** illustrates an optical system in accordance with the principles of the present invention.

(120) FIG. **105** illustrates a blocking optic in accordance with the principles of the present invention.

(121) FIGS. **106a**, **106b**, and **106c** illustrate a blocking optic system in accordance with the principles of the present invention.

(122) FIG. **107** illustrates a full color image in accordance with the principles of the present invention.

(123) FIGS. **108A** and **108B** illustrate color breakup management in accordance with the principles of the present invention.

(124) FIG. **109** illustrates timing sequences in accordance with the principles of the present invention.

(125) FIG. **110** illustrates timing sequences in accordance with the principles of the present invention.

(126) FIGS. **111a** and **111b** illustrate sequentially displayed images in accordance with the principles of the present invention.

(127) FIG. **112** illustrates a see-through display with rotated components in accordance with the principles of the present invention.

(128) FIG. **113** illustrates an optics module with twisted reflective surfaces in accordance with the principles of the present invention.

(129) FIG. **114** illustrates PCB and see-through optics module positions within a glasses form factor in accordance with the principles of the present invention.

(130) FIG. **115** illustrates PCB and see-through optics module positions within a glasses form factor in accordance with the principles of the present invention.

- (131) FIG. **116** illustrates a PCB and see-through optics module positions within a glasses form factor in accordance with the principles of the present invention.
- (132) FIG. **117** illustrates a user interface in accordance with the principles of the present invention.
- (133) FIG. **118** illustrates a user interface in accordance with the principles of the present invention.
- (134) FIG. **119** illustrates a lens arrangement in accordance with the principles of the present invention.
- (135) FIGS. **120** and **121** illustrate eye imaging systems in accordance with the principles of the present invention.
- (136) FIG. **122** illustrates an identification process in accordance with the principles of the present invention.
- (137) FIGS. **123** and **124** illustrate combiner assemblies in accordance with the principles of the present invention.
- (138) While the invention has been described in connection with certain preferred embodiments, other embodiments would be understood by one of ordinary skill in the art and are encompassed herein.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENT(S)

- (139) Aspects of the present invention relate to head-worn computing (“HWC”) systems. HWC involves, in some instances, a system that mimics the appearance of head-worn glasses or sunglasses. The glasses may be a fully developed computing platform, such as including computer displays presented in each of the lenses of the glasses to the eyes of the user. In embodiments, the lenses and displays may be configured to allow a person wearing the glasses to see the environment through the lenses while also seeing, simultaneously, digital imagery, which forms an overlaid image that is perceived by the person as a digitally augmented image of the environment, or augmented reality (“AR”).
- (140) HWC involves more than just placing a computing system on a person's head. The system may need to be designed as a lightweight, compact and fully functional computer display, such as wherein the computer display includes a high resolution digital display that provides a high level of emersion comprised of the displayed digital content and the see-through view of the environmental surroundings. User interfaces and control systems suited to the HWC device may be required that are unlike those used for a more conventional computer such as a laptop. For the HWC and associated systems to be most effective, the glasses may be equipped with sensors to determine environmental conditions, geographic location, relative positioning to other points of interest, objects identified by imaging and movement by the user or other users in a connected group, and the like. The HWC may then change the mode of operation to match the conditions, location, positioning, movements, and the like, in a method generally referred to as a contextually aware HWC. The glasses also may need to be connected, wirelessly or otherwise, to other systems either locally or through a network. Controlling the glasses may be achieved through the use of an external device, automatically through contextually gathered information, through user gestures captured by the glasses sensors, and the like. Each technique may be further refined depending on the software application being used in the glasses. The glasses may further be used to control or coordinate with external devices that are associated with the glasses.
- (141) Referring to FIG. **1**, an overview of the HWC system **100** is presented. As shown, the HWC system **100** comprises a HWC **102**, which in this instance is configured as glasses to be worn on the head with sensors such that the HWC **102** is aware of the objects and conditions in the environment **114**. In this instance, the HWC **102** also receives and interprets control inputs such as gestures and movements **116**. The HWC **102** may communicate with external user interfaces **104**. The external user interfaces **104** may provide a physical user interface to take control instructions from a user of the HWC **102** and the external user interfaces **104** and the HWC **102** may communicate bi-directionally to affect the user's command and provide feedback to the external device **108**. The HWC **102** may also communicate bi-directionally with externally controlled or

coordinated local devices **108**. For example, an external user interface **104** may be used in connection with the HWC **102** to control an externally controlled or coordinated local device **108**. The externally controlled or coordinated local device **108** may provide feedback to the HWC **102** and a customized GUI may be presented in the HWC **102** based on the type of device or specifically identified device **108**. The HWC **102** may also interact with remote devices and information sources **112** through a network connection **110**. Again, the external user interface **104** may be used in connection with the HWC **102** to control or otherwise interact with any of the remote devices **108** and information sources **112** in a similar way as when the external user interfaces **104** are used to control or otherwise interact with the externally controlled or coordinated local devices **108**. Similarly, HWC **102** may interpret gestures **116** (e.g. captured from forward, downward, upward, rearward facing sensors such as camera(s), range finders, IR sensors, etc.) or environmental conditions sensed in the environment **114** to control either local or remote devices **108** or **112**.

(142) We will now describe each of the main elements depicted on FIG. **1** in more detail; however, these descriptions are intended to provide general guidance and should not be construed as limiting. Additional description of each element may also be further described herein.

(143) The HWC **102** is a computing platform intended to be worn on a person's head. The HWC **102** may take many different forms to fit many different functional requirements. In some situations, the HWC **102** will be designed in the form of conventional glasses. The glasses may or may not have active computer graphics displays. In situations where the HWC **102** has integrated computer displays the displays may be configured as see-through displays such that the digital imagery can be overlaid with respect to the user's view of the environment **114**. There are a number of see-through optical designs that may be used, including ones that have a reflective display (e.g. LCoS, DLP), emissive displays (e.g. OLED, LED), hologram, TIR waveguides, and the like. In embodiments, lighting systems used in connection with the display optics may be solid state lighting systems, such as LED, OLED, quantum dot, quantum dot LED, etc. In addition, the optical configuration may be monocular or binocular. It may also include vision corrective optical components. In embodiments, the optics may be packaged as contact lenses. In other embodiments, the HWC **102** may be in the form of a helmet with a see-through shield, sunglasses, safety glasses, goggles, a mask, fire helmet with see-through shield, police helmet with see through shield, military helmet with see-through shield, utility form customized to a certain work task (e.g. inventory control, logistics, repair, maintenance, etc.), and the like.

(144) The HWC **102** may also have a number of integrated computing facilities, such as an integrated processor, integrated power management, communication structures (e.g. cell net, WiFi, Bluetooth, local area connections, mesh connections, remote connections (e.g. client server, etc.)), and the like. The HWC **102** may also have a number of positional awareness sensors, such as GPS, electronic compass, altimeter, tilt sensor, IMU, and the like. It may also have other sensors such as a camera, rangefinder, hyper-spectral camera, Geiger counter, microphone, spectral illumination detector, temperature sensor, chemical sensor, biologic sensor, moisture sensor, ultrasonic sensor, and the like.

(145) The HWC **102** may also have integrated control technologies. The integrated control technologies may be contextual based control, passive control, active control, user control, and the like. For example, the HWC **102** may have an integrated sensor (e.g. camera) that captures user hand or body gestures **116** such that the integrated processing system can interpret the gestures and generate control commands for the HWC **102**. In another example, the HWC **102** may have sensors that detect movement (e.g. a nod, head shake, and the like) including accelerometers, gyros and other inertial measurements, where the integrated processor may interpret the movement and generate a control command in response. The HWC **102** may also automatically control itself based on measured or perceived environmental conditions. For example, if it is bright in the environment the HWC **102** may increase the brightness or contrast of the displayed image. In embodiments, the

integrated control technologies may be mounted on the HWC **102** such that a user can interact with it directly. For example, the HWC **102** may have a button(s), touch capacitive interface, and the like.

(146) As described herein, the HWC **102** may be in communication with external user interfaces **104**. The external user interfaces may come in many different forms. For example, a cell phone screen may be adapted to take user input for control of an aspect of the HWC **102**. The external user interface may be a dedicated UI, such as a keyboard, touch surface, button(s), joy stick, and the like. In embodiments, the external controller may be integrated into another device such as a ring, watch, bike, car, and the like. In each case, the external user interface **104** may include sensors (e.g. IMU, accelerometers, compass, altimeter, and the like) to provide additional input for controlling the HWC **102**.

(147) As described herein, the HWC **102** may control or coordinate with other local devices **108**. The external devices **108** may be an audio device, visual device, vehicle, cell phone, computer, and the like. For instance, the local external device **108** may be another HWC **102**, where information may then be exchanged between the separate HWCs **108**.

(148) Similar to the way the HWC **102** may control or coordinate with local devices **106**, the HWC **102** may control or coordinate with remote devices **112**, such as the HWC **102** communicating with the remote devices **112** through a network **110**. Again, the form of the remote device **112** may have many forms. Included in these forms is another HWC **102**. For example, each HWC **102** may communicate its GPS position such that all the HWCs **102** know where all of HWC **102** are located.

(149) FIG. 2 illustrates a HWC **102** with an optical system that includes an upper optical module **202** and a lower optical module **204**. While the upper and lower optical modules **202** and **204** will generally be described as separate modules, it should be understood that this is illustrative only and the present invention includes other physical configurations, such as that when the two modules are combined into a single module or where the elements making up the two modules are configured into more than two modules. In embodiments, the upper module **202** includes a computer controlled display (e.g. LCoS, DLP, OLED, etc.) and image light delivery optics. In embodiments, the lower module includes eye delivery optics that are configured to receive the upper module's image light and deliver the image light to the eye of a wearer of the HWC. In FIG. 2, it should be noted that while the upper and lower optical modules **202** and **204** are illustrated in one side of the HWC such that image light can be delivered to one eye of the wearer, that it is envisioned by the present invention that embodiments will contain two image light delivery systems, one for each eye.

(150) FIG. 3b illustrates an upper optical module **202** in accordance with the principles of the present invention. In this embodiment, the upper optical module **202** includes a DLP (also known as DMD or digital micromirror device) computer operated display **304** which includes pixels comprised of rotatable mirrors (such as, for example, the DLP3000 available from Texas Instruments), polarized light source **302**, $\frac{1}{4}$ wave retarder film **308**, reflective polarizer **310** and a field lens **312**. The polarized light source **302** provides substantially uniform polarized light that is generally directed towards the reflective polarizer **310**. The reflective polarizer reflects light of one polarization state (e.g. S polarized light) and transmits light of the other polarization state (e.g. P polarized light). The polarized light source **302** and the reflective polarizer **310** are oriented so that the polarized light from the polarized light source **302** is reflected generally towards the DLP **304**. The light then passes through the $\frac{1}{4}$ wave film **308** once before illuminating the pixels of the DLP **304** and then again after being reflected by the pixels of the DLP **304**. In passing through the $\frac{1}{4}$ wave film **308** twice, the light is converted from one polarization state to the other polarization state (e.g. the light is converted from S to P polarized light). The light then passes through the reflective polarizer **310**. In the event that the DLP pixel(s) are in the "on" state (i.e. the mirrors are positioned to reflect light towards the field lens **312**, the "on" pixels reflect the light generally

along the optical axis and into the field lens **312**. This light that is reflected by “on” pixels and which is directed generally along the optical axis of the field lens **312** will be referred to as image light **316**. The image light **316** then passes through the field lens to be used by a lower optical module **204**.

(151) The light that is provided by the polarized light source **302**, which is subsequently reflected by the reflective polarizer **310** before it reflects from the DLP **304**, will generally be referred to as illumination light. The light that is reflected by the “off” pixels of the DLP **304** is reflected at a different angle than the light reflected by the “on” pixels, so that the light from the “off” pixels is generally directed away from the optical axis of the field lens **312** and toward the side of the upper optical module **202** as shown in FIG. **3**. The light that is reflected by the “off” pixels of the DLP **304** will be referred to as dark state light **314**.

(152) The DLP **304** operates as a computer controlled display and is generally thought of as a MEMs device. The DLP pixels are comprised of small mirrors that can be directed. The mirrors generally flip from one angle to another angle. The two angles are generally referred to as states. When light is used to illuminate the DLP the mirrors will reflect the light in a direction depending on the state. In embodiments herein, we generally refer to the two states as “on” and “off,” which is intended to depict the condition of a display pixel. “On” pixels will be seen by a viewer of the display as emitting light because the light is directed along the optical axis and into the field lens and the associated remainder of the display system. “Off” pixels will be seen by a viewer of the display as not emitting light because the light from these pixels is directed to the side of the optical housing and into a light trap or light dump where the light is absorbed. The pattern of “on” and “off” pixels produces image light that is perceived by a viewer of the display as a computer generated image. Full color images can be presented to a user by sequentially providing illumination light with complimentary colors such as red, green and blue. Where the sequence is presented in a recurring cycle that is faster than the user can perceive as separate images and as a result the user perceives a full color image comprised of the sum of the sequential images. Bright pixels in the image are provided by pixels that remain in the “on” state for the entire time of the cycle, while dimmer pixels in the image are provided by pixels that switch between the “on” state and “off” state within the time of the cycle, or frame time when in a video sequence of images.

(153) FIG. **3a** shows an illustration of a system for a DLP **304** in which the unpolarized light source **350** is pointed directly at the DLP **304**. In this case, the angle required for the illumination light is such that the field lens **352** must be positioned substantially distant from the DLP **304** to avoid the illumination light from being clipped by the field lens **352**. The large distance between the field lens **352** and the DLP **304** along with the straight path of the dark state light **354**, means that the light trap for the dark state light **354** is also located at a substantial distance from the DLP. For these reasons, this configuration is larger in size compared to the upper optics module **202** of the preferred embodiments.

(154) The configuration illustrated in FIG. **3b** can be lightweight and compact such that it fits into a small portion of a HWC. For example, the upper modules **202** illustrated herein can be physically adapted to mount in an upper frame of a HWC such that the image light can be directed into a lower optical module **204** for presentation of digital content to a wearer's eye. The package of components that combine to generate the image light (i.e. the polarized light source **302**, DLP **304**, reflective polarizer **310** and $\frac{1}{4}$ wave film **308**) is very light and is compact. The height of the system, excluding the field lens, may be less than 8 mm. The width (i.e. from front to back) may be less than 8 mm. The weight may be less than 2 grams. The compactness of this upper optical module **202** allows for a compact mechanical design of the HWC and the light weight nature of these embodiments help make the HWC lightweight to provide for a HWC that is comfortable for a wearer of the HWC.

(155) The configuration illustrated in FIG. **3b** can produce sharp contrast, high brightness and deep blacks, especially when compared to LCD or LCoS displays used in HWC. The “on” and “off”

states of the DLP provide for a strong differentiator in the light reflection path representing an “on” pixel and an “off” pixel. As will be discussed in more detail below, the dark state light from the “off” pixel reflections can be managed to reduce stray light in the display system to produce images with high contrast.

(156) FIG. 4 illustrates another embodiment of an upper optical module **202** in accordance with the principles of the present invention. This embodiment includes a light source **404**, but in this case, the light source can provide unpolarized illumination light. The illumination light from the light source **404** is directed into a TIR wedge **418** such that the illumination light is incident on an internal surface of the TIR wedge **418** (shown as the angled lower surface of the TRI wedge **418** in FIG. 4) at an angle that is beyond the critical angle as defined by Eqn 1.

Critical angle= $\arcsin(1/n)$ Eqn 1

(157) Where the critical angle is the angle beyond which the illumination light is reflected from the internal surface when the internal surface comprises an interface from a solid with a higher refractive index (n) to air with a refractive index of 1 (e.g. for an interface of acrylic, with a refractive index of n=1.5, to air, the critical angle is 41.8 degrees; for an interface of polycarbonate, with a refractive index of n=1.59, to air the critical angle is 38.9 degrees). Consequently, the TIR wedge **418** is associated with a thin air gap **408** along the internal surface to create an interface between a solid with a higher refractive index and air. By choosing the angle of the light source **404** relative to the DLP **402** in correspondence to the angle of the internal surface of the TIR wedge **418**, illumination light is turned toward the DLP **402** at an angle suitable for providing image light **414** as reflected from “on” pixels. Wherein, the illumination light is provided to the DLP **402** at approximately twice the angle of the pixel mirrors in the DLP **402** that are in the “on” state, such that after reflecting from the pixel mirrors, the image light **414** is directed generally along the optical axis of the field lens. Depending on the state of the DLP pixels, the illumination light from “on” pixels may be reflected as image light **414** which is directed towards a field lens and a lower optical module **204**, while illumination light reflected from “off” pixels (generally referred to herein as “dark” state light, “off” pixel light or “off” state light) **410** is directed in a separate direction, which may be trapped and not used for the image that is ultimately presented to the wearer's eye.

(158) The light trap for the dark state light **410** may be located along the optical axis defined by the direction of the dark state light **410** and in the side of the housing, with the function of absorbing the dark state light. To this end, the light trap may be comprised of an area outside of the cone of image light **414** from the “on” pixels. The light trap is typically made up of materials that absorb light including coatings of black paints or other light absorbing materials to prevent light scattering from the dark state light degrading the image perceived by the user. In addition, the light trap may be recessed into the wall of the housing or include masks or guards to block scattered light and prevent the light trap from being viewed adjacent to the displayed image.

(159) The embodiment of FIG. 4 also includes a corrective wedge **420** to correct the effect of refraction of the image light **414** as it exits the TIR wedge **418**. By including the corrective wedge **420** and providing a thin air gap **408** (e.g. 25 micron), the image light from the “on” pixels can be maintained generally in a direction along the optical axis of the field lens (i.e. the same direction as that defined by the image light **414**) so it passes into the field lens and the lower optical module **204**. As shown in FIG. 4, the image light **414** from the “on” pixels exits the corrective wedge **420** generally perpendicular to the surface of the corrective wedge **420** while the dark state light exits at an oblique angle. As a result, the direction of the image light **414** from the “on” pixels is largely unaffected by refraction as it exits from the surface of the corrective wedge **420**. In contrast, the dark state light **410** is substantially changed in direction by refraction when the dark state light **410** exits the corrective wedge **420**.

(160) The embodiment illustrated in FIG. 4 has the similar advantages of those discussed in connection with the embodiment of FIG. 3b. The dimensions and weight of the upper module **202**

depicted in FIG. 4 may be approximately 8×8 mm with a weight of less than 3 grams. A difference in overall performance between the configuration illustrated in FIGS. 3b and the configuration illustrated in FIG. 4 is that the embodiment of FIG. 4 doesn't require the use of polarized light as supplied by the light source 404. This can be an advantage in some situations as will be discussed in more detail below (e.g. increased see-through transparency of the HWC optics from the user's perspective). Polarized light may be used in connection with the embodiment depicted in FIG. 4, in embodiments. An additional advantage of the embodiment of FIG. 4 compared to the embodiment shown in FIG. 3b is that the dark state light (shown as DLP off light 410) is directed at a steeper angle away from the optical axis of the image light 414 due to the added refraction encountered when the dark state light 410 exits the corrective wedge 420. This steeper angle of the dark state light 410 allows for the light trap to be positioned closer to the DLP 402 so that the overall size of the upper module 202 can be reduced. The light trap can also be made larger since the light trap doesn't interfere with the field lens, thereby the efficiency of the light trap can be increased and as a result, stray light can be reduced and the contrast of the image perceived by the user can be increased. FIG. 4a illustrates the embodiment described in connection with FIG. 4 with an example set of corresponding angles at the various surfaces with the reflected angles of a ray of light passing through the upper optical module 202. In this example, the DLP mirrors are provided at 17 degrees to the surface of the DLP device. The angles of the TIR wedge are selected in correspondence to one another to provide TIR reflected illumination light at the correct angle for the DLP mirrors while allowing the image light and dark state light to pass through the thin air gap, various combinations of angles are possible to achieve this.

(161) FIG. 5 illustrates yet another embodiment of an upper optical module 202 in accordance with the principles of the present invention. As with the embodiment shown in FIG. 4, the embodiment shown in FIG. 5 does not require the use of polarized light. Polarized light may be used in connection with this embodiment, but it is not required. The optical module 202 depicted in FIG. 5 is similar to that presented in connection with FIG. 4; however, the embodiment of FIG. 5 includes an off light redirection wedge 502. As can be seen from the illustration, the off light redirection wedge 502 allows the image light 414 to continue generally along the optical axis toward the field lens and into the lower optical module 204 (as illustrated). However, the off light 504 is redirected substantially toward the side of the corrective wedge 420 where it passes into the light trap. This configuration may allow further height compactness in the HWC because the light trap (not illustrated) that is intended to absorb the off light 504 can be positioned laterally adjacent the upper optical module 202 as opposed to below it. In the embodiment depicted in FIG. 5 there is a thin air gap between the TIR wedge 418 and the corrective wedge 420 (similar to the embodiment of FIG. 4). There is also a thin air gap between the corrective wedge 420 and the off light redirection wedge 502. There may be HWC mechanical configurations that warrant the positioning of a light trap for the dark state light elsewhere and the illustration depicted in FIG. 5 should be considered illustrative of the concept that the off light can be redirected to create compactness of the overall HWC. FIG. 5a illustrates an example of the embodiment described in connection with FIG. 5 with the addition of more details on the relative angles at the various surfaces and a light ray trace for image light and a light ray trace for dark light are shown as it passes through the upper optical module 202. Again, various combinations of angles are possible.

(162) FIG. 4b shows an illustration of a further embodiment in which a solid transparent matched set of wedges 456 is provided with a reflective polarizer 450 at the interface between the wedges. Wherein the interface between the wedges in the wedge set 456 is provided at an angle so that illumination light 452 from the polarized light source 458 is reflected at the proper angle (e.g. 34 degrees for a 17 degree DLP mirror) for the DLP mirror “on” state so that the reflected image light 414 is provided along the optical axis of the field lens. The general geometry of the wedges in the wedge set 456 is similar to that shown in FIGS. 4 and 4a. A quarter wave film 454 is provided on the DLP 402 surface so that the illumination light 452 is one polarization state (e.g. S polarization

state) while in passing through the quarter wave film **454**, reflecting from the DLP mirror and passing back through the quarter wave film **454**, the image light **414** is converted to the other polarization state (e.g. P polarization state). The reflective polarizer is oriented such that the illumination light **452** with its polarization state is reflected and the image light **414** with its other polarization state is transmitted. Since the dark state light from the “off pixels **410**” also passes through the quarter wave film **454** twice, it is also the other polarization state (e.g. P polarization state) so that it is transmitted by the reflective polarizer **450**.

(163) The angles of the faces of the wedge set **450** correspond to the needed angles to provide illumination light **452** at the angle needed by the DLP mirrors when in the “on” state so that the reflected image light **414** is reflected from the DLP along the optical axis of the field lens. The wedge set **456** provides an interior interface where a reflective polarizer film can be located to redirect the illumination light **452** toward the mirrors of the DLP **402**. The wedge set also provides a matched wedge on the opposite side of the reflective polarizer **450** so that the image light **414** from the “on” pixels exits the wedge set **450** substantially perpendicular to the exit surface, while the dark state light from the “off” pixels **410** exits at an oblique angle to the exit surface. As a result, the image light **414** is substantially unrefracted upon exiting the wedge set **456**, while the dark state light from the “off” pixels **410** is substantially refracted upon exiting the wedge set **456** as shown in FIG. **4b**.

(164) By providing a solid transparent matched wedge set, the flatness of the interface is reduced, because variations in the flatness have a negligible effect as long as they are within the cone angle of the illuminating light **452**. Which can be $f \#2.2$ with a 26 degree cone angle. In a preferred embodiment, the reflective polarizer is bonded between the matched internal surfaces of the wedge set **456** using an optical adhesive so that Fresnel reflections at the interfaces on either side of the reflective polarizer **450** are reduced. The optical adhesive can be matched in refractive index to the material of the wedge set **456** and the pieces of the wedge set **456** can be all made from the same material such as BK7 glass or cast acrylic. Wherein the wedge material can be selected to have low birefringence as well to reduce non-uniformities in brightness. The wedge set **456** and the quarter wave film **454** can also be bonded to the DLP **402** to further reduce Fresnel reflections at the DLP interface losses. In addition, since the image light **414** is substantially normal to the exit surface of the wedge set **456**, the flatness of the surface is not critical to maintain the wavefront of the image light **414** so that high image quality can be obtained in the displayed image without requiring very tightly toleranced flatness on the exit surface.

(165) A yet further embodiment of the invention that is not illustrated, combines the embodiments illustrated in FIGS. **4b** and FIG. **5**. In this embodiment, the wedge set **456** is comprised of three wedges with the general geometry of the wedges in the wedge set corresponding to that shown in FIGS. **5** and **5a**. A reflective polarizer is bonded between the first and second wedges similar to that shown in FIG. **4b**, however, a third wedge is provided similar to the embodiment of FIG. **5**. Wherein there is an angled thin air gap between the second and third wedges so that the dark state light is reflected by TIR toward the side of the second wedge where it is absorbed in a light trap. This embodiment, like the embodiment shown in FIG. **4b**, uses a polarized light source as has been previously described. The difference in this embodiment is that the image light is transmitted through the reflective polarizer and is transmitted through the angled thin air gap so that it exits normal to the exit surface of the third wedge.

(166) FIG. **5b** illustrates an upper optical module **202** with a dark light trap **514a**. As described in connection with FIGS. **4** and **4a**, image light can be generated from a DLP when using a TIR and corrective lens configuration. The upper module may be mounted in a HWC housing **510** and the housing **510** may include a dark light trap **514a**. The dark light trap **514a** is generally positioned/constructed/formed in a position that is optically aligned with the dark light optical axis **512**. As illustrated, the dark light trap may have depth such that the trap internally reflects dark light in an attempt to further absorb the light and prevent the dark light from combining with the

image light that passes through the field lens. The dark light trap may be of a shape and depth such that it absorbs the dark light. In addition, the dark light trap **514b**, in embodiments, may be made of light absorbing materials or coated with light absorbing materials. In embodiments, the recessed light trap **514a** may include baffles to block a view of the dark state light. This may be combined with black surfaces and textured or fibrous surfaces to help absorb the light. The baffles can be part of the light trap, associated with the housing, or field lens, etc.

(167) FIG. **5c** illustrates another embodiment with a light trap **514b**. As can be seen in the illustration, the shape of the trap is configured to enhance internal reflections within the light trap **514b** to increase the absorption of the dark light **512**. FIG. **5d** illustrates another embodiment with a light trap **514c**. As can be seen in the illustration, the shape of the trap **514c** is configured to enhance internal reflections to increase the absorption of the dark light **512**.

(168) FIG. **5e** illustrates another embodiment of an upper optical module **202** with a dark light trap **514d**. This embodiment of upper module **202** includes an off light reflection wedge **502**, as illustrated and described in connection with the embodiment of FIGS. **5** and **5a**. As can be seen in FIG. **5e**, the light trap **514d** is positioned along the optical path of the dark light **512**. The dark light trap **514d** may be configured as described in other embodiments herein. The embodiment of the light trap **514d** illustrated in FIG. **5e** includes a black area on the side wall of the wedge, wherein the side wall is located substantially away from the optical axis of the image light **414**. In addition, baffles **5252** may be added to one or more edges of the field lens **312** to block the view of the light trap **514d** adjacent to the displayed image seen by the user.

(169) FIG. **6** illustrates a combination of an upper optical module **202** with a lower optical module **204**. In this embodiment, the image light projected from the upper optical module **202** may or may not be polarized. The image light is reflected off a flat combiner element **602** such that it is directed towards the user's eye. Wherein, the combiner element **602** is a partial mirror that reflects image light while transmitting a substantial portion of light from the environment so the user can look through the combiner element and see the environment surrounding the HWC.

(170) The combiner **602** may include a holographic pattern, to form a holographic mirror. If a monochrome image is desired, there may be a single wavelength reflection design for the holographic pattern on the surface of the combiner **602**. If the intention is to have multiple colors reflected from the surface of the combiner **602**, a multiple wavelength holographic mirror may be included on the combiner surface. For example, in a three-color embodiment, where red, green and blue pixels are generated in the image light, the holographic mirror may be reflective to wavelengths substantially matching the wavelengths of the red, green and blue light provided by the light source. This configuration can be used as a wavelength specific mirror where pre-determined wavelengths of light from the image light are reflected to the user's eye. This configuration may also be made such that substantially all other wavelengths in the visible pass through the combiner element **602** so the user has a substantially clear view of the surroundings when looking through the combiner element **602**. The transparency between the user's eye and the surrounding may be approximately 80% when using a combiner that is a holographic mirror. Wherein holographic mirrors can be made using lasers to produce interference patterns in the holographic material of the combiner where the wavelengths of the lasers correspond to the wavelengths of light that are subsequently reflected by the holographic mirror.

(171) In another embodiment, the combiner element **602** may include a notch mirror comprised of a multilayer coated substrate wherein the coating is designed to substantially reflect the wavelengths of light provided by the light source and substantially transmit the remaining wavelengths in the visible spectrum. For example, in the case where red, green and blue light is provided by the light source to enable full color images to be provided to the user, the notch mirror is a tristimulus notch mirror wherein the multilayer coating is designed to reflect narrow bands of red, green and blue light that are matched to the what is provided by the light source and the remaining visible wavelengths are transmitted through the coating to enable a view of the

environment through the combiner. In another example where monochrome images are provided to the user, the notch mirror is designed to reflect a single narrow band of light that is matched to the wavelength range of the light provided by the light source while transmitting the remaining visible wavelengths to enable a see-thru view of the environment. The combiner **602** with the notch mirror would operate, from the user's perspective, in a manner similar to the combiner that includes a holographic pattern on the combiner element **602**. The combiner, with the tristimulus notch mirror, would reflect the "on" pixels to the eye because of the match between the reflective wavelengths of the notch mirror and the color of the image light, and the wearer would be able to see with high clarity the surroundings. The transparency between the user's eye and the surrounding may be approximately 80% when using the tristimulus notch mirror. In addition, the image provided by the upper optical module **202** with the notch mirror combiner can provide higher contrast images than the holographic mirror combiner due to less scattering of the imaging light by the combiner.

(172) Light can escape through the combiner **602** and may produce face glow as the light is generally directed downward onto the cheek of the user. When using a holographic mirror combiner or a tristimulus notch mirror combiner, the escaping light can be trapped to avoid face glow. In embodiments, if the image light is polarized before the combiner, a linear polarizer can be laminated, or otherwise associated, to the combiner, with the transmission axis of the polarizer oriented relative to the polarized image light so that any escaping image light is absorbed by the polarizer. In embodiments, the image light would be polarized to provide S polarized light to the combiner for better reflection. As a result, the linear polarizer on the combiner would be oriented to absorb S polarized light and pass P polarized light. This provides the preferred orientation of polarized sunglasses as well.

(173) If the image light is unpolarized, a microlouvered film such as a privacy filter can be used to absorb the escaping image light while providing the user with a see-thru view of the environment. In this case, the absorbance or transmittance of the microlouvered film is dependent on the angle of the light. Where steep angle light is absorbed and light at less of an angle is transmitted. For this reason, in an embodiment, the combiner with the microlouver film is angled at greater than 45 degrees to the optical axis of the image light (e.g. the combiner can be oriented at 50 degrees so the image light from the file lens is incident on the combiner at an oblique angle).

(174) FIG. 7 illustrates an embodiment of a combiner element **602** at various angles when the combiner element **602** includes a holographic mirror. Normally, a mirrored surface reflects light at an angle equal to the angle that the light is incident to the mirrored surface. Typically, this necessitates that the combiner element be at 45 degrees, **602a**, if the light is presented vertically to the combiner so the light can be reflected horizontally towards the wearer's eye. In embodiments, the incident light can be presented at angles other than vertical to enable the mirror surface to be oriented at other than 45 degrees, but in all cases wherein a mirrored surface is employed (including the tristimulus notch mirror described previously), the incident angle equals the reflected angle. As a result, increasing the angle of the combiner **602a** requires that the incident image light be presented to the combiner **602a** at a different angle which positions the upper optical module **202** to the left of the combiner as shown in FIG. 7. In contrast, a holographic mirror combiner, included in embodiments, can be made such that light is reflected at a different angle from the angle that the light is incident onto the holographic mirrored surface. This allows freedom to select the angle of the combiner element **602b** independent of the angle of the incident image light and the angle of the light reflected into the wearer's eye. In embodiments, the angle of the combiner element **602b** is greater than 45 degrees (shown in FIG. 7) as this allows a more laterally compact HWC design. The increased angle of the combiner element **602b** decreases the front to back width of the lower optical module **204** and may allow for a thinner HWC display (i.e. the furthest element from the wearer's eye can be closer to the wearer's face).

(175) FIG. 8 illustrates another embodiment of a lower optical module **204**. In this embodiment, polarized image light provided by the upper optical module **202**, is directed into the lower optical

module **204**. The image light reflects off a polarized mirror **804** and is directed to a focusing partially reflective mirror **802**, which is adapted to reflect the polarized light. An optical element such as a $\frac{1}{4}$ wave film located between the polarized mirror **804** and the partially reflective mirror **802**, is used to change the polarization state of the image light such that the light reflected by the partially reflective mirror **802** is transmitted by the polarized mirror **804** to present image light to the eye of the wearer. The user can also see through the polarized mirror **804** and the partially reflective mirror **802** to see the surrounding environment. As a result, the user perceives a combined image comprised of the displayed image light overlaid onto the see-thru view of the environment.

(176) While many of the embodiments of the present invention have been referred to as upper and lower modules containing certain optical components, it should be understood that the image light and dark light production and management functions described in connection with the upper module may be arranged to direct light in other directions (e.g. upward, sideward, etc.). In embodiments, it may be preferred to mount the upper module **202** above the wearer's eye, in which case the image light would be directed downward. In other embodiments it may be preferred to produce light from the side of the wearer's eye, or from below the wearer's eye. In addition, the lower optical module is generally configured to deliver the image light to the wearer's eye and allow the wearer to see through the lower optical module, which may be accomplished through a variety of optical components.

(177) FIG. **8a** illustrates an embodiment of the present invention where the upper optical module **202** is arranged to direct image light into a TIR waveguide **810**. In this embodiment, the upper optical module **202** is positioned above the wearer's eye **812** and the light is directed horizontally into the TIR waveguide **810**. The TIR waveguide is designed to internally reflect the image light in a series of downward TIR reflections until it reaches the portion in front of the wearer's eye, where the light passes out of the TIR waveguide **812** into the wearer's eye. In this embodiment, an outer shield **814** is positioned in front of the TIR waveguide **810**.

(178) FIG. **8b** illustrates an embodiment of the present invention where the upper optical module **202** is arranged to direct image light into a TIR waveguide **818**. In this embodiment, the upper optical module **202** is arranged on the side of the TIR waveguide **818**. For example, the upper optical module may be positioned in the arm or near the arm of the HWC when configured as a pair of head worn glasses. The TIR waveguide **818** is designed to internally reflect the image light in a series of TIR reflections until it reaches the portion in front of the wearer's eye, where the light passes out of the TIR waveguide **812** into the wearer's eye.

(179) FIG. **8c** illustrates yet further embodiments of the present invention where an upper optical module **202** is directing polarized image light into an optical guide **828** where the image light passes through a polarized reflector **824**, changes polarization state upon reflection of the optical element **822** which includes a $\frac{1}{4}$ wave film for example and then is reflected by the polarized reflector **824** towards the wearer's eye, due to the change in polarization of the image light. The upper optical module **202** may be positioned to direct light to a mirror **820**, to position the upper optical module **202** laterally, in other embodiments, the upper optical module **202** may direct the image light directly towards the polarized reflector **824**. It should be understood that the present invention comprises other optical arrangements intended to direct image light into the wearer's eye.

(180) Another aspect of the present invention relates to eye imaging. In embodiments, a camera is used in connection with an upper optical module **202** such that the wearer's eye can be imaged using pixels in the "off" state on the DLP. FIG. **9** illustrates a system where the eye imaging camera **802** is mounted and angled such that the field of view of the eye imaging camera **802** is redirected toward the wearer's eye by the mirror pixels of the DLP **402** that are in the "off" state. In this way, the eye imaging camera **802** can be used to image the wearer's eye along the same optical axis as the displayed image that is presented to the wearer. Wherein, image light that is presented to the wearer's eye illuminates the wearer's eye so that the eye can be imaged by the eye imaging camera

802. In the process, the light reflected by the eye passes back through the optical train of the lower optical module **204** and a portion of the upper optical module to where the light is reflected by the “off” pixels of the DLP **402** toward the eye imaging camera **802**.

(181) In embodiments, the eye imaging camera may image the wearer's eye at a moment in time where there are enough “off” pixels to achieve the required eye image resolution. In another embodiment, the eye imaging camera collects eye image information from “off” pixels over time and forms a time lapsed image. In another embodiment, a modified image is presented to the user wherein enough “off” state pixels are included that the camera can obtain the desired resolution and brightness for imaging the wearer's eye and the eye image capture is synchronized with the presentation of the modified image.

(182) The eye imaging system may be used for security systems. The HWC may not allow access to the HWC or other system if the eye is not recognized (e.g. through eye characteristics including retina or iris characteristics, etc.). The HWC may be used to provide constant security access in some embodiments. For example, the eye security confirmation may be a continuous, near-continuous, real-time, quasi real-time, periodic, etc. process so the wearer is effectively constantly being verified as known. In embodiments, the HWC may be worn and eye security tracked for access to other computer systems.

(183) The eye imaging system may be used for control of the HWC. For example, a blink, wink, or particular eye movement may be used as a control mechanism for a software application operating on the HWC or associated device.

(184) The eye imaging system may be used in a process that determines how or when the HWC **102** delivers digitally displayed content to the wearer. For example, the eye imaging system may determine that the user is looking in a direction and then HWC may change the resolution in an area of the display or provide some content that is associated with something in the environment that the user may be looking at. Alternatively, the eye imaging system may identify different user's and change the displayed content or enabled features provided to the user. User's may be identified from a database of users eye characteristics either located on the HWC **102** or remotely located on the network **110** or on a server **112**. In addition, the HWC may identify a primary user or a group of primary users from eye characteristics wherein the primary user(s) are provided with an enhanced set of features and all other user's are provided with a different set of features. Thus in this use case, the HWC **102** uses identified eye characteristics to either enable features or not and eye characteristics need only be analyzed in comparison to a relatively small database of individual eye characteristics.

(185) FIG. **10** illustrates a light source that may be used in association with the upper optics module **202** (e.g. polarized light source if the light from the solid state light source is polarized such as polarized light source **302** and **458**), and light source **404**. In embodiments, to provide a uniform surface of light **1008** to be directed into the upper optical module **202** and towards the DLP of the upper optical module, either directly or indirectly, the solid state light source **1002** may be projected into a backlighting optical system **1004**. The solid state light source **1002** may be one or more LEDs, laser diodes, OLEDs. In embodiments, the backlighting optical system **1004** includes an extended section with a length/distance ratio of greater than 3, wherein the light undergoes multiple reflections from the sidewalls to mix or homogenize the light as supplied by the solid state light source **1002**. The backlighting optical system **1004** can also include structures on the surface opposite (on the left side as shown in FIG. **10**) to where the uniform light **1008** exits the backlight **1004** to change the direction of the light toward the DLP **302** and the reflective polarizer **310** or the DLP **402** and the TIR wedge **418**. The backlighting optical system **1004** may also include structures to collimate the uniform light **1008** to provide light to the DLP with a smaller angular distribution or narrower cone angle. Diffusers or polarizers can be used on the entrance or exit surface of the backlighting optical system. Diffusers can be used to spread or uniformize the exiting light from the backlight to improve the uniformity or increase the angular spread of the uniform light **1008**.

Elliptical diffusers that diffuse the light more in some directions and less in others can be used to improve the uniformity or spread of the uniform light **1008** in directions orthogonal to the optical axis of the uniform light **1008**. Linear polarizers can be used to convert unpolarized light as supplied by the solid state light source **1002** to polarized light so the uniform light **1008** is polarized with a desired polarization state. A reflective polarizer can be used on the exit surface of the backlight **1004** to polarize the uniform light **1008** to the desired polarization state, while reflecting the other polarization state back into the backlight where it is recycled by multiple reflections within the backlight **1004** and at the solid state light source **1002**. Therefore by including a reflective polarizer at the exit surface of the backlight **1004**, the efficiency of the polarized light source is improved.

(186) FIGS. **10a** and **10b** show illustrations of structures in backlight optical systems **1004** that can be used to change the direction of the light provided to the entrance face **1045** by the light source and then collimates the light in a direction lateral to the optical axis of the exiting uniform light **1008**. Structure **1060** includes an angled sawtooth pattern in a transparent waveguide wherein the left edge of each sawtooth clips the steep angle rays of light thereby limiting the angle of the light being redirected. The steep surface at the right (as shown) of each sawtooth then redirects the light so that it reflects off the left angled surface of each sawtooth and is directed toward the exit surface **1040**. The sawtooth surfaces shown on the lower surface in FIGS. **10a** and **10b**, can be smooth and coated (e.g. with an aluminum coating or a dielectric mirror coating) to provide a high level of reflectivity without scattering. Structure **1050** includes a curved face on the left side (as shown) to focus the rays after they pass through the exit surface **1040**, thereby providing a mechanism for collimating the uniform light **1008**. In a further embodiment, a diffuser can be provided between the solid state light source **1002** and the entrance face **1045** to homogenize the light provided by the solid state light source **1002**. In yet a further embodiment, a polarizer can be used between the diffuser and the entrance face **1045** of the backlight **1004** to provide a polarized light source. Because the sawtooth pattern provides smooth reflective surfaces, the polarization state of the light can be preserved from the entrance face **1045** to the exit face **1040**. In this embodiment, the light entering the backlight from the solid state light source **1002** passes through the polarizer so that it is polarized with the desired polarization state. If the polarizer is an absorptive linear polarizer, the light of the desired polarization state is transmitted while the light of the other polarization state is absorbed. If the polarizer is a reflective polarizer, the light of the desired polarization state is transmitted into the backlight **1004** while the light of the other polarization state is reflected back into the solid state light source **1002** where it can be recycled as previously described, to increase the efficiency of the polarized light source.

(187) FIG. **11a** illustrates a light source **1100** that may be used in association with the upper optics module **202**. In embodiments, the light source **1100** may provide light to a backlighting optical system **1004** as described above in connection with FIG. **10**. In embodiments, the light source **1100** includes a tristimulus notch filter **1102**. The tristimulus notch filter **1102** has narrow band pass filters for three wavelengths, as indicated in FIG. **11c** in a transmission graph **1108**. The graph shown in FIG. **11b**, as **1104** illustrates an output of three different colored LEDs. One can see that the bandwidths of emission are narrow, but they have long tails. The tristimulus notch filter **1102** can be used in connection with such LEDs to provide a light source **1100** that emits narrow filtered wavelengths of light as shown in FIG. **11d** as the transmission graph **1110**. Wherein the clipping effects of the tristimulus notch filter **1102** can be seen to have cut the tails from the LED emission graph **1104** to provide narrower wavelength bands of light to the upper optical module **202**. The light source **1100** can be used in connection with a combiner **602** with a holographic mirror or tristimulus notch mirror to provide narrow bands of light that are reflected toward the wearer's eye with less waste light that does not get reflected by the combiner, thereby improving efficiency and reducing escaping light that can cause faceglow.

(188) FIG. **12a** illustrates another light source **1200** that may be used in association with the upper

optics module **202**. In embodiments, the light source **1200** may provide light to a backlighting optical system **1004** as described above in connection with FIG. **10**. In embodiments, the light source **1200** includes a quantum dot cover glass **1202**. Where the quantum dots absorb light of a shorter wavelength and emit light of a longer wavelength (FIG. **12b** shows an example wherein a UV spectrum **1202** applied to a quantum dot results in the quantum dot emitting a narrow band shown as a PL spectrum **1204**) that is dependent on the material makeup and size of the quantum dot. As a result, quantum dots in the quantum dot cover glass **1202** can be tailored to provide one or more bands of narrow bandwidth light (e.g. red, green and blue emissions dependent on the different quantum dots included as illustrated in the graph shown in FIG. **12c** where three different quantum dots are used. In embodiments, the LED driver light emits UV light, deep blue or blue light. For sequential illumination of different colors, multiple light sources **1200** would be used where each light source **1200** would include a quantum dot cover glass **1202** with a quantum dot selected to emit at one of the desired colors. The light source **1100** can be used in connection with a combiner **602** with a holographic mirror or tristimulus notch mirror to provide narrow transmission bands of light that are reflected toward the wearer's eye with less waste light that does not get reflected.

(189) Another aspect of the present invention relates to the generation of peripheral image lighting effects for a person wearing a HWC. In embodiments, a solid state lighting system (e.g. LED, OLED, etc), or other lighting system, may be included inside the optical elements of an lower optical module **204**. The solid state lighting system may be arranged such that lighting effects outside of a field of view (FOV) of the presented digital content is presented to create an emersive effect for the person wearing the HWC. To this end, the lighting effects may be presented to any portion of the HWC that is visible to the wearer. The solid state lighting system may be digitally controlled by an integrated processor on the HWC. In embodiments, the integrated processor will control the lighting effects in coordination with digital content that is presented within the FOV of the HWC. For example, a movie, picture, game, or other content, may be displayed or playing within the FOV of the HWC. The content may show a bomb blast on the right side of the FOV and at the same moment, the solid state lighting system inside of the upper module optics may flash quickly in concert with the FOV image effect. The effect may not be fast, it may be more persistent to indicate, for example, a general glow or color on one side of the user. The solid state lighting system may be color controlled, with red, green and blue LEDs, for example, such that color control can be coordinated with the digitally presented content within the field of view.

(190) FIG. **13a** illustrates optical components of a lower optical module **204** together with an outer lens **1302**. FIG. **13a** also shows an embodiment including effects LED's **1308a** and **1308b**. FIG. **13a** illustrates image light **1312**, as described herein elsewhere, directed into the upper optical module where it will reflect off of the combiner element **1304**, as described herein elsewhere. The combiner element **1304** in this embodiment is angled towards the wearer's eye at the top of the module and away from the wearer's eye at the bottom of the module, as also illustrated and described in connection with FIG. **8** (e.g. at a 45 degree angle). The image light **1312** provided by an upper optical module **202** (not shown in FIG. **13a**) reflects off of the combiner element **1304** towards the collimating mirror **1310**, away from the wearer's eye, as described herein elsewhere. The image light **1312** then reflects and focuses off of the collimating mirror **1304**, passes back through the combiner element **1304**, and is directed into the wearer's eye. The wearer can also view the surrounding environment through the transparency of the combiner element **1304**, collimating mirror **1310**, and outer lens **1302** (if it is included). As described herein elsewhere, various surfaces are polarized to create the optical path for the image light and to provide transparency of the elements such that the wearer can view the surrounding environment. The wearer will generally perceive that the image light forms an image in the FOV **1305**. In embodiments, the outer lens **1302** may be included. The outer lens **1302** is an outer lens that may or may not be corrective and it may be designed to conceal the lower optical module components in an effort to make the HWC

appear to be in a form similar to standard glasses or sunglasses.

(191) In the embodiment illustrated in FIG. **13a**, the effects LEDs **1308a** and **1308b** are positioned at the sides of the combiner element **1304** and the outer lens **1302** and/or the collimating mirror **1310**. In embodiments, the effects LEDs **1308a** are positioned within the confines defined by the combiner element **1304** and the outer lens **1302** and/or the collimating mirror. The effects LEDs **1308a** and **1308b** are also positioned outside of the FOV **1305**. In this arrangement, the effects LEDs **1308a** and **1308b** can provide lighting effects within the lower optical module outside of the FOV **1305**. In embodiments the light emitted from the effects LEDs **1308a** and **1308b** may be polarized such that the light passes through the combiner element **1304** toward the wearer's eye and does not pass through the outer lens **1302** and/or the collimating mirror **1310**. This arrangement provides peripheral lighting effects to the wearer in a more private setting by not transmitting the lighting effects through the front of the HWC into the surrounding environment. However, in other embodiments, the effects LEDs **1308a** and **1308b** may be unpolarized so the lighting effects provided are made to be purposefully viewable by others in the environment for entertainment such as giving the effect of the wearer's eye glowing in correspondence to the image content being viewed by the wearer.

(192) FIG. **13b** illustrates a cross section of the embodiment described in connection with FIG. **13a**. As illustrated, the effects LED **1308a** is located in the upper-front area inside of the optical components of the lower optical module. It should be understood that the effects LED **1308a** position in the described embodiments is only illustrative and alternate placements are encompassed by the present invention. Additionally, in embodiments, there may be one or more effects LEDs **1308a** in each of the two sides of HWC to provide peripheral lighting effects near one or both eyes of the wearer.

(193) FIG. **13c** illustrates an embodiment where the combiner element **1304** is angled away from the eye at the top and towards the eye at the bottom (e.g. in accordance with the holographic or notch filter embodiments described herein). In this embodiment, the effects LED **1308a** is located on the outer lens **1302** side of the combiner element **1304** to provide a concealed appearance of the lighting effects. As with other embodiments, the effects LED **1308a** of FIG. **13c** may include a polarizer such that the emitted light can pass through a polarized element associated with the combiner element **1304** and be blocked by a polarized element associated with the outer lens **1302**.

(194) Another aspect of the present invention relates to the mitigation of light escaping from the space between the wearer's face and the HWC itself. Another aspect of the present invention relates to maintaining a controlled lighting environment in proximity to the wearer's eyes. In embodiments, both the maintenance of the lighting environment and the mitigation of light escape are accomplished by including a removable and replaceable flexible shield for the HWC. Wherein the removable and replaceable shield can be provided for one eye or both eyes in correspondence to the use of the displays for each eye. For example, in a night vision application, the display to only one eye could be used for night vision while the display to the other eye is turned off to provide good see-thru when moving between areas where visible light is available and dark areas where night vision enhancement is needed.

(195) FIG. **14a** illustrates a removable and replaceable flexible eye cover **1402** with an opening **1408** that can be attached and removed quickly from the HWC **102** through the use of magnets. Other attachment methods may be used, but for illustration of the present invention we will focus on a magnet implementation. In embodiments, magnets may be included in the eye cover **1402** and magnets of an opposite polarity may be included (e.g. embedded) in the frame of the HWC **102**. The magnets of the two elements would attract quite strongly with the opposite polarity configuration. In another embodiment, one of the elements may have a magnet and the other side may have metal for the attraction. In embodiments, the eye cover **1402** is a flexible elastomeric shield. In embodiments, the eye cover **1402** may be an elastomeric bellows design to accommodate flexibility and more closely align with the wearer's face. FIG. **14b** illustrates a removable and

replaceable flexible eye cover **1404** that is adapted as a single eye cover. In embodiments, a single eye cover may be used for each side of the HWC to cover both eyes of the wearer. In embodiments, the single eye cover may be used in connection with a HWC that includes only one computer display for one eye. These configurations prevent light that is generated and directed generally towards the wearer's face by covering the space between the wearer's face and the HWC. The opening **1408** allows the wearer to look through the opening **1408** to view the displayed content and the surrounding environment through the front of the HWC. The image light in the lower optical module **204** can be prevented from emitting from the front of the HWC through internal optics polarization schemes, as described herein, for example.

(196) FIG. **14c** illustrates another embodiment of a light suppression system. In this embodiment, the eye cover **1410** may be similar to the eye cover **1402**, but eye cover **1410** includes a front light shield **1412**. The front light shield **1412** may be opaque to prevent light from escaping the front lens of the HWC. In other embodiments, the front light shield **1412** is polarized to prevent light from escaping the front lens. In a polarized arrangement, in embodiments, the internal optical elements of the HWC (e.g. of the lower optical module **204**) may polarize light transmitted towards the front of the HWC and the front light shield **1412** may be polarized to prevent the light from transmitting through the front light shield **1412**.

(197) In embodiments, an opaque front light shield **1412** may be included and the digital content may include images of the surrounding environment such that the wearer can visualize the surrounding environment. One eye may be presented with night vision environmental imagery and this eye's surrounding environment optical path may be covered using an opaque front light shield **1412**. In other embodiments, this arrangement may be associated with both eyes.

(198) Another aspect of the present invention relates to automatically configuring the lighting system(s) used in the HWC **102**. In embodiments, the display lighting and/or effects lighting, as described herein, may be controlled in a manner suitable for when an eye cover **1408** is attached or removed from the HWC **102**. For example, at night, when the light in the environment is low, the lighting system(s) in the HWC may go into a low light mode to further control any amounts of stray light escaping from the HWC and the areas around the HWC. Covert operations at night, while using night vision or standard vision, may require a solution which prevents as much escaping light as possible so a user may clip on the eye cover(s) **1408** and then the HWC may go into a low light mode. The low light mode may, in some embodiments, only go into a low light mode when the eye cover **1408** is attached if the HWC identifies that the environment is in low light conditions (e.g. through environment light level sensor detection). In embodiments, the low light level may be determined to be at an intermediate point between full and low light dependent on environmental conditions.

(199) Another aspect of the present invention relates to automatically controlling the type of content displayed in the HWC when eye covers **1408** are attached or removed from the HWC. In embodiments, when the eye cover(s) **1408** is attached to the HWC, the displayed content may be restricted in amount or in color amounts. For example, the display(s) may go into a simple content delivery mode to restrict the amount of information displayed. This may be done to reduce the amount of light produced by the display(s). In an embodiment, the display(s) may change from color displays to monochrome displays to reduce the amount of light produced. In an embodiment, the monochrome lighting may be red to limit the impact on the wearer's eyes to maintain an ability to see better in the dark.

(200) Referring to FIG. **15**, we now turn to describe a particular external user interface **104**, referred to generally as a pen **1500**. The pen **1500** is a specially designed external user interface **104** and can operate as a user interface, such as to many different styles of HWC **102**. The pen **1500** generally follows the form of a conventional pen, which is a familiar user handled device and creates an intuitive physical interface for many of the operations to be carried out in the HWC system **100**. The pen **1500** may be one of several user interfaces **104** used in connection with

controlling operations within the HWC system **100**. For example, the HWC **102** may watch for and interpret hand gestures **116** as control signals, where the pen **1500** may also be used as a user interface with the same HWC **102**. Similarly, a remote keyboard may be used as an external user interface **104** in concert with the pen **1500**. The combination of user interfaces or the use of just one control system generally depends on the operation(s) being executed in the HWC's system **100**. (201) While the pen **1500** may follow the general form of a conventional pen, it contains numerous technologies that enable it to function as an external user interface **104**. FIG. **15** illustrates technologies comprised in the pen **1500**. As can be seen, the pen **1500** may include a camera **1508**, which is arranged to view through lens **1502**. The camera may then be focused, such as through lens **1502**, to image a surface upon which a user is writing or making other movements to interact with the HWC **102**. There are situations where the pen **1500** will also have an ink, graphite, or other system such that what is being written can be seen on the writing surface. There are other situations where the pen **1500** does not have such a physical writing system so there is no deposit on the writing surface, where the pen would only be communicating data or commands to the HWC **102**. The lens configuration is described in greater detail herein. The function of the camera is to capture information from an unstructured writing surface such that pen strokes can be interpreted as intended by the user. To assist in the predication of the intended stroke path, the pen **1500** may include a sensor, such as an IMU **1512**. Of course, the IMU could be included in the pen **1500** in its separate parts (e.g. gyro, accelerometer, etc.) or an IMU could be included as a single unit. In this instance, the IMU **1512** is used to measure and predict the motion of the pen **1500**. In turn, the integrated microprocessor **1510** would take the IMU information and camera information as inputs and process the information to form a prediction of the pen tip movement.

(202) The pen **1500** may also include a pressure monitoring system **1504**, such as to measure the pressure exerted on the lens **1502**. As will be described in greater detail herein, the pressure measurement can be used to predict the user's intention for changing the weight of a line, type of a line, type of brush, click, double click, and the like. In embodiments, the pressure sensor may be constructed using any force or pressure measurement sensor located behind the lens **1502**, including for example, a resistive sensor, a current sensor, a capacitive sensor, a voltage sensor such as a piezoelectric sensor, and the like.

(203) The pen **1500** may also include a communications module **1518**, such as for bi-directional communication with the HWC **102**. In embodiments, the communications module **1518** may be a short distance communication module (e.g. Bluetooth). The communications module **1518** may be security matched to the HWC **102**. The communications module **1518** maybe arranged to communicate data and commands to and from the microprocessor **1510** of the pen **1500**. The microprocessor **1510** may be programmed to interpret data generated from the camera **1508**, IMU **1512**, and pressure sensor **1504**, and the like, and then pass a command onto the HWC **102** through the communications module **1518**, for example. In another embodiment, the data collected from any of the input sources (e.g. camera **1508**, IMU **1512**, pressure sensor **1504**) by the microprocessor may be communicated by the communication module **1518** to the HWC **102**, and the HWC **102** may perform data processing and prediction of the user's intention when using the pen **1500**. In yet another embodiment, the data may be further passed on through a network **110** to a remote device **112**, such as a server, for the data processing and prediction. The commands may then be communicated back to the HWC **102** for execution (e.g. display writing in the glasses display, make a selection within the UI of the glasses display, control a remote external device **112**, control a local external device **108**), and the like. The pen may also include memory **1514** for long or short term uses.

(204) The pen **1500** may also include a number of physical user interfaces, such as quick launch buttons **1522**, a touch sensor **1520**, and the like. The quick launch buttons **1522** may be adapted to provide the user with a fast way of jumping to a software application in the HWC system **100**. For example, the user may be a frequent user of communication software packages (e.g. email, text,

Twitter, Instagram, Facebook, Google+, and the like), and the user may program a quick launch button **1522** to command the HWC **102** to launch an application. The pen **1500** may be provided with several quick launch buttons **1522**, which may be user programmable or factory programmable. The quick launch button **1522** may be programmed to perform an operation. For example, one of the buttons may be programmed to clear the digital display of the HWC **102**. This would create a fast way for the user to clear the screens on the HWC **102** for any reason, such as for example to better view the environment. The quick launch button functionality will be discussed in further detail below. The touch sensor **1520** may be used to take gesture style input from the user. For example, the user may be able to take a single finger and run it across the touch sensor **1520** to affect a page scroll.

(205) The pen **1500** may also include a laser pointer **1524**. The laser pointer **1524** may be coordinated with the IMU **1512** to coordinate gestures and laser pointing. For example, a user may use the laser **1524** in a presentation to help with guiding the audience with the interpretation of graphics and the IMU **1512** may, either simultaneously or when the laser **1524** is off, interpret the user's gestures as commands or data input.

(206) FIGS. **16A-C** illustrate several embodiments of lens and camera arrangements **1600** for the pen **1500**. One aspect relates to maintaining a constant distance between the camera and the writing surface to enable the writing surface to be kept in focus for better tracking of movements of the pen **1500** over the writing surface. Another aspect relates to maintaining an angled surface following the circumference of the writing tip of the pen **1500** such that the pen **1500** can be rolled or partially rolled in the user's hand to create the feel and freedom of a conventional writing instrument.

(207) FIG. **16A** illustrates an embodiment of the writing lens end of the pen **1500**. The configuration includes a ball lens **1604**, a camera or image capture surface **1602**, and a domed cover lens **1608**. In this arrangement, the camera views the writing surface through the ball lens **1604** and dome cover lens **1608**. The ball lens **1604** causes the camera to focus such that the camera views the writing surface when the pen **1500** is held in the hand in a natural writing position, such as with the pen **1500** in contact with a writing surface. In embodiments, the ball lens **1604** should be separated from the writing surface to obtain the highest resolution of the writing surface at the camera **1602**. In embodiments, the ball lens **1604** is separated by approximately 1 to 3 mm. In this configuration, the domed cover lens **1608** provides a surface that can keep the ball lens **1604** separated from the writing surface at a constant distance, such as substantially independent of the angle used to write on the writing surface. For instance, in embodiments the field of view of the camera in this arrangement would be approximately 60 degrees.

(208) The domed cover lens, or other lens **1608** used to physically interact with the writing surface, will be transparent or transmissive within the active bandwidth of the camera **1602**. In embodiments, the domed cover lens **1608** may be spherical or other shape and comprised of glass, plastic, sapphire, diamond, and the like. In other embodiments where low resolution imaging of the surface is acceptable. The pen **1500** can omit the domed cover lens **1608** and the ball lens **1604** can be in direct contact with the surface.

(209) FIG. **16B** illustrates another structure where the construction is somewhat similar to that described in connection with FIG. **16A**; however this embodiment does not use a dome cover lens **1608**, but instead uses a spacer **1610** to maintain a predictable distance between the ball lens **1604** and the writing surface, wherein the spacer may be spherical, cylindrical, tubular or other shape that provides spacing while allowing for an image to be obtained by the camera **1602** through the lens **1604**. In a preferred embodiment, the spacer **1610** is transparent. In addition, while the spacer **1610** is shown as spherical, other shapes such as an oval, doughnut shape, half sphere, cone, cylinder or other form may be used.

(210) FIG. **16C** illustrates yet another embodiment, where the structure includes a post **1614**, such as running through the center of the lensed end of the pen **1500**. The post **1614** may be an ink

deposition system (e.g. ink cartridge), graphite deposition system (e.g. graphite holder), or a dummy post whose purpose is mainly only that of alignment. The selection of the post type is dependent on the pen's use. For instance, in the event the user wants to use the pen **1500** as a conventional ink depositing pen as well as a fully functional external user interface **104**, the ink system post would be the best selection. If there is no need for the 'writing' to be visible on the writing surface, the selection would be the dummy post. The embodiment of FIG. **16C** includes camera(s) **1602** and an associated lens **1612**, where the camera **1602** and lens **1612** are positioned to capture the writing surface without substantial interference from the post **1614**. In embodiments, the pen **1500** may include multiple cameras **1602** and lenses **1612** such that more or all of the circumference of the tip **1614** can be used as an input system. In an embodiment, the pen **1500** includes a contoured grip that keeps the pen aligned in the user's hand so that the camera **1602** and lens **1612** remains pointed at the surface.

(211) Another aspect of the pen **1500** relates to sensing the force applied by the user to the writing surface with the pen **1500**. The force measurement may be used in a number of ways. For example, the force measurement may be used as a discrete value, or discontinuous event tracking, and compared against a threshold in a process to determine a user's intent. The user may want the force interpreted as a 'click' in the selection of an object, for instance. The user may intend multiple force exertions interpreted as multiple clicks. There may be times when the user holds the pen **1500** in a certain position or holds a certain portion of the pen **1500** (e.g. a button or touch pad) while clicking to affect a certain operation (e.g. a 'right click'). In embodiments, the force measurement may be used to track force and force trends. The force trends may be tracked and compared to threshold limits, for example. There may be one such threshold limit, multiple limits, groups of related limits, and the like. For example, when the force measurement indicates a fairly constant force that generally falls within a range of related threshold values, the microprocessor **1510** may interpret the force trend as an indication that the user desires to maintain the current writing style, writing tip type, line weight, brush type, and the like. In the event that the force trend appears to have gone outside of a set of threshold values intentionally, the microprocessor may interpret the action as an indication that the user wants to change the current writing style, writing tip type, line weight, brush type, and the like. Once the microprocessor has made a determination of the user's intent, a change in the current writing style, writing tip type, line weight, brush type, and the like may be executed. In embodiments, the change may be noted to the user (e.g. in a display of the HWC **102**), and the user may be presented with an opportunity to accept the change.

(212) FIG. **17A** illustrates an embodiment of a force sensing surface tip **1700** of a pen **1500**. The force sensing surface tip **1700** comprises a surface connection tip **1702** (e.g. a lens as described herein elsewhere) in connection with a force or pressure monitoring system **1504**. As a user uses the pen **1500** to write on a surface or simulate writing on a surface the force monitoring system **1504** measures the force or pressure the user applies to the writing surface and the force monitoring system communicates data to the microprocessor **1510** for processing. In this configuration, the microprocessor **1510** receives force data from the force monitoring system **1504** and processes the data to make predictions of the user's intent in applying the particular force that is currently being applied. In embodiments, the processing may be provided at a location other than on the pen (e.g. at a server in the HWC system **100**, on the HWC **102**). For clarity, when reference is made herein to processing information on the microprocessor **1510**, the processing of information contemplates processing the information at a location other than on the pen. The microprocessor **1510** may be programmed with force threshold(s), force signature(s), force signature library and/or other characteristics intended to guide an inference program in determining the user's intentions based on the measured force or pressure. The microprocessor **1510** may be further programmed to make inferences from the force measurements as to whether the user has attempted to initiate a discrete action (e.g. a user interface selection 'click') or is performing a constant action (e.g. writing within a particular writing style). The inferencing process is important as it causes the pen **1500** to act as

an intuitive external user interface **104**.

(213) FIG. **17B** illustrates a force **1708** versus time **1710** trend chart with a single threshold **1718**. The threshold **1718** may be set at a level that indicates a discrete force exertion indicative of a user's desire to cause an action (e.g. select an object in a GUI). Event **1712**, for example, may be interpreted as a click or selection command because the force quickly increased from below the threshold **1718** to above the threshold **1718**. The event **1714** may be interpreted as a double click because the force quickly increased above the threshold **1718**, decreased below the threshold **1718** and then essentially repeated quickly. The user may also cause the force to go above the threshold **1718** and hold for a period indicating that the user is intending to select an object in the GUI (e.g. a GUI presented in the display of the HWC **102**) and 'hold' for a further operation (e.g. moving the object).

(214) While a threshold value may be used to assist in the interpretation of the user's intention, a signature force event trend may also be used. The threshold and signature may be used in combination or either method may be used alone. For example, a single-click signature may be represented by a certain force trend signature or set of signatures. The single-click signature(s) may require that the trend meet a criteria of a rise time between λ any y values, a hold time of between a and b values and a fall time of between c and d values, for example. Signatures may be stored for a variety of functions such as click, double click, right click, hold, move, etc. The microprocessor **1510** may compare the real-time force or pressure tracking against the signatures from a signature library to make a decision and issue a command to the software application executing in the GUI.

(215) FIG. **17C** illustrates a force **1708** versus time **1710** trend chart with multiple thresholds **1718**. By way of example, the force trend is plotted on the chart with several pen force or pressure events. As noted, there are both presumably intentional events **1720** and presumably non-intentional events **1722**. The two thresholds **1718** of FIG. **4C** create three zones of force: a lower, middle and higher range. The beginning of the trend indicates that the user is placing a lower zone amount of force. This may mean that the user is writing with a given line weight and does not intend to change the weight, the user is writing. Then the trend shows a significant increase **1720** in force into the middle force range. This force change appears, from the trend to have been sudden and thereafter it is sustained. The microprocessor **1510** may interpret this as an intentional change and as a result change the operation in accordance with preset rules (e.g. change line width, increase line weight, etc.). The trend then continues with a second apparently intentional event **1720** into the higher-force range. During the performance in the higher-force range, the force dips below the upper threshold **1718**. This may indicate an unintentional force change and the microprocessor may detect the change in range however not affect a change in the operations being coordinated by the pen **1500**. As indicated above, the trend analysis may be done with thresholds and/or signatures.

(216) Generally, in the present disclosure, instrument stroke parameter changes may be referred to as a change in line type, line weight, tip type, brush type, brush width, brush pressure, color, and other forms of writing, coloring, painting, and the like.

(217) Another aspect of the pen **1500** relates to selecting an operating mode for the pen **1500** dependent on contextual information and/or selection interface(s). The pen **1500** may have several operating modes. For instance, the pen **1500** may have a writing mode where the user interface(s) of the pen **1500** (e.g. the writing surface end, quick launch buttons **1522**, touch sensor **1520**, motion based gesture, and the like) is optimized or selected for tasks associated with writing. As another example, the pen **1500** may have a wand mode where the user interface(s) of the pen is optimized or selected for tasks associated with software or device control (e.g. the HWC **102**, external local device, remote device **112**, and the like). The pen **1500**, by way of another example, may have a presentation mode where the user interface(s) is optimized or selected to assist a user with giving a presentation (e.g. pointing with the laser pointer **1524** while using the button(s) **1522** and/or gestures to control the presentation or applications relating to the presentation). The pen may, for example, have a mode that is optimized or selected for a particular device that a user is attempting

to control. The pen **1500** may have a number of other modes and an aspect of the present invention relates to selecting such modes.

(218) FIG. **18A** illustrates an automatic user interface(s) mode selection based on contextual information. The microprocessor **1510** may be programmed with IMU thresholds **1814** and **1812**. The thresholds **1814** and **1812** may be used as indications of upper and lower bounds of an angle **1804** and **1802** of the pen **1500** for certain expected positions during certain predicted modes. When the microprocessor **1510** determines that the pen **1500** is being held or otherwise positioned within angles **1802** corresponding to writing thresholds **1814**, for example, the microprocessor **1510** may then institute a writing mode for the pen's user interfaces. Similarly, if the microprocessor **1510** determines (e.g. through the IMU **1512**) that the pen is being held at an angle **1804** that falls between the predetermined wand thresholds **1812**, the microprocessor may institute a wand mode for the pen's user interface. Both of these examples may be referred to as context based user interface mode selection as the mode selection is based on contextual information (e.g. position) collected automatically and then used through an automatic evaluation process to automatically select the pen's user interface(s) mode.

(219) As with other examples presented herein, the microprocessor **1510** may monitor the contextual trend (e.g. the angle of the pen over time) in an effort to decide whether to stay in a mode or change modes. For example, through signatures, thresholds, trend analysis, and the like, the microprocessor may determine that a change is an unintentional change and therefore no user interface mode change is desired.

(220) FIG. **18B** illustrates an automatic user interface(s) mode selection based on contextual information. In this example, the pen **1500** is monitoring (e.g. through its microprocessor) whether or not the camera at the writing surface end **1508** is imaging a writing surface in close proximity to the writing surface end of the pen **1500**. If the pen **1500** determines that a writing surface is within a predetermined relatively short distance, the pen **1500** may decide that a writing surface is present **1820** and the pen may go into a writing mode user interface(s) mode. In the event that the pen **1500** does not detect a relatively close writing surface **1822**, the pen may predict that the pen is not currently being used to as a writing instrument and the pen may go into a non-writing user interface(s) mode.

(221) FIG. **18C** illustrates a manual user interface(s) mode selection. The user interface(s) mode may be selected based on a twist of a section **1824** of the pen **1500** housing, clicking an end button **1828**, pressing a quick launch button **1522**, interacting with touch sensor **1520**, detecting a predetermined action at the pressure monitoring system (e.g. a click), detecting a gesture (e.g. detected by the IMU), etc. The manual mode selection may involve selecting an item in a GUI associated with the pen **1500** (e.g. an image presented in the display of HWC **102**).

(222) In embodiments, a confirmation selection may be presented to the user in the event a mode is going to change. The presentation may be physical (e.g. a vibration in the pen **1500**), through a GUI, through a light indicator, etc.

(223) FIG. **19** illustrates a couple pen use-scenarios **1900** and **1901**. There are many use scenarios and we have presented a couple in connection with FIG. **19** as a way of illustrating use scenarios to further the understanding of the reader. As such, the use-scenarios should be considered illustrative and non-limiting.

(224) Use scenario **1900** is a writing scenario where the pen **1500** is used as a writing instrument. In this example, quick launch button **122A** is pressed to launch a note application **1910** in the GUI **1908** of the HWC **102** display **1904**. Once the quick launch button **122A** is pressed, the HWC **102** launches the note program **1910** and puts the pen into a writing mode. The user uses the pen **1500** to scribe symbols **1902** on a writing surface, the pen records the scribing and transmits the scribing to the HWC **102** where symbols representing the scribing are displayed **1912** within the note application **1910**.

(225) Use scenario **1901** is a gesture scenario where the pen **1500** is used as a gesture capture and

command device. In this example, the quick launch button **122B** is activated and the pen **1500** activates a wand mode such that an application launched on the HWC **102** can be controlled. Here, the user sees an application chooser **1918** in the display(s) of the HWC **102** where different software applications can be chosen by the user. The user gestures (e.g. swipes, spins, turns, etc.) with the pen to cause the application chooser **1918** to move from application to application. Once the correct application is identified (e.g. highlighted) in the chooser **1918**, the user may gesture or click or otherwise interact with the pen **1500** such that the identified application is selected and launched. Once an application is launched, the wand mode may be used to scroll, rotate, change applications, select items, initiate processes, and the like, for example.

(226) In an embodiment, the quick launch button **122A** may be activated and the HWC **102** may launch an application chooser presenting to the user a set of applications. For example, the quick launch button may launch a chooser to show all communication programs (e.g. SMS, Twitter, Instagram, Facebook, email, etc.) available for selection such that the user can select the program the user wants and then go into a writing mode. By way of further example, the launcher may bring up selections for various other groups that are related or categorized as generally being selected at a given time (e.g. Microsoft Office products, communication products, productivity products, note products, organizational products, and the like).

(227) FIG. **20** illustrates yet another embodiment of the present invention. FIG. **2000** illustrates a watchband clip on controller **2000**. The watchband clip on controller may be a controller used to control the HWC **102** or devices in the HWC system **100**. The watchband clip on controller **2000** has a fastener **2018** (e.g. rotatable clip) that is mechanically adapted to attach to a watchband, as illustrated at **2004**.

(228) The watchband controller **2000** may have quick launch interfaces **2008** (e.g. to launch applications and choosers as described herein), a touch pad **2014** (e.g. to be used as a touch style mouse for GUI control in a HWC **102** display) and a display **2012**. The clip **2018** may be adapted to fit a wide range of watchbands so it can be used in connection with a watch that is independently selected for its function. The clip, in embodiments, is rotatable such that a user can position it in a desirable manner. In embodiments the clip may be a flexible strap. In embodiments, the flexible strap may be adapted to be stretched to attach to a hand, wrist, finger, device, weapon, and the like.

(229) In embodiments, the watchband controller may be configured as a removable and replacable watchband. For example, the controller may be incorporated into a band with a certain width, segment spacing's, etc. such that the watchband, with its incorporated controller, can be attached to a watch body. The attachment, in embodiments, may be mechanically adapted to attach with a pin upon which the watchband rotates. In embodiments, the watchband controller may be electrically connected to the watch and/or watch body such that the watch, watch body and/or the watchband controller can communicate data between them.

(230) The watchband controller may have 3-axis motion monitoring (e.g. through an IMU, accelerometers, magnetometers, gyroscopes, etc.) to capture user motion. The user motion may then be interpreted for gesture control.

(231) In embodiments, the watchband controller may comprise fitness sensors and a fitness computer. The sensors may track heart rate, calories burned, strides, distance covered, and the like. The data may then be compared against performance goals and/or standards for user feedback.

(232) Another aspect of the present invention relates to visual display techniques relating to micro Doppler ("mD") target tracking signatures ("mD signatures"). mD is a radar technique that uses a series of angle dependent electromagnetic pulses that are broadcast into an environment and return pulses are captured. Changes between the broadcast pulse and return pulse are indicative of changes in the shape, distance and angular location of objects or targets in the environment. These changes provide signals that can be used to track a target and identify the target through the mD signature. Each target or target type has a unique mD signature. Shifts in the radar pattern can be analyzed in the time domain and frequency domain based on mD techniques to derive information

about the types of targets present (e.g. whether people are present), the motion of the targets and the relative angular location of the targets and the distance to the targets. By selecting a frequency used for the mD pulse relative to known objects in the environment, the pulse can penetrate the known objects to enable information about targets to be gathered even when the targets are visually blocked by the known objects. For example, pulse frequencies can be used that will penetrate concrete buildings to enable people to be identified inside the building. Multiple pulse frequencies can be used as well in the mD radar to enable different types of information to be gathered about the objects in the environment. In addition, the mD radar information can be combined with other information such as distance measurements or images captured of the environment that are analyzed jointly to provide improved object identification and improved target identification and tracking. In embodiments, the analysis can be performed on the HWC or the information can be transmitted to a remote network for analysis and results transmitted back to the HWC. Distance measurements can be provided by laser range finding, structured lighting, stereoscopic depth maps or sonar measurements. Images of the environment can be captured using one or more cameras capable of capturing images from visible, ultraviolet or infrared light. The mD radar can be attached to the HWC, located adjacently (e.g. in a vehicle) and associated wirelessly with the HWC or located remotely. Maps or other previously determined information about the environment can also be used in the analysis of the mD radar information. Embodiments of the present invention relate to visualizing the mD signatures in useful ways.

(233) FIG. **21** illustrates a FOV **2102** of a HWC **102** from a wearer's perspective. The wearer, as described herein elsewhere, has a see-through FOV **2102** wherein the wearer views adjacent surroundings, such as the buildings illustrated in FIG. **21**. The wearer, as described herein elsewhere, can also see displayed digital content presented within a portion of the FOV **2102**. The embodiment illustrated in FIG. **21** is indicating that the wearer can see the buildings and other surrounding elements in the environment and digital content representing traces, or travel paths, of bullets being fired by different people in the area. The surroundings are viewed through the transparency of the FOV **2102**. The traces are presented via the digital computer display, as described herein elsewhere. In embodiments, the trace presented is based on a mD signature that is collected and communicated to the HWC in real time. The mD radar itself may be on or near the wearer of the HWC **102** or it may be located remote from the wearer. In embodiments, the mD radar scans the area, tracks and identifies targets, such as bullets, and communicates traces, based on locations, to the HWC **102**.

(234) There are several traces **2108** and **2104** presented to the wearer in the embodiment illustrated in FIG. **21**. The traces communicated from the mD radar may be associated with GPS locations and the GPS locations may be associated with objects in the environment, such as people, buildings, vehicles, etc, both in latitude and longitude perspective and an elevation perspective. The locations may be used as markers for the HWC such that the traces, as presented in the FOV, can be associated, or fixed in space relative to the markers. For example, if the friendly fire trace **2108** is determined, by the mD radar, to have originated from the upper right window of the building on the left, as illustrated in FIG. **21**, then a virtual marker may be set on or near the window. When the HWC views, through its camera or other sensor, for example, the building's window, the trace may then virtually anchor with the virtual marker on the window. Similarly, a marker may be set near the termination position or other flight position of the friendly fire trace **2108**, such as the upper left window of the center building on the right, as illustrated in FIG. **21**. This technique fixes in space the trace such that the trace appears fixed to the environmental positions independent of where the wearer is looking. So, for example, as the wearer's head turns, the trace appears fixed to the marked locations.

(235) In embodiments, certain user positions may be known and thus identified in the FOV. For example, the shooter of the friendly fire trace **2108** may be from a known friendly combatant and as such his location may be known. The position may be known based on his GPS location based

on a mobile communication system on him, such as another HWC **102**. In other embodiments, the friendly combatant may be marked by another friendly. For example, if the friendly position in the environment is known through visual contact or communicated information, a wearer of the HWC **102** may use a gesture or external user interface **104** to mark the location. If a friendly combatant location is known the originating position of the friendly fire trace **2108** may be color coded or otherwise distinguished from unidentified traces on the displayed digital content. Similarly, enemy fire traces **2104** may be color coded or otherwise distinguished on the displayed digital content. In embodiments, there may be an additional distinguished appearance on the displayed digital content for unknown traces.

(236) In addition to situationally associated trace appearance, the trace colors or appearance may be different from the originating position to the terminating position. This path appearance change may be based on the mD signature. The mD signature may indicate that the bullet, for example, is slowing as it propagates and this slowing pattern may be reflected in the FOV **2102** as a color or pattern change. This can create an intuitive understanding of where the shooter is located. For example, the originating color may be red, indicative of high speed, and it may change over the course of the trace to yellow, indicative of a slowing trace. This pattern changing may also be different for a friendly, enemy and unknown combatant. The enemy may go blue to green for a friendly trace, for example.

(237) FIG. **21** illustrates an embodiment where the user sees the environment through the FOV and may also see color coded traces, which are dependent on bullet speed and combatant type, where the traces are fixed in environmental positions independent on the wearer's perspective. Other information, such as distance, range, range rings, time of day, date, engagement type (e.g. hold, stop firing, back away, etc.) may also be displayed in the FOV.

(238) Another aspect of the present invention relates to mD radar techniques that trace and identify targets through other objects, such as walls (referred to generally as through wall mD), and visualization techniques related therewith. FIG. **22** illustrates a through wall mD visualization technique according to the principles of the present invention. As described herein elsewhere, the mD radar scanning the environment may be local or remote from the wearer of a HWC **102**. The mD radar may identify a target (e.g. a person) that is visible **2204** and then track the target as he goes behind a wall **2208**. The tracking may then be presented to the wearer of a HWC **102** such that digital content reflective of the target and the target's movement, even behind the wall, is presented in the FOV **2202** of the HWC **102**. In embodiments, the target, when out of visible sight, may be represented by an avatar in the FOV to provide the wearer with imagery representing the target.

(239) mD target recognition methods can identify the identity of a target based on the vibrations and other small movements of the target. This can provide a personal signature for the target. In the case of humans, this may result in a personal identification of a target that has been previously characterized. The cardio, heart beat, lung expansion and other small movements within the body may be unique to a person and if those attributes are pre-identified they may be matched in real time to provide a personal identification of a person in the FOV **2202**. The person's mD signatures may be determined based on the position of the person. For example, the database of personal mD signature attributes may include mD signatures for a person standing, sitting, laying down, running, walking, jumping, etc. This may improve the accuracy of the personal data match when a target is tracked through mD signature techniques in the field. In the event a person is personally identified, a specific indication of the person's identity may be presented in the FOV **2202**. The indication may be a color, shape, shade, name, indication of the type of person (e.g. enemy, friendly, etc.), etc. to provide the wearer with intuitive real time information about the person being tracked. This may be very useful in a situation where there is more than one person in an area of the person being tracked. If just one person in the area is personally identified, that person or the avatar of that person can be presented differently than other people in the area.

(240) FIG. **23** illustrates an mD scanned environment **2300**. An mD radar may scan an environment

in an attempt to identify objects in the environment. In this embodiment, the mD scanned environment reveals two vehicles **2302a** and **2302b**, an enemy combatant **2309**, two friendly combatants **2308a** and **2308b** and a shot trace **2318**. Each of these objects may be personally identified or type identified. For example, the vehicles **2302a** and **2302b** may be identified through the mD signatures as a tank and heavy truck. The enemy combatant **2309** may be identified as a type (e.g. enemy combatant) or more personally (e.g. by name). The friendly combatants may be identified as a type (e.g. friendly combatant) or more personally (e.g. by name). The shot trace **2318** may be characterized by type of projectile or weapon type for the projectile, for example.

(241) FIG. **23a** illustrates two separate HWC **102** FOV display techniques according to the principles of the present invention. FOV **2312** illustrates a map view **2310** where the mD scanned environment is presented. Here, the wearer has a perspective on the mapped area so he can understand all tracked targets in the area. This allows the wearer to traverse the area with knowledge of the targets. FOV **2312** illustrates a heads-up view to provide the wearer with an augmented reality style view of the environment that is in proximity of the wearer.

(242) An aspect of the present invention relates to suppression of extraneous or stray light. As discussed herein elsewhere, eyeglow and faceglow are two such artifacts that develop from such light. Eyeglow and faceglow can be caused by image light escaping from the optics module. The escaping light is then visible, particularly in dark environments when the user is viewing bright displayed images with the HWC. Light that escapes through the front of the HWC is visible as eyeglow as it that light that is visible in the region of the user's eyes. Eyeglow can appear in the form of a small version of the displayed image that the user is viewing. Light that escapes from the bottom of the HWC shines onto the user's face, cheek or chest so that these portions of the user appear to glow. Eyeglow and faceglow can both increase the visibility of the user and highlight the use of the HWC, which may be viewed negatively by the user. As such, reducing eyeglow and faceglow is advantageous. In combat situations (e.g. the mD trace presentation scenarios described herein) and certain gaming situations, the suppression of extraneous or stray light is very important.

(243) The disclosure relating to FIG. **6** shows an example where a portion of the image light passes through the combiner **602** such that the light shines onto the user's face, thereby illuminating a portion of the user's face in what is generally referred to herein as faceglow. Faceglow be caused by any portion of light from the HWC that illuminates the user's face.

(244) An example of the source for the faceglow light can come from wide cone angle light associated with the image light incident onto the combiner **602**. Where the combiner can include a holographic mirror or a notch mirror in which the narrow bands of high reflectivity are matched to wavelengths of light by the light source. The wide cone angle associated with the image light corresponds with the field of view provided by the HWC. Typically the reflectivity of holographic mirrors and notch mirrors is reduced as the cone angle of the incident light is increased above 8 degrees. As a result, for a field of view of 30 degrees, substantial image light can pass through the combiner and cause faceglow.

(245) FIG. **24** shows an illustration of a light trap **2410** for the faceglow light. In this embodiment, an extension of the outer shield lens of the HWC is coated with a light absorbing material in the region where the converging light responsible for faceglow is absorbed in a light trap **2410**. The light absorbing material can be black or it can be a filter designed to absorb only the specific wavelengths of light provided by the light source(s) in the HWC. In addition, the surface of the light trap **2410** may be textured or fibrous to further improve the absorption.

(246) FIG. **25** illustrates an optical system for a HWC that includes an outer absorptive polarizer **2520** to block the faceglow light. In this embodiment, the image light is polarized and as a result the light responsible for faceglow is similarly polarized. The absorptive polarizer is oriented with a transmission axis such that the faceglow light is absorbed and not transmitted. In this case, the rest of the imaging system in the HWC may not require polarized image light and the image light may be polarized at any point before the combiner. In embodiments, the transmission axis of the

absorptive polarizer **2520** is oriented vertically so that external glare from water (S polarized light) is absorbed and correspondingly, the polarization of the image light is selected to be horizontal (S polarization). Consequently, image light that passes through the combiner **602** and is then incident onto the absorptive polarizer **2520**, is absorbed. In FIG. **25** the absorptive polarizer **2520** is shown outside the shield lens, alternatively the absorptive polarizer **2520** can be located inside the shield lens.

(247) FIG. **26** illustrates an optical system for a HWC that includes a film with an absorptive notch filter **2620**. In this case, the absorptive notch filter absorbs narrow bands of light that are selected to match the light provided by the optical system's light source. As a result, the absorptive notch filter is opaque with respect to the faceglow light and is transparent to the remainder of the wavelengths included in the visible spectrum so that the user has a clear view of the surrounding environment. A triple notch filter suitable for this approach is available from Iridian Spectral Technologies, Ottawa, ON: <http://www.ilphotonics.com/cdv2/Iridian-Interference%20Filters/New%20filters/Triple%20Notch%20Filter.pdf>.

(248) In embodiments, the combiner **602** may include a notch mirror coating to reflect the wavelengths of light in the image light and a notch filter **2620** can be selected in correspondence to the wavelengths of light provided by the light source and the narrow bands of high reflectivity provided by the notch mirror. In this way, image light that is not reflected by the notch mirror is absorbed by the notch filter **2620**. In embodiments of the invention the light source can provide one narrow band of light for a monochrome imaging or three narrow bands of light for full color imaging. The notch mirror and associated notch filter would then each provide one narrow band or three narrow bands of high reflectivity and absorption respectively.

(249) FIG. **27** includes a microlouver film **2750** to block the faceglow light. Microlouver film is sold by 3M as ALCF-P, for example and is typically used as a privacy filter for computer. See http://multimedia.3m.com/mws/mediawebserver?mwsid=SSSSSuH8gc7nZxtUoY_x1Y_eevUqel7zHvTSevTSeSSSSSS--&fn=ALCF-P_ABR2_Control_Film_DS.pdf The microlouver film transmits light within a somewhat narrow angle (e.g. 30 degrees of normal and absorbs light beyond 30 degrees of normal). In FIG. **27**, the microlouver film **2750** is positioned such that the faceglow light **2758** is incident beyond 30 degrees from normal while the see-through light **2755** is incident within 30 degrees of normal to the microlouver film **2750**. As such, the faceglow light **2758** is absorbed by the microlouver film and the see-through light **2755** is transmitted so that the user has a bright see-thru view of the surrounding environment.

(250) We now turn back to a description of eye imaging technologies. Aspects of the present invention relate to various methods of imaging the eye of a person wearing the HWC **102**. In embodiments, technologies for imaging the eye using an optical path involving the “off” state and “no power” state, which is described in detail below, are described. In embodiments, technologies for imaging the eye with optical configurations that do not involve reflecting the eye image off of DLP mirrors is described. In embodiments, unstructured light, structured light, or controlled lighting conditions, are used to predict the eye's position based on the light reflected off of the front of the wearer's eye. In embodiments, a reflection of a presented digital content image is captured as it reflects off of the wearer's eye and the reflected image may be processed to determine the quality (e.g. sharpness) of the image presented. In embodiments, the image may then be adjusted (e.g. focused differently) to increase the quality of the image presented based on the image reflection.

(251) FIGS. **28a**, **28b** and **28c** show illustrations of the various positions of the DLP mirrors. FIG. **28a** shows the DLP mirrors in the “on” state **2815**. With the mirror in the “on” state **2815**, illumination light **2810** is reflected along an optical axis **2820** that extends into the lower optical module **204**. FIG. **28b** shows the DLP mirrors in the “off” state **2825**. With the mirror in the “off” state **2825**, illumination light **2810** is reflected along an optical axis **2830** that is substantially to the side of optical axis **2820** so that the “off” state light is directed toward a dark light trap as has been described herein elsewhere. FIG. **28c** shows the DLP mirrors in a third position, which occurs

when no power is applied to the DLP. This “no power” state differs from the “on” and “off” states in that the mirror edges are not in contact with the substrate and as such are less accurately positioned. FIG. **28c** shows all of the DLP mirrors in the “no power” state **2835**. The “no power” state is achieved by simultaneously setting the voltage to zero for the “on” contact and “off” contact for a DLP mirror, as a result, the mirror returns to a no stress position where the DLP mirror is in the plane of the DLP platform as shown in FIG. **28c**. Although not normally done, it is also possible to apply the “no power” state to individual DLP mirrors. When the DLP mirrors are in the “no power” state they do not contribute image content. Instead, as shown in FIG. **28c**, when the DLP mirrors are in the “no power” state, the illumination light **2810** is reflected along an optical axis **2840** that is between the optical axes **2820** and **2830** that are respectively associated with the “on” and “off” states and as such this light doesn't contribute to the displayed image as a bright or dark pixel. This light can however contribute scattered light into the lower optical module **204** and as a result the displayed image contrast can be reduced or artifacts can be created in the image that detract from the image content. Consequently, it is generally desirable, in embodiments, to limit the time associated with the “no power” state to times when images are not displayed or to reduce the time associated with having DLP mirrors in the “no power” state so that the affect of the scattered light is reduced.

(252) FIG. **29** shows an embodiment of the invention that can be used for displaying digital content images to a wearer of the HWC **102** and capturing images of the wearer's eye. In this embodiment, light from the eye **2971** passes back through the optics in the lower module **204**, the solid corrective wedge **2966**, at least a portion of the light passes through the partially reflective layer **2960**, the solid illumination wedge **2964** and is reflected by a plurality of DLP mirrors on the DLP **2955** that are in the “no power” state. The reflected light then passes back through the illumination wedge **2964** and at least a portion of the light is reflected by the partially reflective layer **2960** and the light is captured by the camera **2980**.

(253) For comparison, illuminating light rays **2973** from the light source **2958** are also shown being reflected by the partially reflective layer **2960**. Where the angle of the illuminating light **2973** is such that the DLP mirrors, when in the “on” state, reflect the illuminating light **2973** to form image light **2969** that substantially shares the same optical axis as the light from the wearer's eye **2971**. In this way, images of the wearer's eye are captured in a field of view that overlaps the field of view for the displayed image content. In contrast, light reflected by DLP mirrors in the “off” state form dark light **2975** which is directed substantially to the side of the image light **2969** and the light from eye **2971**. Dark light **2975** is directed toward a light trap **2962** that absorbs the dark light to improve the contrast of the displayed image as has been described above in this specification.

(254) [In an embodiment, partially reflective layer **2960** is a reflective polarizer. The light that is reflected from the eye **2971** can then be polarized prior to entering the corrective wedge **2966** (e.g with an absorptive polarizer between the upper module **202** and the lower module **204**), with a polarization orientation relative to the reflective polarizer that enables the light reflected from the eye **2971** to substantially be transmitted by the reflective polarizer. A quarter wave retarder layer **2957** is then included adjacent to the DLP **2955** (as previously disclosed in FIG. **3b**) so that the light reflected from the eye **2971** passes through the quarter wave retarder layer **2957** once before being reflected by the plurality of DLP mirrors in the “no power” state and then passes through a second time after being reflected. By passing through the quarter wave retarder layer **2957** twice, the polarization state of the light from the eye **2971** is reversed, such that when it is incident upon the reflective polarizer, the light from the eye **2971** is then substantially reflected toward the camera **2980**. By using a partially reflective layer **2960** that is a reflective polarizer and polarizing the light from the eye **2971** prior to entering the corrective wedge **2964**, losses attributed to the partially reflective layer **2960** are reduced.

(255) FIG. **28c** shows the case wherein the DLP mirrors are simultaneously in the “no power” state, this mode of operation can be particularly useful when the HWC **102** is first put onto the head of

the wearer. When the HWC **102** is first put onto the head of the wearer, it is not necessary to display an image yet. As a result, the DLP can be in a “no power” state for all the DLP mirrors and an image of the wearer's eyes can be captured. The captured image of the wearer's eye can then be compared to a database, using iris identification techniques, or other eye pattern identification techniques to determine, for example, the identity of the wearer.

(256) In a further embodiment illustrated by FIG. **29** all of the DLP mirrors are put into the “no power” state for a portion of a frame time (e.g. 50% of a frame time for the displayed digital content image) and the capture of the eye image is synchronized to occur at the same time and for the same duration. By reducing the time that the DLP mirrors are in the “no power” state, the time where light is scattered by the DLP mirrors being in the “no power” state is reduced such that the wearer doesn't perceive a change in the displayed image quality. This is possible because the DLP mirrors have a response time on the order of microseconds while typical frame times for a displayed image are on the order of 0.016 seconds. This method of capturing images of the wearer's eye can be used periodically to capture repetitive images of the wearer's eye. For example, eye images could be captured for 50% of the frame time of every 10^{sup}.th frame displayed to the wearer. In another example, eye images could be captured for 10% of the frame time of every frame displayed to the wearer.

(257) Alternately, the “no power” state can be applied to a subset of the DLP mirrors (e.g. 10% of the DLP mirrors) within while another subset is in busy generating image light for content to be displayed. This enables the capture of an eye image(s) during the display of digital content to the wearer. The DLP mirrors used for eye imaging can, for example, be distributed randomly across the area of the DLP to minimize the impact on the quality of the digital content being displayed to the wearer. To improve the displayed image perceived by the wearer, the individual DLP mirrors put into the “no power” state for capturing each eye image, can be varied over time such as in a random pattern, for example. In yet a further embodiment, the DLP mirrors put into the “no power” state for eye imaging may be coordinated with the digital content in such a way that the “no power” mirrors are taken from a portion of the image that requires less resolution.

(258) In the embodiments of the invention as illustrated in FIGS. **9** and **29**, in both cases the reflective surfaces provided by the DLP mirrors do not preserve the wavefront of the light from the wearer's eye so that the image quality of captured image of the eye is somewhat limited. It may still be useful in certain embodiments, but it is somewhat limited. This is due to the DLP mirrors not being constrained to be on the same plane. In the embodiment illustrated in FIG. **9**, the DLP mirrors are tilted so that they form rows of DLP mirrors that share common planes. In the embodiment illustrated in FIG. **29**, the individual DLP mirrors are not accurately positioned to be in the same plane since they are not in contact with the substrate. Examples of advantages of the embodiments associated with FIG. **29** are: first, the camera **2980** can be located between the DLP **2955** and the illumination light source **2958** to provide a more compact upper module **202**. Second, the polarization state of the light reflected from the eye **2971** can be the same as that of the image light **2969** so that the optical path of the light reflected from the eye and the image light can be the same in the lower module **204**.

(259) FIG. **30** shows an illustration of an embodiment for displaying images to the wearer and simultaneously capturing images of the wearer's eye, wherein light from the eye **2971** is reflected towards a camera **3080** by the partially reflective layer **2960**. The partially reflective layer **2960** can be an optically flat layer such that the wavefront of the light from the eye **2971** is preserved and as a result, higher quality images of the wearer's eye can be captured. In addition, since the DLP **2955** is not included in the optical path for the light from the eye **2971**, and the eye imaging process shown in FIG. **30** does not interfere with the displayed image, images of the wearer's eye can be captured independently (e.g. with independent of timing, impact on resolution, or pixel count used in the image light) from the displayed images.

(260) In the embodiment illustrated in FIG. **30**, the partially reflective layer **2960** is a reflective

polarizer, the illuminating light **2973** is polarized, the light from the eye **2971** is polarized and the camera **3080** is located behind a polarizer **3085**. The polarization axis of the illuminating light **2973** and the polarization axis of the light from the eye are oriented perpendicular to the transmission axis of the reflective polarizer so that they are both substantially reflected by the reflective polarizer. The illumination light **2973** passes through a quarter wave layer **2957** before being reflected by the DLP mirrors in the DLP **2955**. The reflected light passes back through the quarter wave layer **2957** so that the polarization states of the image light **2969** and dark light **2975** are reversed in comparison to the illumination light **2973**. As such, the image light **2969** and dark light **2975** are substantially transmitted by the reflective polarizer. Where the DLP mirrors in the “on” state provide the image light **2969** along an optical axis that extends into the lower optical module **204** to display an image to the wearer. At the same time, DLP mirrors in the “off” state provide the dark light **2975** along an optical axis that extends to the side of the upper optics module **202**. In the region of the corrective wedge **2966** where the dark light **2975** is incident on the side of the upper optics module **202**, an absorptive polarizer **3085** is positioned with its transmission axis perpendicular to the polarization axis of the dark light and parallel to the polarization axis of the light from the eye so that the dark light **2975** is absorbed and the light from the eye **2971** is transmitted to the camera **3080**.

(261) FIG. **31** shows an illustration of another embodiment of a system for displaying images and simultaneously capturing image of the wearer's eye that is similar to the one shown in FIG. **30**. The difference in the system shown in FIG. **31** is that the light from the eye **2971** is subjected to multiple reflections before being captured by the camera **3180**. To enable the multiple reflections, a mirror **3187** is provided behind the absorptive polarizer **3185**. Therefore, the light from the eye **2971** is polarized prior to entering the corrective wedge **2966** with a polarization axis that is perpendicular to the transmission axis of the reflective polarizer that comprises the partially reflective layer **2960**. In this way, the light from the eye **2971** is reflected first by the reflective polarizer, reflected second by the mirror **3187** and reflected third by the reflective polarizer before being captured by the camera **3180**. While the light from the eye **2971** passes through the absorptive polarizer **3185** twice, since the polarization axis of the light from the eye **2971** is oriented parallel to the polarization axis of the light from the eye **2971**, it is substantially transmitted by the absorptive polarizer **3185**. As with the system described in connection with FIG. **30**, the system shown in FIG. **31** includes an optically flat partially reflective layer **2960** that preserves the wavefront of the light from the eye **2971** so that higher quality images of the wearer's eye can be captured. Also, since the DLP **2955** is not included in the optical path for the light reflected from the eye **2971** and the eye imaging process shown in FIG. **31** does not interfere with the displayed image, images of the wearer's eye can be captured independently from the displayed images.

(262) FIG. **32** shows an illustration of a system for displaying images and simultaneously capturing images of the wearer's eye that includes a beam splitter plate **3212** comprised of a reflective polarizer, which is held in air between the light source **2958**, the DLP **2955** and the camera **3280**. The illumination light **2973** and the light from the eye **2971** are both polarized with polarization axes that are perpendicular to the transmission axis of the reflective polarizer. As a result, both the illumination light **2973** and the light from the eye **2971** are substantially reflected by the reflective polarizer. The illumination light **2973** is reflected toward the DLP **2955** by the reflective polarizer and split into image light **2969** and dark light **3275** depending on whether the individual DLP mirrors are respectively in the “on” state or the “off” state. By passing through the quarter wave layer **2957** twice, the polarization state of the illumination light **2973** is reversed in comparison to the polarization state of the image light **2969** and the dark light **3275**. As a result, the image light **2969** and the dark light **3275** are then substantially transmitted by the reflective polarizer. The absorptive polarizer **3285** at the side of the beam splitter plate **3212** has a transmission axis that is perpendicular to the polarization axis of the dark light **3275** and parallel to the polarization axis of

the light from the eye **2971** so that the dark light **3275** is absorbed and the light from the eye **2971** is transmitted to the camera **3280**. As in the system shown in FIG. **30**, the system shown in FIG. **31** includes an optically flat beam splitter plate **3212** that preserves the wavefront of the light from the eye **2971** so that higher quality images of the wearer's eye can be captured. Also, since the DLP **2955** is not included in the optical path for the light from the eye **2971** and the eye imaging process shown in FIG. **31** does not interfere with the displayed image, images of the wearer's eye can be captured independently from the displayed images.

(263) Eye imaging systems where the polarization state of the light from the eye **2971** needs to be opposite to that of the image light **2969** (as shown in FIGS. **30**, **31** and **32**), need to be used with lower modules **204** that include combiners that will reflect both polarization states. As such, these upper modules **202** are best suited for use with the lower modules **204** that include combiners that are reflective regardless of polarization state, examples of these lower modules are shown in FIGS. **6**, **8a**, **8b**, **8c** and **24-27**.

(264) In a further embodiment shown in FIG. **33**, the partially reflective layer **3360** is comprised of a reflective polarizer on the side facing the illumination light **2973** and a short pass dichroic mirror on the side facing the light from the eye **3371** and the camera **3080**. Where the short pass dichroic mirror is a dielectric mirror coating that transmits visible light and reflects infrared light. The partially reflective layer **3360** can be comprised of a reflective polarizer bonded to the inner surface of the illumination wedge **2964** and a short pass dielectric mirror coating on the opposing inner surface of the corrective wedge **2966**, wherein the illumination wedge **2964** and the corrective wedge **2966** are then optically bonded together. Alternatively, the partially reflective layer **3360** can be comprised of a thin substrate that has a reflective polarizer bonded to one side and a short pass dichroic mirror coating on the other side, where the partially reflective layer **3360** is then bonded between the illumination wedge **2964** and the corrective wedge **2966**. In this embodiment, an infrared light is included to illuminate the eye so that the light from the eye and the images captured of the eye are substantially comprised of infrared light. The wavelength of the infrared light is then matched to the reflecting wavelength of the shortpass dichroic mirror and the wavelength that the camera can capture images, for example an 800 nm wavelength can be used. In this way, the short pass dichroic mirror transmits the image light and reflects the light from the eye. The camera **3080** is then positioned at the side of the corrective wedge **2966** in the area of the absorbing light trap **3382**, which is provided to absorb the dark light **2975**. By positioning the camera **3080** in a depression in the absorbing light trap **3382**, scattering of the dark light **2975** by the camera **3080** can be reduced so that higher contrast images can be displayed to the wearer. An advantage of this embodiment is that the light from the eye need not be polarized, which can simplify the optical system and increase efficiency for the eye imaging system.

(265) In yet another embodiment shown in FIG. **32a** a beam splitter plate **3222** is comprised of a reflective polarizer on the side facing the illumination light **2973** and a short pass dichroic mirror on the side facing the light from the eye **3271** and the camera **3280**. An absorbing surface **3295** is provided to trap the dark light **3275** and the camera **3280** is positioned in an opening in the absorbing surface **3295**. In this way the system of FIG. **32** can be made to function with unpolarized light from the eye **3271**.

(266) In embodiments directed to capturing images of the wearer's eye, light to illuminate the wearer's eye can be provided by several different sources including: light from the displayed image (i.e. image light); light from the environment that passes through the combiner or other optics; light provided by a dedicated eye light, etc. FIGS. **34** and **34a** show illustrations of dedicated eye illumination lights **3420**. FIG. **34** shows an illustration from a side view in which the dedicated illumination eye light **3420** is positioned at a corner of the combiner **3410** so that it doesn't interfere with the image light **3415**. The dedicated eye illumination light **3420** is pointed so that the eye illumination light **3425** illuminates the eyebox **3427** where the eye **3430** is located when the wearer is viewing displayed images provided by the image light **3415**. FIG. **34a** shows an illustration from

the perspective of the eye of the wearer to show how the dedicated eye illumination light **3420** is positioned at the corner of the combiner **3410**. While the dedicated eye illumination light **3420** is shown at the upper left corner of the combiner **3410**, other positions along one of the edges of the combiner **3410**, or other optical or mechanical components, are possible as well. In other embodiments, more than one dedicated eye light **3420** with different positions can be used. In an embodiment, the dedicated eye light **3420** is an infrared light that is not visible by the wearer (e.g. 800 nm) so that the eye illumination light **3425** doesn't interfere with the displayed image perceived by the wearer.

(267) FIG. **35** shows a series of illustrations of captured eye images that show the eye glint (i.e. light that reflects off the front of the eye) produced by a dedicated eye light. In this embodiment of the invention, captured images of the wearer's eye are analyzed to determine the relative positions of the iris **3550**, pupil, or other portion of the eye, and the eye glint **3560**. The eye glint is a reflected image of the dedicated eye light **3420** when the dedicated light is used. FIG. **35** illustrates the relative positions of the iris **3550** and the eye glint **3560** for a variety of eye positions. By providing a dedicated eye light **3420** in a fixed position, combined with the fact that the human eye is essentially spherical, or at least a reliably repeatable shape, the eye glint provides a fixed reference point against which the determined position of the iris can be compared to determine where the wearer is looking, either within the displayed image or within the see-through view of the surrounding environment. By positioning the dedicated eye light **3420** at a corner of the combiner **3410**, the eye glint **3560** is formed away from the iris **3550** in the captured images. As a result, the positions of the iris and the eye glint can be determined more easily and more accurately during the analysis of the captured images, since they do not interfere with one another. In a further embodiment, the combiner includes an associated cut filter that prevents infrared light from the environment from entering the HWC and the camera is an infrared camera, so that the eye glint is only provided by light from the dedicated eye light. For example, the combiner can include a low pass filter that passes visible light while absorbing infrared light and the camera can include a high pass filter that absorbs visible light while passing infrared light.

(268) In an embodiment of the eye imaging system, the lens for the camera is designed to take into account the optics associated with the upper module **202** and the lower module **204**. This is accomplished by designing the camera to include the optics in the upper module **202** and optics in the lower module **204**, so that a high MTF image is produced, at the image sensor in the camera, of the wearer's eye. In yet a further embodiment, the camera lens is provided with a large depth of field to eliminate the need for focusing the camera to enable sharp image of the eye to be captured. Where a large depth of field is typically provided by a high $f/\#$ lens (e.g. $f/\# > 5$). In this case, the reduced light gathering associated with high $f/\#$ lenses is compensated by the inclusion of a dedicated eye light to enable a bright image of the eye to be captured. Further, the brightness of the dedicated eye light can be modulated and synchronized with the capture of eye images so that the dedicated eye light has a reduced duty cycle and the brightness of infrared light on the wearer's eye is reduced.

(269) In a further embodiment, FIG. **36a** shows an illustration of an eye image that is used to identify the wearer of the HWC. In this case, an image of the wearer's eye **3611** is captured and analyzed for patterns of identifiable features **3612**. The patterns are then compared to a database of eye images to determine the identity of the wearer. After the identity of the wearer has been verified, the operating mode of the HWC and the types of images, applications, and information to be displayed can be adjusted and controlled in correspondence to the determined identity of the wearer. Examples of adjustments to the operating mode depending on who the wearer is determined to be or not be include: making different operating modes or feature sets available, shutting down or sending a message to an external network, allowing guest features and applications to run, etc.

(270) FIG. **36b** is an illustration of another embodiment using eye imaging, in which the sharpness

of the displayed image is determined based on the eye glint produced by the reflection of the displayed image from the wearer's eye surface. By capturing images of the wearer's eye **3611**, an eye glint **3622**, which is a small version of the displayed image can be captured and analyzed for sharpness. If the displayed image is determined to not be sharp, then an automated adjustment to the focus of the HWC optics can be performed to improve the sharpness. This ability to perform a measurement of the sharpness of a displayed image at the surface of the wearer's eye can provide a very accurate measurement of image quality. Having the ability to measure and automatically adjust the focus of displayed images can be very useful in augmented reality imaging where the focus distance of the displayed image can be varied in response to changes in the environment or changes in the method of use by the wearer.

(271) An aspect of the present invention relates to controlling the HWC **102** through interpretations of eye imagery. In embodiments, eye-imaging technologies, such as those described herein, are used to capture an eye image or series of eye images for processing. The image(s) may be process to determine a user intended action, an HWC predetermined reaction, or other action. For example, the imagery may be interpreted as an affirmative user control action for an application on the HWC **102**. Or, the imagery may cause, for example, the HWC **102** to react in a pre-determined way such that the HWC **102** is operating safely, intuitively, etc.

(272) FIG. **37** illustrates a eye imagery process that involves imaging the HWC **102** wearer's eye(s) and processing the images (e.g. through eye imaging technologies described herein) to determine in what position **3702** the eye is relative to it's neutral or forward looking position and/or the FOV **3708**. The process may involve a calibration step where the user is instructed, through guidance provided in the FOV of the HWC **102**, to look in certain directions such that a more accurate prediction of the eye position relative to areas of the FOV can be made. In the event the wearer's eye is determined to be looking towards the right side of the FOV **3708** (as illustrated in FIG. **37**, the eye is looking out of the page) a virtual target line may be established to project what in the environment the wearer may be looking towards or at. The virtual target line may be used in connection with an image captured by camera on the HWC **102** that images the surrounding environment in front of the wearer. In embodiments, the field of view of the camera capturing the surrounding environment matches, or can be matched (e.g. digitally), to the FOV **3708** such that making the comparison is made more clear. For example, with the camera capturing the image of the surroundings in an angle that matches the FOV **3708** the virtual line can be processed (e.g. in **2d** or **3d**, depending on the camera images capabilities and/or the processing of the images) by projecting what surrounding environment objects align with the virtual target line. In the event there are multiple objects along the virtual target line, focal planes may be established corresponding to each of the objects such that digital content may be placed in an area in the FOV **3708** that aligns with the virtual target line and falls at a focal plane of an intersecting object. The user then may see the digital content when he focuses on the object in the environment, which is at the same focal plane. In embodiments, objects in line with the virtual target line may be established by comparison to mapped information of the surroundings.

(273) In embodiments, the digital content that is in line with the virtual target line may not be displayed in the FOV until the eye position is in the right position. This may be a predetermined process. For example, the system may be set up such that a particular piece of digital content (e.g. an advertisement, guidance information, object information, etc.) will appear in the event that the wearer looks at a certain object(s) in the environment. A virtual target line(s) may be developed that virtually connects the wearer's eye with an object(s) in the environment (e.g. a building, portion of a building, mark on a building, gps location, etc.) and the virtual target line may be continually updated depending on the position and viewing direction of the wearer (e.g. as determined through GPS, e-compass, IMU, etc.) and the position of the object. When the virtual target line suggests that the wearer's pupil is substantially aligned with the virtual target line or about to be aligned with the virtual target line, the digital content may be displayed in the FOV **3704**.

(274) In embodiments, the time spent looking along the virtual target line and/or a particular portion of the FOV **3708** may indicate that the wearer is interested in an object in the environment and/or digital content being displayed. In the event there is no digital content being displayed at the time a predetermined period of time is spent looking at a direction, digital content may be presented in the area of the FOV **3708**. The time spent looking at an object may be interpreted as a command to display information about the object, for example. In other embodiments, the content may not relate to the object and may be presented because of the indication that the person is relatively inactive. In embodiments, the digital content may be positioned in proximity to the virtual target line, but not in-line with it such that the wearer's view of the surroundings are not obstructed but information can augment the wearer's view of the surroundings. In embodiments, the time spent looking along a target line in the direction of displayed digital content may be an indication of interest in the digital content. This may be used as a conversion event in advertising. For example, an advertiser may pay more for an add placement if the wearer of the HWC **102** looks at a displayed advertisement for a certain period of time. As such, in embodiments, the time spent looking at the advertisement, as assessed by comparing eye position with the content placement, target line or other appropriate position may be used to determine a rate of conversion or other compensation amount due for the presentation.

(275) An aspect of the invention relates to removing content from the FOV of the HWC **102** when the wearer of the HWC **102** apparently wants to view the surrounding environments clearly. FIG. **38** illustrates a situation where eye imagery suggests that the eye has or is moving quickly so the digital content **3804** in the FOV **3808** is removed from the FOV **3808**. In this example, the wearer may be looking quickly to the side indicating that there is something on the side in the environment that has grabbed the wearer's attention. This eye movement **3802** may be captured through eye imaging techniques (e.g. as described herein) and if the movement matches a predetermined movement (e.g. speed, rate, pattern, etc.) the content may be removed from view. In embodiments, the eye movement is used as one input and HWC movements indicated by other sensors (e.g. IMU in the HWC) may be used as another indication. These various sensor movements may be used together to project an event that should cause a change in the content being displayed in the FOV.

(276) Another aspect of the present invention relates to determining a focal plane based on the wearer's eye convergence. Eyes are generally converged slightly and converge more when the person focuses on something very close. This is generally referred to as convergence. In embodiments, convergence is calibrated for the wearer. That is, the wearer may be guided through certain focal plane exercises to determine how much the wearer's eyes converge at various focal planes and at various viewing angles. The convergence information may then be stored in a database for later reference. In embodiments, a general table may be used in the event there is no calibration step or the person skips the calibration step. The two eyes may then be imaged periodically to determine the convergence in an attempt to understand what focal plane the wearer is focused on. In embodiments, the eyes may be imaged to determine a virtual target line and then the eye's convergence may be determined to establish the wearer's focus, and the digital content may be displayed or altered based thereon.

(277) FIG. **39** illustrates a situation where digital content is moved **3902** within one or both of the FOVs **3908** and **3910** to align with the convergence of the eyes as determined by the pupil movement **3904**. By moving the digital content to maintain alignment, in embodiments, the overlapping nature of the content is maintained so the object appears properly to the wearer. This can be important in situations where 3D content is displayed.

(278) An aspect of the present invention relates to controlling the HWC **102** based on events detected through eye imaging. A wearer winking, blinking, moving his eyes in a certain pattern, etc. may, for example, control an application of the HWC **102**. Eye imaging (e.g. as described herein) may be used to monitor the eye(s) of the wearer and once a pre-determined pattern is detected an application control command may be initiated.

(279) An aspect of the invention relates to monitoring the health of a person wearing a HWC **102** by monitoring the wearer's eye(s). Calibrations may be made such that the normal performance, under various conditions (e.g. lighting conditions, image light conditions, etc.) of a wearer's eyes may be documented. The wearer's eyes may then be monitored through eye imaging (e.g. as described herein) for changes in their performance. Changes in performance may be indicative of a health concern (e.g. concussion, brain injury, stroke, loss of blood, etc.). If detected the data indicative of the change or event may be communicated from the HWC **102**.

(280) Aspects of the present invention relate to security and access of computer assets (e.g. the HWC itself and related computer systems) as determined through eye image verification. As discussed herein elsewhere, eye imagery may be compared to known person eye imagery to confirm a person's identity. Eye imagery may also be used to confirm the identity of people wearing the HWCs **102** before allowing them to link together or share files, streams, information, etc.

(281) A variety of use cases for eye imaging are possible based on technologies described herein. An aspect of the present invention relates to the timing of eye image capture. The timing of the capture of the eye image and the frequency of the capture of multiple images of the eye can vary dependent on the use case for the information gathered from the eye image. For example, capturing an eye image to identify the user of the HWC may be required only when the HWC has been turned ON or when the HWC determines that the HWC has been put onto a wearer's head, to control the security of the HWC and the associated information that is displayed to the user. Wherein, the orientation, movement pattern, stress or position of the earhorns (or other portions of the HWC) of the HWC can be used to determine that a person has put the HWC onto their head with the intention to use the HWC. Those same parameters may be monitored in an effort to understand when the HWC is dismounted from the user's head. This may enable a situation where the capture of an eye image for identifying the wearer may be completed only when a change in the wearing status is identified. In a contrasting example, capturing eye images to monitor the health of the wearer may require images to be captured periodically (e.g. every few seconds, minutes, hours, days, etc.). For example, the eye images may be taken in minute intervals when the images are being used to monitor the health of the wearer when detected movements indicate that the wearer is exercising. In a further contrasting example, capturing eye images to monitor the health of the wearer for long-term effects may only require that eye images be captured monthly. Embodiments of the invention relate to selection of the timing and rate of capture of eye images to be in correspondence with the selected use scenario associated with the eye images. These selections may be done automatically, as with the exercise example above where movements indicate exercise, or these selections may be set manually. In a further embodiment, the selection of the timing and rate of eye image capture is adjusted automatically depending on the mode of operation of the HWC. The selection of the timing and rate of eye image capture can further be selected in correspondence with input characteristics associated with the wearer including age and health status, or sensed physical conditions of the wearer including heart rate, chemical makeup of the blood and eye blink rate.

(282) FIG. **40** illustrates an embodiment in which digital content presented in a see-through FOV is positioned based on the speed in which the wearer is moving. When the person is not moving, as measured by sensor(s) in the HWC **102** (e.g. IMU, GPS based tracking, etc.), digital content may be presented at the stationary person content position **4004**. The content position **4004** is indicated as being in the middle of the see-through FOV **4002**; however, this is meant to illustrate that the digital content is positioned within the see-through FOV at a place that is generally desirable knowing that the wearer is not moving and as such the wearer's surrounding see through view can be somewhat obstructed. So, the stationary person content position, or neutral position, may not be centered in the see-through FOV; it may be positioned somewhere in the see-through FOV deemed desirable and the sensor feedback may shift the digital content from the neutral position. The

movement of the digital content for a quickly moving person is also shown in FIG. **40** wherein as the person turns their head to the side, the digital content moves out of the see-through FOV to content position **4008** and then moves back as the person turns their head back. For a slowly moving person, the head movement can be more complex and as such the movement of the digital content in an out of the see-through FOV can follow a path such as that shown by content position **4010**.

(283) In embodiments, the sensor that assesses the wearer's movements may be a GPS sensor, IMU, accelerometer, etc. The content position may be shifted from a neutral position to a position towards a side edge of the field of view as the forward motion increases. The content position may be shifted from a neutral position to a position towards a top or bottom edge of the field of view as the forward motion increases. The content position may shift based on a threshold speed of the assessed motion. The content position may shift linearly based on the speed of the forward motion. The content position may shift non-linearly based on the speed of the forward motion. The content position may shift outside of the field of view. In embodiments, the content is no longer displayed if the speed of movement exceeds a predetermined threshold and will be displayed again once the forward motion slows.

(284) In embodiments, the content position may generally be referred to as shifting; it should be understood that the term shifting encompasses a process where the movement from one position to another within the see-through FOV or out of the FOV is visible to the wearer (e.g. the content appears to slowly or quickly move and the user perceives the movement itself) or the movement from one position to another may not be visible to the wearer (e.g. the content appears to jump in a discontinuous fashion or the content disappears and then reappears in the new position).

(285) Another aspect of the present invention relates to removing the content from the field of view or shifting it to a position within the field of view that increases the wearer's view of the surrounding environment when a sensor causes an alert command to be issued. In embodiments, the alert may be due to a sensor or combination of sensors that sense a condition above a threshold value. For example, if an audio sensor detects a loud sound of a certain pitch, content in the field of view may be removed or shifted to provide a clear view of the surrounding environment for the wearer. In addition to the shifting of the content, in embodiments, an indication of why the content was shifted may be presented in the field of view or provided through audio feedback to the wearer. For instance, if a carbon monoxide sensor detects a high concentration in the area, content in the field of view may be shifted to the side of the field of view or removed from the field of view and an indication may be provided to the wearer that there is a high concentration of carbon monoxide in the area. This new information, when presented in the field of view, may similarly be shifted within or outside of the field of view depending on the movement speed of the wearer.

(286) FIG. **41** illustrates how content may be shifted from a neutral position **4104** to an alert position **4108**. In this embodiment, the content is shifted outside of the see-through FOV **4102**. In other embodiments, the content may be shifted as described herein.

(287) Another aspect of the present invention relates to identification of various vectors or headings related to the HWC **102**, along with sensor inputs, to determine how to position content in the field of view. In embodiments, the speed of movement of the wearer is detected and used as an input for position of the content and, depending on the speed, the content may be positioned with respect to a movement vector or heading (i.e. the direction of the movement), or a sight vector or heading (i.e. the direction of the wearer's sight direction). For example, if the wearer is moving very fast the content may be positioned within the field of view with respect to the movement vector because the wearer is only going to be looking towards the sides of himself periodically and for short periods of time. As another example, if the wearer is moving slowly, the content may be positioned with respect to the sight heading because the user may more freely be shifting his view from side to side.

(288) FIG. **42** illustrates two examples where the movement vector may effect content positioning. Movement vector A **4202** is shorter than movement vector B **4210** indicating that the forward speed

and/or acceleration of movement of the person associated with movement vector A **4202** is lower than the person associated with movement vector B **4210**. Each person is also indicated as having a sight vector or heading **4208** and **4212**. The sight vectors A **4208** and B **4210** are the same from a relative perspective. The white area inside of the black triangle in front of each person is indicative of how much time each person likely spends looking at a direction that is not in line with the movement vector. The time spent looking off angle A **4204** is indicated as being more than that of the time spent looking off angle B **4214**. This may be because the movement vector speed A is lower than movement vector speed B. The faster the person moves forward the more the person tends to look in the forward direction, typically. The FOVs A **4218** and B **4222** illustrate how content may be aligned depending on the movement vectors **4202** and **4210** and sight vectors **4208** and **4212**. FOV A **4218** is illustrated as presenting content in-line with the sight vector **4220**. This may be due to the lower speed of the movement vector A **4202**. This may also be due to the prediction of a larger amount of time spent looking off angle A **4204**. FOV B **4222** is illustrated as presenting content in line with the movement vector **4224**. This may be due to the higher speed of movement vector B **4210**. This may also be due to the prediction of a shorter amount of time spent looking off angle B **4214**.

(289) Another aspect of the present invention relates to damping a rate of content position change within the field of view. As illustrated in FIG. **43**, the sight vector may undergo a rapid change **4304**. This rapid change may be an isolated event or it may be made at or near a time when other sight vector changes are occurring. The wearer's head may be turning back and forth for some reason. In embodiments, the rapid successive changes in sight vector may cause a damped rate of content position change **4308** within the FOV **4302**. For example, the content may be positioned with respect to the sight vector, as described herein, and the rapid change in sight vector may normally cause a rapid content position change; however, since the sight vector is successively changing, the rate of position change with respect to the sight vector may be damped, slowed, or stopped. The position rate change may be altered based on the rate of change of the sight vector, average of the sight vector changes, or otherwise altered.

(290) Another aspect of the present invention relates to simultaneously presenting more than one content in the field of view of a see-through optical system of a HWC **102** and positioning one content with the sight heading and one content with the movement heading. FIG. **44** illustrates two FOV's A **4414** and B **4420**, which correspond respectively to the two identified sight vectors A **4402** and B **4404**. FIG. **44** also illustrates an object in the environment **4408** at a position relative to the sight vectors A **4402** and B **4404**. When the person is looking along sight vector A **4402**, the environment object **4408** can be seen through the field of view A **4414** at position **4412**. As illustrated, sight heading aligned content is presented as TEXT in proximity with the environment object **4412**. At the same time, other content **4418** is presented in the field of view A **4414** at a position aligned in correspondence with the movement vector. As the movement speed increases, the content **4418** may shift as described herein. When the sight vector of the person is sight vector B **4404** the environmental object **4408** is not seen in the field of view B **4420**. As a result, the sight aligned content **4410** is not presented in field of view B **4420**; however, the movement aligned content **4418** is presented and is still dependent on the speed of the motion.

(291) FIG. **45** shows an example set of data for a person moving through an environment over a path that starts with a movement heading of 0 degrees and ends with a movement heading of 114 degrees during which time the speed of movement varies from 0m/sec to 20 m/sec. The sight heading can be seen to vary on either side of the movement heading while moving as the person looks from side to side. Large changes in sight heading occur when the movement speed is 0m/sec when the person is standing still, followed by step changes in movement heading.

(292) Embodiments provide a process for determining the display heading that takes into account the way a user moves through an environment and provides a display heading that makes it easy for the user to find the displayed information while also providing unencumbered see-through views of

the environment in response to different movements, speed of movement or different types of information being displayed.

(293) FIG. **46** illustrates a see-through view as may be seen when using a HWC wherein information is overlaid onto a see-through view of the environment. The tree and the building are actually in the environment and the text is displayed in the see-through display such that it appears overlaid on the environment. In addition to text information such as, for example, instructions and weather information, some augmented reality information is shown that relates to nearby objects in the environment.

(294) In an embodiment, the display heading is determined based on speed of movement. At low speeds, the display heading may be substantially the same as the sight heading while at high speed the display heading may be substantially the same as the movement heading. In embodiments, as long as the user remains stationary, the displayed information is presented directly in front of the user and HWC. However, as the movement speed increases (e.g. above a threshold or continually, etc.) the display heading becomes substantially the same as the movement heading regardless of the direction the user is looking, so that when the user looks in the direction of movement, the displayed information is directly in front of the user and HMD and when the user looks to the side the displayed information is not visible.

(295) Rapid changes in sight heading can be followed by a slower change in the display heading to provide a damped response to head rotation. Alternatively, the display heading can be substantially the time averaged sight heading so that the displayed information is presented at a heading that is in the middle of a series of sight headings over a period of time. In this embodiment, if the user stops moving their head, the display heading gradually becomes the same as the sight heading and the displayed information moves into the display field of view in front of the user and HMD. In embodiments, when there is a high rate of sight heading change, the process delays the effect of the time averaged sight heading on the display heading. In this way, the effect of rapid head movements on display heading is reduced and the positioning of the displayed information within the display field of view is stabilized laterally.

(296) In another embodiment, display heading is determined based on speed of movement where at high-speed, the display heading is substantially the same as the movement heading. At mid-speed the display heading is substantially the same as a time averaged sight heading so that rapid head rotations are damped out and the display heading is in the middle of back and forth head movements.

(297) In yet another embodiment, the type of information being displayed is included in determining how the information should be displayed. Augmented reality information that is connected to objects in the environment is given a display heading that substantially matches the sight heading. In this way, as the user rotates their head, augmented reality information comes into view that is related to objects that are in the see-through view of the environment. At the same time, information that is not connected to objects in the environment is given a display heading that is determined based on the type of movements and speed of movements as previously described in this specification.

(298) In yet a further embodiment, when the speed of movement is determined to be above a threshold, the information displayed is moved downward in the display field of view so that the upper portion of the display field of view has less information or no information displayed to provide the user with an unencumbered see-through view of the environment.

(299) FIGS. **47** and **48** show illustrations of a see-through view including overlaid displayed information. FIG. **47** shows the see-through view immediately after a rapid change in sight heading from the sight heading associated with the see-through view shown in FIG. **46** wherein the change in sight heading comes from a head rotation. In this case, the display heading is delayed. FIG. **48** shows how at a later time, the display heading catches up to the sight heading. The augmented reality information remains in positions within the display field of view where the association with

objects in the environment can be readily made by the user.

(300) FIG. 49 shows an illustration of a see-through view example including overlaid displayed information that has been shifted downward in the display field of view to provide an unencumbered see-through view in the upper portion of the see-through view. At the same time, augmented reality labels have been maintained in locations within the display field of view so they can be readily associated with objects in the environment.

(301) In a further embodiment, in an operating mode such as when the user is moving in an environment, digital content is presented at the side of the user's see-through FOV so that the user can only view the digital content by turning their head. In this case, when the user is looking straight ahead, such as when the movement heading matches the sight heading, the see-through view FOV does not include digital content. The user then accesses the digital content by turning their head to the side whereupon the digital content moves laterally into the user's see-through FOV. In another embodiment, the digital content is ready for presentation and will be presented if an indication for its presentation is received. For example, the information may be ready for presentation and if the sight heading or predetermined position of the HWC 102 is achieved the content may then be presented. The wearer may look to the side and the content may be presented. In another embodiment, the user may cause the content to move into an area in the field of view by looking in a direction for a predetermined period of time, blinking, winking, or displaying some other pattern that can be captured through eye imaging technologies (e.g. as described herein elsewhere).

(302) In yet another embodiment, an operating mode is provided wherein the user can define sight headings wherein the associated see-through FOV includes digital content or does not include digital content. In an example, this operating mode can be used in an office environment where when the user is looking at a wall digital content is provided within the FOV, whereas when the user is looking toward a hallway, the FOV is unencumbered by digital content. In another example, when the user is looking horizontally digital content is provided within the FOV, but when the user looks down (e.g. to look at a desktop or a cellphone) the digital content is removed from the FOV.

(303) Another aspect of the present invention relates to collecting and using eye position and sight heading information. Head worn computing with motion heading, sight heading, and/or eye position prediction (sometimes referred to as "eye heading" herein) may be used to identify what a wearer of the HWC 102 is apparently interested in and the information may be captured and used. In embodiments, the information may be characterized as viewing information because the information apparently relates to what the wearer is looking at. The viewing information may be used to develop a personal profile for the wearer, which may indicate what the wearer tends to look at. The viewing information from several or many HWC's 102 may be captured such that group or crowd viewing trends may be established. For example, if the movement heading and sight heading are known, a prediction of what the wearer is looking at may be made and used to generate a personal profile or portion of a crowd profile. In another embodiment, if the eye heading and location, sight heading and/or movement heading are known, a prediction of what is being looked at may be predicted. The prediction may involve understanding what is in proximity of the wearer and this may be understood by establishing the position of the wearer (e.g. through GPS or other location technology) and establishing what mapped objects are known in the area. The prediction may involve interpreting images captured by the camera or other sensors associated with the HWC 102. For example, if the camera captures an image of a sign and the camera is in-line with the sight heading, the prediction may involve assessing the likelihood that the wearer is viewing the sign. The prediction may involve capturing an image or other sensory information and then performing object recognition analysis to determine what is being viewed. For example, the wearer may be walking down a street and the camera that is in the HWC 102 may capture an image and a processor, either on-board or remote from the HWC 102, may recognize a face, object, marker, image, etc. and it may be determined that the wearer may have been looking at it or towards it.

(304) FIG. 50 illustrates a cross section of an eyeball of a wearer of an HWC with focus points that can be associated with the eye imaging system of the invention. The eyeball **5010** includes an iris **5012** and a retina **5014**. Because the eye imaging system of the invention provides coaxial eye imaging with a display system, images of the eye can be captured from a perspective directly in front of the eye and inline with where the wearer is looking. In embodiments of the invention, the eye imaging system can be focused at the iris **5012** and/or the retina **5014** of the wearer, to capture images of the external surface of the iris **5012** or the internal portions of the eye, which includes the retina **5014**. FIG. 50 shows light rays **5020** and **5025** that are respectively associated with capturing images of the iris **5012** or the retina **5014** wherein the optics associated with the eye imaging system are respectively focused at the iris **5012** or the retina **5014**. Illuminating light can also be provided in the eye imaging system to illuminate the iris **5012** or the retina **5014**. FIG. 51 shows an illustration of an eye including an iris **5130** and a sclera **5125**. In embodiments, the eye imaging system can be used to capture images that include the iris **5130** and portions the sclera **5125**. The images can then be analyzed to determine color, shapes and patterns that are associated with the user. In further embodiments, the focus of the eye imaging system is adjusted to enable images to be captured of the iris **5012** or the retina **5014**. Illuminating light can also be adjusted to illuminate the iris **5012** or to pass through the pupil of the eye to illuminate the retina **5014**. The illuminating light can be visible light to enable capture of colors of the iris **5012** or the retina **5014**, or the illuminating light can be ultraviolet (e.g. 340 nm), near infrared (e.g. 850 nm) or mid-wave infrared (e.g. 5000 nm) light to enable capture of hyperspectral characteristics of the eye.

(305) FIG. 53 illustrates a display system that includes an eye imaging system. The display system includes a polarized light source **2958**, a DLP **2955**, a quarter wave film **2957** and a beam splitter plate **5345**. The eye imaging system includes a camera **3280**, illuminating lights **5355** and beam splitter plate **5345**. Where the beam splitter plate **5345** can be a reflective polarizer on the side facing the polarized light source **2958** and a hot mirror on the side facing the camera **3280**. Wherein the hot mirror reflects infrared light (e.g. wavelengths 700 to 2000 nm) and transmits visible light (e.g. wavelengths 400 to 670 nm). The beam splitter plate **5345** can be comprised of multiple laminated films, a substrate film with coatings or a rigid transparent substrate with films on either side. By providing a reflective polarizer on the one side, the light from the polarized light source **2958** is reflected toward the DLP **2955** where it passes through the quarter wave film **2957** once, is reflected by the DLP mirrors in correspondence with the image content being displayed by the DLP **2955** and then passes back through the quarter wave film **2957**. In so doing, the polarization state of the light from the polarized light source is changed, so that it is transmitted by the reflective polarizer on the beam splitter plate **5345** and the image light **2971** passes into the lower optics module **204** where the image is displayed to the user. At the same time, infrared light **5357** from the illuminating lights **5355** is reflected by the hot mirror so that it passes into the lower optics module **204** where it illuminates the user's eye. Portions of the infrared light **2969** are reflected by the user's eye and this light passes back through the lower optics module **204**, is reflected by the hot mirror on the beam splitter plate **5345** and is captured by the camera **3280**. In this embodiment, the image light **2971** is polarized while the infrared light **5357** and **2969** can be unpolarized. In an embodiment, the illuminating lights **5355** provide two different infrared wavelengths and eye images are captured in pairs, wherein the pairs of eye images are analyzed together to improve the accuracy of identification of the user based on iris analysis.

(306) FIG. 54 shows an illustration of a further embodiment of a display system with an eye imaging system. In addition to the features of FIG. 53, this system includes a second camera **5460**. Wherein the second camera **5460** is provided to capture eye images in the visible wavelengths. Illumination of the eye can be provided by the displayed image or by see-through light from the environment. Portions of the displayed image can be modified to provide improved illumination of the user's eye when images of the eye are to be captured such as by increasing the brightness of the displayed image or increasing the white areas within the displayed image. Further, modified

displayed images can be presented briefly for the purpose of capturing eye images and the display of the modified images can be synchronized with the capture of the eye images. As shown in FIG. 54, visible light 5467 is polarized when it is captured by the second camera 5460 since it passes through the beam splitter 5445 and the beam splitter 5445 is a reflective polarizer on the side facing the second camera 5460. In this eye imaging system, visible eye images can be captured by the second camera 5460 at the same time that infrared eye images are captured by the camera 3280. Wherein, the characteristics of the camera 3280 and the second camera 5460 and the associated respective images captured can be different in terms of resolution and capture rate.

(307) FIGS. 52a and 52b illustrate captured images of eyes where the eyes are illuminated with structured light patterns. In FIG. 52a, an eye 5220 is shown with a projected structured light pattern 5230, where the light pattern is a grid of lines. A light pattern of such as 5230 can be provided by the light source 5355 shown in FIG. 53 by including a diffractive or a refractive device to modify the light 5357 as are known by those skilled in the art. A visible light source can also be included for the second camera 5460 shown in FIG. 54 which can include a diffractive or refractive to modify the light 5467 to provide a light pattern. FIG. 52b illustrates how the structured light pattern of 5230 becomes distorted to 5235 when the user's eye 5225 looks to the side. This distortion comes from the fact that the human eye is not spherical in shape, instead the iris sticks out slightly from the eyeball to form a bump in the area of the iris. As a result, the shape of the eye and the associated shape of the reflected structured light pattern is different depending on which direction the eye is pointed, when images of the eye are captured from a fixed position. Changes in the structured light pattern can subsequently be analyzed in captured eye images to determine the direction that the eye is looking.

(308) The eye imaging system can also be used for the assessment of aspects of health of the user. In this case, information gained from analyzing captured images of the iris 5012 is different from information gained from analyzing captured images of the retina 5014. Where images of the retina 5014 are captured using light 5357 that illuminates the inner portions of the eye including the retina 5014. The light 5357 can be visible light, but in an embodiment, the light 5357 is infrared light (e.g. wavelength 1 to 5 microns) and the camera 3280 is an infrared light sensor (e.g. an InGaAs sensor) or a low resolution infrared image sensor that is used to determine the relative amount of light 5357 that is absorbed, reflected or scattered by the inner portions of the eye. Wherein the majority of the light that is absorbed, reflected or scattered can be attributed to materials in the inner portion of the eye including the retina where there are densely packed blood vessels with thin walls so that the absorption, reflection and scattering are caused by the material makeup of the blood. These measurements can be conducted automatically when the user is wearing the HWC, either at regular intervals, after identified events or when prompted by an external communication. In a preferred embodiment, the illuminating light is near infrared or mid infrared (e.g. 0.7 to 5 microns wavelength) to reduce the chance for thermal damage to the wearer's eye. In another embodiment, the polarizer 3285 is antireflection coated to reduce any reflections from this surface from the light 5357, the light 2969 or the light 3275 and thereby increase the sensitivity of the camera 3280. In a further embodiment, the light source 5355 and the camera 3280 together comprise a spectrometer wherein the relative intensity of the light reflected by the eye is analyzed over a series of narrow wavelengths within the range of wavelengths provided by the light source 5355 to determine a characteristic spectrum of the light that is absorbed, reflected or scattered by the eye. For example, the light source 5355 can provide a broad range of infrared light to illuminate the eye and the camera 3280 can include: a grating to laterally disperse the reflected light from the eye into a series of narrow wavelength bands that are captured by a linear photodetector so that the relative intensity by wavelength can be measured and a characteristic absorbance spectrum for the eye can be determined over the broad range of infrared. In a further example, the light source 5355 can provide a series of narrow wavelengths of light (ultraviolet, visible or infrared) to sequentially illuminate the eye and camera 3280 includes a photodetector that is selected to measure the relative

intensity of the series of narrow wavelengths in a series of sequential measurements that together can be used to determine a characteristic spectrum of the eye. The determined characteristic spectrum is then compared to known characteristic spectra for different materials to determine the material makeup of the eye. In yet another embodiment, the illuminating light **5357** is focused on the retina **5014** and a characteristic spectrum of the retina **5014** is determined and the spectrum is compared to known spectra for materials that may be present in the user's blood. For example, in the visible wavelengths 540 nm is useful for detecting hemoglobin and 660 nm is useful for differentiating oxygenated hemoglobin. In a further example, in the infrared, a wide variety of materials can be identified as is known by those skilled in the art, including: glucose, urea, alcohol and controlled substances. FIG. **55** shows a series of example spectrum for a variety of controlled substances as measured using a form of infrared spectroscopy (ThermoScientific Application Note 51242, by C. Petty, B. Garland and the Mesa Police Department Forensic Laboratory, which is hereby incorporated by reference herein). FIG. **56** shows an infrared absorbance spectrum for glucose (Hewlett Packard Company 1999, G. Hopkins, G. Mauze; "In-vivo NIR Diffuse-reflectance Tissue Spectroscopy of Human Subjects," which is hereby incorporated by reference herein). U.S. Pat. No. 6,675,030, which is hereby incorporated by reference herein, provides a near infrared blood glucose monitoring system that includes infrared scans of a body part such as a foot. United States Patent publication 2006/0183986, which is hereby incorporated by reference herein, provides a blood glucose monitoring system including a light measurement of the retina. Embodiments of the present invention provide methods for automatic measurements of specific materials in the user's blood by illuminating at one or more narrow wavelengths into the iris of the wearer's eye and measuring the relative intensity of the light reflected by the eye to identify the relative absorbance spectrum and comparing the measured absorbance spectrum with known absorbance spectra for the specific material, such as illuminating at 540 and 660 nm to determine the level of hemoglobin present in the user's blood.

(309) Another aspect of the present invention relates to collecting and using eye position and sight heading information. Head worn computing with motion heading, sight heading, and/or eye position prediction (sometimes referred to as "eye heading" herein) may be used to identify what a wearer of the HWC **102** is apparently interested in and the information may be captured and used. In embodiments, the information may be characterized as viewing information because the information apparently relates to what the wearer is looking at. The viewing information may be used to develop a personal profile for the wearer, which may indicate what the wearer tends to look at. The viewing information from several or many HWC's **102** may be captured such that group or crowd viewing trends may be established. For example, if the movement heading and sight heading are known, a prediction of what the wearer is looking at may be made and used to generate a personal profile or portion of a crowd profile. In another embodiment, if the eye heading and location, sight heading and/or movement heading are known, a prediction of what is being looked at may be predicted. The prediction may involve understanding what is in proximity of the wearer and this may be understood by establishing the position of the wearer (e.g. through GPS or other location technology) and establishing what mapped objects are known in the area. The prediction may involve interpreting images captured by the camera or other sensors associated with the HWC **102**. For example, if the camera captures an image of a sign and the camera is in-line with the sight heading, the prediction may involve assessing the likelihood that the wearer is viewing the sign. The prediction may involve capturing an image or other sensory information and then performing object recognition analysis to determine what is being viewed. For example, the wearer may be walking down a street and the camera that is in the HWC **102** may capture an image and a processor, either on-board or remote from the HWC **102**, may recognize a face, object, marker, image, etc. and it may be determined that the wearer may have been looking at it or towards it.

(310) FIG. **57** illustrates a scene where a person is walking with a HWC **102** mounted on his head. In this scene, the person's geo-spatial location **5704** is known through a GPS sensor, which could

be another location system, and his movement heading, sight heading **5714** and eye heading **5702** are known and can be recorded (e.g. through systems described herein). There are objects and a person in the scene. Person **5712** may be recognized by the wearer's HWC **102** system, the person may be mapped (e.g. the person's GPS location may be known or recognized), or otherwise known. The person may be wearing a garment or device that is recognizable. For example, the garment may be of a certain style and the HWC may recognize the style and record it's viewing. The scene also includes a mapped object **5718** and a recognized object **5720**. As the wearer moves through the scene, the sight and/or eye headings may be recorded and communicated from the HWC **102**. In embodiments, the time that the sight and/or eye heading maintains a particular position may be recorded. For example, if a person appears to look at an object or person for a predetermined period of time (e.g. 2 seconds or longer), the information may be communicated as gaze persistence information as an indication that the person may have been interested in the object.

(311) In embodiments, sight headings may be used in conjunction with eye headings or eye and/or sight headings may be used alone. Sight headings can do a good job of predicting what direction a wearer is looking because many times the eyes are looking forward, in the same general direction as the sight heading. In other situations, eye headings may be a more desirable metric because the eye and sight headings are not always aligned. In embodiments herein examples may be provided with the term “eye/sight” heading, which indicates that either or both eye heading and sight heading may be used in the example.

(312) FIG. **58** illustrates a system for receiving, developing and using movement heading, sight heading, eye heading and/or persistence information from HWC(s) **102**. The server **5804** may receive heading or gaze persistence information, which is noted as persistence information **5802**, for processing and/or use. The heading and/or gaze persistence information may be used to generate a personal profile **5808** and/or a group profile **5810**. The personal profile **5718** may reflect the wearer's general viewing tendencies and interests. The group profile **5810** may be an assemblage of different wearer's heading and persistence information to create impressions of general group viewing tendencies and interests. The group profile **5810** may be broken into different groups based on other information such as gender, likes, dislikes, biographical information, etc. such that certain groups can be distinguished from other groups. This may be useful in advertising because an advertiser may be interested in what a male adult sports goer is generally looking at as oppose to a younger female. The profiles **5808** and **5810** and raw heading and persistence information may be used by retailers **5814**, advertisers **5818**, trainers, etc. For example, an advertiser may have an advertisement posted in an environment and may be interested in knowing how many people look at the advertisement, how long they look at it and where they go after looking at it. This information may be used as conversion information to assess the value of the advertisement and thus the payment to be received for the advertisement.

(313) In embodiments, the process involves collecting eye and/or sight heading information from a plurality of head-worn computers that come into proximity with an object in an environment. For example, a number of people may be walking through an area and each of the people may be wearing a head worn computer with the ability to track the position of the wearer's eye(s) as well as possibly the wearer's sight and movement headings. The various HWC wearing individuals may then walk, ride, or otherwise come into proximity with some object in the environment (e.g. a store, sign, person, vehicle, box, bag, etc.). When each person passes by or otherwise comes near the object, the eye imaging system may determine if the person is looking towards the object. All of the eye/sight heading information may be collected and used to form impressions of how the crowd reacted to the object. A store may be running a sale and so the store may put out a sign indicating such. The storeowners and managers may be very interested to know if anyone is looking at their sign. The sign may be set as the object of interest in the area and as people navigate near the sign, possibly determined by their GPS locations, the eye/sight heading determination system may record information relative to the environment and the sign. Once, or as, the eye/sight heading

information is collected and associations between the eye headings and the sign are determined, feedback may be sent back to the storeowner, managers, advertiser, etc. as an indication of how well their sign is attracting people. In embodiments, the sign's effectiveness at attracting people's attention, as indicated through the eye/sight headings, may be considered a conversion metric and impact the economic value of the sign and/or the signs placement.

(314) In embodiments, a map of the environment with the object may be generated by mapping the locations and movement paths of the people in the crowd as they navigate by the object (e.g. the sign). Layered on this map may be an indication of the various eye/sight headings. This may be useful in indicating where people were in relation to the object when they viewed the object. The map may also have an indication of how long people looked at the object from the various positions in the environment and where they went after seeing the object.

(315) In embodiments, the process involves collecting a plurality of eye/sight headings from a head-worn computer, wherein each of the plurality of eye/sight headings is associated with a different pre-determined object in an environment. This technology may be used to determine which of the different objects attracts more of the person's attention. For example, if there are three objects placed in an environment and a person enters the environment navigating his way through it, he may look at one or more of the objects and his eye/sight heading may persist on one or more objects longer than others. This may be used in making or refining the person's personal attention profile and/or it may be used in connection with other such people's data on the same or similar objects to determine an impression of how the population or crowd reacts to the objects. Testing advertisements in this way may provide good feedback of its effectiveness.

(316) In embodiments, the process may involve capturing eye/sight headings once there is substantial alignment between the eye/sight heading and an object of interest. For example, the person with the HWC may be navigating through an environment and once the HWC detects substantial alignment or the projected occurrence of an upcoming substantial alignment between the eye/sight heading and the object of interest, the occurrence and/or persistence may be recorded for use.

(317) In embodiments, the process may involve collecting eye/sight heading information from a head-worn computer and collecting a captured image from the head-worn computer that was taken at substantially the same time as the eye/sight heading information was captured. These two pieces of information may be used in conjunction to gain an understanding of what the wearer was looking at and possibly interested in. The process may further involve associating the eye/sight heading information with an object, person, or other thing found in the captured image. This may involve processing the captured image looking for objects or patterns. In embodiments, gaze time or persistence may be measured and used in conjunction with the image processing. The process may still involve object and/or pattern recognition, but it may also involve attempting to identify what the person gazed at for the period of time by more particularly identifying a portion of the image in conjunction with image processing.

(318) In embodiments, the process may involve setting a pre-determined eye/sight heading from a pre-determined geospatial location and using them as triggers. In the event that a head worn computer enters the geospatial location and an eye/sight heading associated with the head worn computer aligns with the pre-determined eye/sight heading, the system may collect the fact that there was an apparent alignment and/or the system may record information identifying how long the eye/sight heading remains substantially aligned with the pre-determined eye/sight heading to form a persistence statistic. This may eliminate or reduce the need for image processing as the triggers can be used without having to image the area. In other embodiments, image capture and processing is performed in conjunction with the triggers. In embodiments, the triggers may be a series a geospatial locations with corresponding eye/sight headings such that many spots can be used as triggers that indicate when a person entered an area in proximity to an object of interest and/or when that person actually appeared to look at the object.

(319) In embodiments, eye imaging may be used to capture images of both eyes of the wearer in order to determine the amount of convergence of the eyes (e.g. through technologies described herein elsewhere) to get an understanding of what focal plane is being concentrated on by the wearer. For example, if the convergence measurement suggests that the focal plane is within 15 feet of the wearer, than, even though the eye/sight headings may align with an object that is more than 15 feet away it may be determined that the wearer was not looking at the object. If the object were within the 15 foot suggested focal plane, the determination may be that the wearer was looking at the object. FIG. 59 illustrates an environmentally position locked digital content **5912** that is indicative of a person's location **5902**. In this disclosure the term “BlueForce” is generally used to indicate team members or members for which geo-spatial locations are known and can be used. In embodiments, “Blue Force” is a term to indicate members of a tactical arms team (e.g. a police force, secret service force, security force, military force, national security force, intelligence force, etc.). In many embodiments herein one member may be referred to as the primary or first BlueForce member and it is this member, in many described embodiments, that is wearing the HWC. It should be understood that this terminology is to help the reader and make for clear presentations of the various situations and that other members of the Blueforce, or other people, may have HWC's **102** and have similar capabilities. In this embodiment, a first person is wearing a head-worn computer **102** that has a see through field of view (“FOV”) **5914**. The first person can see through the FOV to view the surrounding environment through the FOV and digital content can also be presented in the FOV such that the first person can view the actual surroundings, through the FOV, in a digitally augmented view. The other BlueForce person's location is known and is indicated at a position inside of a building at point **5902**. This location is known in three dimensions, longitude, latitude and altitude, which may have been determined by GPS along with an altimeter associated with the other Blueforce person. Similarly, the location of the first person wearing the HWC **102** is also known, as indicated in FIG. 59 as point **5908**. In this embodiment, the compass heading **5910** of the first person is also known. With the compass heading **5910** known, the angle in which the first person is viewing the surroundings can be estimated. A virtual target line between the location of the first person **5908** and the other person's location **5902** can be established in three dimensional space and emanating from the HWC **102** proximate the FOV **5914**. The three dimensionally oriented virtual target line can then be used to present environmentally position locked digital content in the FOV **5914**, which is indicative of the other person's location **5902**. The environmentally position locked digital content **5902** can be positioned within the FOV **5914** such that the first person, who is wearing the HWC **102**, perceives the content **5902** as locked in position within the environment and marking the location of the other person **5902**.

(320) The three dimensionally positioned virtual target line can be recalculated periodically (e.g. every millisecond, second, minute, etc.) to reposition the environmentally position locked content **5912** to remain in-line with the virtual target line. This can create the illusion that the content **5912** is staying positioned within the environment at a point that is associated with the other person's location **5902** independent of the location of the first person **5908** wearing the HWC **102** and independent of the compass heading of the HWC **102**.

(321) In embodiments, the environmentally locked digital content **5912** may be positioned with an object **5904** that is between the first person's location **5908** and the other person's location **5902**. The virtual target line may intersect the object **5904** before intersecting with the other person's location **5902**. In embodiments, the environmentally locked digital content **5912** may be associated with the object intersection point **5904**. In embodiments, the intersecting object **5904** may be identified by comparing the two person's locations **5902** and **5908** with obstructions identified on a map. In embodiments the intersecting object **5904** may be identified by processing images captured from a camera, or other sensor, associated with the HWC **102**. In embodiments, the digital content **5912** has an appearance that is indicative of being at the location of the other person **5902**, at the

location of the intersecting object **5904** to provide a more clear indication of the position of the other person's position **5902** in the FOV **5914**.

(322) FIG. **60** illustrates how and where digital content may be positioned within the FOV **6008** based on a virtual target line between the location of the first person **5908**, who's wearing the HWC **102**, and the other person **5902**. In addition to positioning the content in a position within the FOV **6008** that is in-line with the virtual target line, the digital content may be presented such that it comes into focus by the first person when the first person focuses at a certain plane or distance in the environment. Presented object A **6018** is digitally generated content that is presented as an image at content position A **6012**. The position **6012** is based on the virtual target line. The presented object A **6018** is presented not only along the virtual target line but also at a focal plane B **6014** such that the content at position A **6012** in the FOV **6008** comes into focus by the first person when the first person's eye **6002** focuses at something in the surrounding environment at the focal plane B **6014** distance. Setting the focal plane of the presented content provides content that does not come into focus until the eye **6002** focuses at the set focal plane. In embodiments, this allows the content at position A to be presented without when the HWC's compass is indicative of the first person looking in the direction of the other person **5902** but it will only come into focus when the first person focuses on in the direction of the other person **5902** and at the focal plane of the other person **5902**.

(323) Presented object B **6020** is aligned with a different virtual target line than presented object A **6018**. Presented object B **6020** is also presented at content position B **6004** at a different focal plane than the content position A **6012**. Presented content B **6020** is presented at a further focal plane, which is indicative that the other person **5902** is physically located at a further distance. If the focal planes are sufficiently different, the content at position A will come into focus at a different time than the content at position B because the two focal planes require different focus from the eye **6002**.

(324) FIG. **61** illustrates several BlueForce members at locations with various points of view from the first person's perspective. In embodiments, the relative positions, distances and obstacles may cause the digital content indicative of the other person's location to be altered. For example, if the other person can be seen by the first person through the first person's FOV, the digital content may be locked at the location of the other person and the digital content may be of a type that indicates the other person's position is being actively marked and tracked. If the other person is in relatively close proximity, but cannot be seen by the first person, the digital content may be locked to an intersecting object or area and the digital content may indicate that the actual location of the other person cannot be seen but the mark is generally tracking the other persons general position. If the other person is not within a pre-determined proximity or is otherwise more significantly obscured from the first person's view, the digital content may generally indicate a direction or area where the other person is located and the digital content may indicate that the other person's location is not closely identified or tracked by the digital content, but that the other person is in the general area.

(325) Continuing to refer to FIG. **61**, several BlueForce members are presented at various positions within an area where the first person is located. The primary BlueForce member **6102** (also referred to generally as the first person, or the person wherein the HWC with the FOV for example purposes) can directly see the BlueForce member in the open field **6104**. In embodiments, the digital content provided in the FOV of the primary BlueForce member may be based on a virtual target line and virtually locked in an environment position that is indicative of the open field position of the BlueForce member **6104**. The digital content may also indicate that the location of the open field BlueForce member is marked and is being tracked. The digital content may change forms if the Blue Force member becomes obscured from the vision of the primary BlueForce member or otherwise becomes unavailable for direct viewing.

(326) BlueForce member **6108** is obscured from the primary BlueForce member's **6102** view by an obstacle that is in close proximity to the obscured member **6108**. As depicted, the obscured member

6108 is in a building but close to one of the front walls. In this situation, the digital content provided in the FOV of the primary member **6102** may be indicative of the general position of the obscured member **6108** and the digital content may indicate that, while the other person's location is fairly well marked, it is obscured so it is not as precise as if the person was in direct view. In addition, the digital content may be virtually positionally locked to some feature on the outside of the building that the obscured member is in. This may make the environmental locking more stable and also provide an indication that the location of the person is somewhat unknown.

(327) BlueForce member **6110** is obscured by multiple obstacles. The member **6110** is in a building and there is another building **6112** in between the primary member **6102** and the obscured member **6110**. In this situation, the digital content in the FOV of the primary member will be spatially quite short of the actual obscured member and as such the digital content may need to be presented in a way that indicates that the obscured member **6110** is in a general direction but that the digital marker is not a reliable source of information for the particular location of obscured member **6110**.

(328) FIG. **62** illustrates yet another method for positioning digital content within the FOV of a HWC where the digital content is intended to indicate a position of another person. This embodiment is similar to the embodiment described in connection with FIG. **62** herein. The main additional element in this embodiment is the additional step of verifying the distance between the first person **5908**, the one wearing the HWC with the FOV digital content presentation of location, and the other person at location **5902**. Here, the range finder may be included in the HWC and measure a distance at an angle that is represented by the virtual target line. In the event that the range finder finds an object obstructing the path of the virtual target line, the digital content presentation in the FOV may indicate such (e.g. as described herein elsewhere). In the event that the range finder confirms that there is a person or object at the end of the prescribed distance and angle defined by the virtual target line, the digital content may represent that the proper location has been marked, as described herein elsewhere.

(329) Another aspect of the present invention relates to predicting the movement of BlueForce members to maintain proper virtual marking of the BlueForce member locations. FIG. **63** illustrates a situation where the primary BlueForce member **6302** is tracking the locations of the other BlueForce members through an augmented environment using a HWC **102**, as described herein elsewhere (e.g. as described in connection with the above figures). The primary BlueForce member **6302** may have knowledge of the tactical movement plan **6308**. The tactical movement plan may be maintained locally (e.g. on the HWCs **102** with sharing of the plan between the BlueForce members) or remotely (e.g. on a server and communicated to the HWC's **102**, or communicated to a subset of HWC's **102** for HWC **102** sharing). In this case, the tactical plan involves the BlueForce group generally moving in the direction of the arrow **6308**. The tactical plan may influence the presentations of digital content in the FOV of the HWC **102** of the primary BlueForce member. For example, the tactical plan may assist in the prediction of the location of the other BlueForce member and the virtual target line may be adjusted accordingly. In embodiments, the area in the tactical movement plan may be shaded or colored or otherwise marked with digital content in the FOV such that the primary BlueForce member can manage his activities with respect to the tactical plan. For example, he may be made aware that one or more BlueForce members are moving towards the tactical path **6308**. He may also be made aware of movements in the tactical path that do not appear associated with BlueForce members.

(330) FIG. **63** also illustrates that internal IMU sensors in the HWCs worn by the BlueForce members may provide guidance on the movement of the members **6304**. This may be helpful in identifying when a GPS location should be updated and hence updating the position of the virtual marker in the FOV. This may also be helpful in assessing the validity of the GPS location. For example, if the GPS location has not updated but there is significant IMU sensor activity, the system may call into question the accuracy of the identified location. The IMU information may also be useful to help track the position of a member in the event the GPS information is

unavailable. For example, dead reckoning may be used if the GPS signal is lost and the virtual marker in the FOV may indicate both indicated movements of the team member and indicate that the location identification is not ideal. The current tactical plan **6308** may be updated periodically and the updated plans may further refine what is presented in the FOV of the HWC **102**.

(331) FIG. **64** illustrates a BlueForce tracking system in accordance with the principles of the present invention. In embodiments, the BlueForce HWC's **102** may have directional antenna's that emit relatively low power directional RF signals such that other BlueForce members within the range of the relatively low power signal can receive and assess it's direction and/or distance based on the strength and varying strength of the signals. In embodiments, the tracking of such RF signals can be used to alter the presentation of the virtual markers of persons locations within the FOV of HWC **102**.

(332) Another aspect of the present invention relates to monitoring the health of BlueForce members. Each BlueForce member may be automatically monitored for health and stress events. For example, the members may have a watchband as described herein elsewhere or other wearable biometric monitoring device and the device may continually monitor the biometric information and predict health concerns or stress events. As another example, the eye imaging systems described herein elsewhere may be used to monitor pupil dilatations as compared to normal conditions to predict head trauma. Each eye may be imaged to check for differences in pupil dilation for indications of head trauma. As another example, an IMU in the HWC **102** may monitor a person's walking gate looking for changes in pattern, which may be an indication of head or other trauma. Biometric feedback from a member indicative of a health or stress concern may be uploaded to a server for sharing with other members or the information may be shared with local members, for example. Once shared, the digital content in the FOF that indicates the location of the person having the health or stress event may include an indication of the health event.

(333) FIG. **65** illustrates a situation where the primary BlueForce member **6502** is monitoring the location of the BlueForce member **6504** that has had a health event and caused a health alert to be transmitted from the HWC **102**. As described herein elsewhere, the FOV of the HWC **102** of the primary BlueForce member may include an indication of the location of the BlueForce member with the health concern **6504**. The digital content in the FOV may also include an indication of the health condition in association with the location indication. In embodiments, non-biometric sensors (e.g. IMU, camera, ranger finder, accelerometer, altimeter, etc.) may be used to provide health and/or situational conditions to the BlueForce team or other local or remote persons interested in the information. For example, if one of the BlueForce members is detected as quickly hitting the ground from a standing position an alert may be sent as an indication of a fall, the person is in trouble and had to drop down, was shot, etc.

(334) Another aspect of the present invention relates to virtually marking various prior acts and events. For example, as depicted in FIG. **66**, the techniques described herein elsewhere may be used to construct a virtual prior movement path **6604** of a BlueForce member. The virtual path may be displayed as digital content in the FOV of the primary BlueForce member **6602** using methods described herein elsewhere. As the BlueForce member moved along the path **6604** he may have virtually placed an event marker **6608** such that when another member views the location the mark can be displayed as digital content. For example, the BlueForce member may inspect and clear an area and then use an external user interface or gesture to indicate that the area has been cleared and then the location would be virtually marked and shared with BlueForce members. Then, when someone wants to understand if the location was inspected he can view the location's information. As indicated herein elsewhere, if the location is visible to the member, the digital content may be displayed in a way that indicates the specific location and if the location is not visible from the person's perspective, the digital content may be somewhat different in that it may not specifically mark the location.

(335) Turning back to optical configurations, another aspect of the present invention relates to an

optical configuration that provides digitally displayed content to an eye of a person wearing a head-worn display (e.g. as used in a HWC **102**) and allows the person to see through the display such that the digital content is perceived by the person as augmenting the see through view of the surrounding environment. The optical configuration may have a variable transmission optical element that is in-line with the person's see-through view such that the transmission of the see-through view can be increased and decreased. This may be helpful in situations where a person wants or would be better served with a high transmission see-through view and when, in the same HWC **102**, the person wants or would be better served with less see-through transmission. The lower see-through transmission may be used in bright conditions and/or in conditions where higher contrast for the digitally presented content is desirable. The optical system may also have a camera that images the surrounding environment by receiving reflected light from the surrounding environment off of an optical element that is in-line with the person's see-through view of the surrounding. In embodiments, the camera may further be aligned in a dark light trap such that light reflected and/or transmitted in the direction of the camera that is not captured by the camera is trapped to reduce stray light.

(336) In embodiments, a HWC **102** is provided that includes a camera that is coaxially aligned with the direction that the user is looking. FIG. **67** shows an illustration of an optical system **6715** that includes an absorptive polarizer **6737** and a camera **6739**. The image source **6710** can include light sources, displays and reflective surfaces as well as one or more lenses **6720**. Image light **6750** is provided by the image source **6710** wherein, a portion of the image light **6750** is reflected toward the user's eye **6730** by a partially reflective combiner **6735**. At the same time, a portion of the image light **6750** may be transmitted by the combiner **6735** such that it is incident onto the absorptive polarizer **6737**. In this embodiment, the image light **6750** is polarized light with the polarization state of the image light **6750** oriented relative to the transmission axis of the absorptive polarizer **6737** such that the incident image light **6750** is absorbed by the absorptive polarizer **6737**. In this way, faceglow produced by escaping image light **6750** is reduced. In embodiments, the absorptive polarizer **6737** includes an antireflection coating to reduce reflections from the surface of the absorptive polarizer **6737**.

(337) FIG. **67** further shows a camera **6739** for capturing images of the environment in the direction that the user is looking. The camera **6739** is positioned behind the absorptive polarizer **6737** and below the combiner **6735** so that a portion of light from the environment **6770** is reflected by the combiner **6735** toward the camera **6739**. Light from the environment **6770** can be unpolarized so that a portion of the light from the environment **6770** that is reflected by the combiner **6735** passes through the absorptive polarizer **6737** and it is this light that is captured by the camera **6739**. As a result, the light captured by the camera will have a polarization state that is opposite that of the image light **6750**. In addition, the camera **6739** is aligned relative to the combiner **6735** such that the field of view associated with the camera **6739** is coaxial to the display field of view provided by image light **6750**. At the same time, a portion of scene light **6760** from the environment is transmitted by the combiner **6735** to provide a see-through view of the environment to the user's eye **6730**. Where the display field of view associated with the image light **6750** is typically coincident to the see-through field of view associated with the scene light **6760** and thereby the see through field of view and the field of view of the camera **6739** are at least partially coaxial. By attaching the camera **6739** to the lower portion of the optical system **6715**, the field of view of the camera **6739** as shown by the light from the environment **6770** moves as the user moves their head so that images captured by the camera **6739** correspond to the area of the environment that the user is looking at. By coaxially aligning the camera field of view with the displayed image and the user's view of the scene, augmented reality images with improved alignment to objects in the scene can be provided. This is because the captured images from the camera **6739** provide an accurate representation of the user's perspective view of the scene. As an example, when the user sees an object in the scene as being located in the middle of the see-

through view of the HWC, the object will be located in the middle of the image captured by the camera and any augmented reality imagery that is to be associated with the object can be located in the middle of the displayed image. As the user moves their head, the relative position of the object as seen in the see-through view of the scene will change and the position of the augmented reality imagery can be changed within the displayed image in a corresponding manner. When a camera **6739** is provided for each of the user's eyes, an accurate representation of the 3D view of the scene can be provided as well. This is an important advantage provided by the invention because images captured by a camera located in the frame of the HWC (e.g. between the eyes or at the corners) capture images that are laterally offset from the user's perspective of the scene and as a result it is difficult to align augmented reality images with objects in the scene as seen from the user's perspective.

(338) In the optical system **6715** shown in FIG. **67**, the absorptive polarizer **6737** simultaneously functions as a light trap for escaping image light **6750**, a light blocker of the image light **6750** for the camera **6739** and a window for light from the environment **6770** to the camera **6739**. This is possible because the polarization state of the image light **6750** is perpendicular to the transmission axis of the absorptive polarizer **6737** while the light from the environment **6770** is unpolarized so that a portion of the light from the environment **6770** that is the opposite polarization state to the image light is transmitted by the absorptive polarizer **6737**. The combiner **6735** can be any partially reflective surface including a simple partial mirror, a notch mirror and a holographic mirror. The reflectivity of the combiner **6735** can be selected to be greater than 50% (e.g. 55% reflectivity and 45% transmission over the visible wavelength spectral band) whereby a majority of the image light **6750** will be reflected toward the user's eye **6730** and a majority of light from the environment **6770** will be reflected toward the camera **6739**, this system will provide a brighter displayed image, a brighter captured image with a dimmer see-through view of the environment. Alternatively, the reflectivity of the combiner **6735** can be selected to be less than 50% (e.g. 20% reflectivity and 80% transmission over the visible wavelength spectral band) whereby the majority of the image light **6750** will be transmitted by the combiner **6735** and a majority of light from the environment **6770** will be transmitted to the user's eye **6730**, this system will provide a brighter see-through view of the environment, while providing a dimmer displayed image and a dimmer captured image. As such, the system can be designed to favor the anticipated use by the user.

(339) In embodiments, the combiner **6735** is planar with an optical flatness that is sufficient to enable a sharp displayed image and a sharp captured image, such as a flatness of less than 20 waves of light within the visible wavelengths. However, in embodiments, the combiner **6735** may be curved in which case the displayed image and the captured image will both be distorted and this distortion will have to be digitally corrected by the associated image processing system. In the case of the displayed image, the image is digitally distorted by the image processing system in a direction that is opposite to the distortion that is caused by the curved combiner so the two distortions cancel one another and as a result the user sees an undistorted displayed image. In the case of the captured image, the captured image is digitally distorted after capture to cancel out the distortion caused by the curved combiner so that the image appears to be undistorted after image processing.

(340) In embodiments, the combiner **6735** is an adjustable partial mirror in which the reflectivity can be changed by the user or automatically to better function within different environmental conditions or different use cases. The adjustable partial mirror can be an electrically controllable mirror such as for example, the e-Transflector that can be obtained from Kent Optronics (<http://www.kentoptronics.com/mirror.html>) where the reflectivity can be adjusted based on an applied voltage. The adjustable partial mirror can also be a fast switchable mirror (e.g. a switching time of less than 0.03 seconds) wherein the perceived transparency is derived from the duty cycle of the mirror rapidly switching between a reflecting state and a transmitting state. In embodiments, the images captured by the camera **6739** can be synchronized to occur when the fast switchable

mirror is in the reflecting state to provide an increased amount of light to the camera **6739** during image capture. As such, an adjustable partial mirror allows for the transmissivity of the partial mirror to be changed corresponding to the environmental conditions, e.g. the transmissivity can be low when the environment is bright and the transmissivity can be high when the environment is dim.

(341) In a further embodiment, the combiner **6735** includes a hot mirror coating on the side facing the camera **6739** wherein visible wavelength light is substantially transmitted while a spectral wavelength band of infrared light is substantially reflected and the camera **6739** captures images that include at least a portion of the infrared wavelength light. In these embodiments, the image light **6750** includes visible wavelength light and a portion of the visible wavelength light is transmitted by the combiner **6735**, where it is then absorbed by the absorptive polarizer **6737**. A portion of the scene light **6760** is comprised of visible wavelength light and this is also transmitted by the combiner **6735**, to provide the user with a see-through view of the environment. The light from the environment **6770** is comprised of visible wavelength light and infrared wavelength light. A portion of the visible wavelength light along with substantially all of the infrared wavelength light within the spectral wavelength band associated with the hot mirror, is reflected by the combiner **6735** toward the camera **6739** thereby passing through the absorptive polarizer **6737**. In embodiments, the camera **6739** is selected to include an image sensor that is sensitive to infrared wavelengths of light and the absorptive polarizer **6737** is selected to substantially transmit infrared wavelengths of light of both polarization states (e.g. ITOS XP44 polarizer which transmits both polarization states of light with wavelengths above 750 nm: see <http://www.itos.de/english/polarisatoren/linear/linear.php>) so that an increased % of infrared light is captured by the camera **6739**. In these embodiments, the absorptive polarizer **6737** functions as a light trap for the escaping image light **6750** and thereby blocking the image light **6750** that is in the visible wavelengths from the camera **6739** while simultaneously acting as a window for infrared wavelength light from the environment **6770** for the camera **6739**.

(342) By coaxially aligning the camera field of view with the displayed image and the user's view of the scene, augmented reality images with improved alignment to objects in the scene can be provided. This is because the captured images from the camera provide an accurate representation of the user's perspective view of the scene. In embodiments, the camera that is coaxially aligned with the user's view captures an image of the scene, the processor then identifies an object in the captured image and identifies a field of view position for the object, which can be compared to the displayed field of view correlated position so digital content is then displayed relative to the position of the object.

(343) Another aspect of the present invention relates to an optical assembly that uses a reflective display where the reflective display is illuminated with a front light arranged to direct the illumination at angles around 90 degrees from the active reflective surface of the reflective display. In embodiments, the optical configuration is light weight, small and produces a high quality image in a head-worn see-through display.

(344) FIG. **68** provides a cross sectional illustration of the compact optical display assembly for a HWC **102** according to principles of the present invention along with illustrative light rays to show how the light passes through the assembly. The display assembly is comprised of upper optics and lower optics. The upper optics include a reflective image source **6810**, a quarter wave film **6815**, a field lens **6820**, a reflective polarizer **6830** and a polarized light source **6850**. The upper optics convert illumination light **6837** into image light **6835**. The lower optics comprise a beam splitter plate **6870** and a rotationally curved partial mirror **6860**. The lower optics deliver the image light to a user who is wearing the HWC **102**. The compact optical display assembly provides the user with image light **6835** that conveys a displayed image along with scene light **6865** that provides a see-through view of the environment so that user sees the displayed image overlaid onto the view of the environment.

(345) In the upper optics, linearly polarized light is provided by the polarized light source **6850**. Where the polarized light source **6850** can include one or more lights such as LEDs, QLEDs, laser diodes, fluorescent lights, etc. The polarized light source **6850** can also include a backlight assembly with light scattering surfaces or diffusers to spread the light uniformly across the output area of the polarized light source. Light control films or light control structures can be included as well to control the distribution of the light (also known as the cone angle) that is provided by the polarized light source **6850**. The light control films can include, for example, diffusers, elliptical diffusers, prism films and lenticular lens arrays. The light control structures can include prism arrays, lenticular lenses, cylindrical lenses, Fresnel lenses, refractive lenses, diffractive lenses or other structures that control the angular distribution of the illumination light **6837**. The output surface of the polarized light source **6850** is a polarizer film to ensure that the illumination light **6837** provided to the upper optics is linearly polarized.

(346) The illumination light **6837** provided by the polarized light source **6850** is reflected by a reflective polarizer **6830**. Where the polarizer on the output surface of the polarized light source **6850** and the reflective polarizer **6830** are oriented so that their respective transmission axes are perpendicular to one another. As a result, the majority of the illumination light **6837** provided by the polarized light source **6850** is reflected by the reflective polarizer **6830**. In addition, the reflective polarizer **6830** is angled so that the illumination light **6837** is reflected toward the reflective image source **6810** thereby illuminating the reflective image source **6810** as shown in FIG. **68**.

(347) The illumination light **6837** passes through a field lens **6820** and is then incident onto the reflective image source **6810**. The illumination light **6837** is then reflected by the reflective image source (otherwise referred to as a reflective display herein elsewhere) **6810**. Wherein the reflective image source **6810** can comprise a liquid crystal on silicon (LCOS) display, a ferroelectric liquid crystal on silicon (FLCSO) display, a reflective liquid crystal display, a cholesteric liquid crystal display, a bistable nematic liquid crystal display, or other such reflective display. The display can be a monochrome reflective display that is used with sequential red/green/blue illumination light **6837** or a full color display that is used with white illumination light **6837**. The reflective image source **6810** locally changes the polarization state of the illumination light **6837** in correspondence to the pixel by pixel image content that is displayed by the reflective image source **6810** thereby forming image light **6835**. Wherein if the reflective image source **6810** is a normally white display, the areas of the image light **6835** that correspond to bright areas of the image content end up with a polarization state that is opposite to the polarization state of the illumination light and dark areas of the image light **6835** end up with a polarization state that is the same as the illumination light **6837** (it should be noted that the invention can be used with normally black displays which provide an opposite effect on polarization in the image light). As such, the image light **6835** as initially reflected by the reflective image source **6810** has a mixed polarization state pixel by pixel. The image light **6835** then passes through the field lens **6820** which modifies the distribution of the image light **6835** while preserving the wavefront to match the requirements (such as for example, magnification and focus) of the lower optics. As the image light **6835** passes through the reflective polarizer **6830**, the bright areas of the image light **6835** that have a polarization state that is opposite to the illumination light **6837** are transmitted through the reflective polarizer **6830** and the dark areas of the image light **6835** that have the same polarization state as the illumination light **6837** are reflected back toward the polarized light source **6850**, as a result, the image light **6835** after passing through the reflective polarizer **6830** is linearly polarized with a single polarization state in all the pixels of the image but now with different intensities pixel by pixel. Thus the reflective polarizer **6830** acts first as a reflector for the illumination light **6837** and then second as an analyzer polarizer for the image light **6835**.

(348) As such, the optical axis of the illumination light **6837** is coincident with the optical axis of the image light **6835** between the reflective polarizer **6830** and the reflective image source **6810**.

The illumination light **6837** and the image light **6835** both pass through the field lens **6820**, but in opposite directions. Wherein the field lens acts to expand the illumination light **6837** so it illuminates the entire active area of the reflective image source **6810** and also to expand the image light **6835** so it fills the eyebox **6882** after passing through the rest of the compact optical display system. By overlapping the portion of the compact optical display assembly associated with the illumination light **6837** with the portion of the compact optical display assembly associated with the image light **6835**, the overall size of the compact optical display assembly is reduced. Given that the focal length associated with the field lens **6820** requires some space in the compact optical display assembly, the reflective polarizer **6830** and the polarized light source **6850** are located in space that would otherwise be unused so the overall size of the display assembly is more compact. (349) The reflective polarizer **6830** can be a relatively thin film (e.g. 80 microns) or thin plate (e.g. 0.2 mm) as shown in FIG. **68**. The reflective polarizer **6830** can be a wiregrid polarizer such as is available from Asahi Kasei under the name WGF, or a multilayer dielectric film polarizer such as is available from 3M under the name DBEF. As previously described, the reflective polarizer **6830** has two functions. First, the reflective polarizer **6830** reflects the illumination light **6837** provided by the polarized light source **6850** and redirects the illumination light **6837** toward the reflective image source **6810**. Second, the reflective polarizer **6830** acts as an analyzer polarizer to the image light **6835** thereby converting the mixed polarization state of the image light **6835** above the reflective polarizer **6830** to linearly polarized light with a single polarization state below the reflective polarizer **6830**. While the illumination light **6837** incident on the reflective polarizer **6830** is incident on a relatively small portion of the reflective polarizer **6830**, the image light **6835** is incident on the majority of the area of the reflective polarizer **6830**. Consequently, the reflective polarizer **6830** extends at least across the entire area of the field lens **6820** and may extend across the entire area between the field lens **6820** and the beam splitter **6870** as shown in FIG. **68**. In addition, the reflective polarizer **6830** is angled at least in the portion where the illumination light **6837** is incident to redirect the illumination light **6837** toward the reflective image source **6810**. However, since reflective polarizers (such as a wiregrid polarizer) can be relatively insensitive to the incident angle, in a preferred embodiment, the reflective polarizer **6830** is a flat surface angled to redirect the illumination light **6837** toward the reflective image source **6810** wherein the flat surface extends substantially across the entire area between the field lens **6820** and the beam splitter **6870** in one continuously flat surface to make manufacturing easier. The thin film or thin plate of the reflective polarizer **6870** can be retained at the edges to position it at the desired angle and to make the surface flat.

(350) The systems and methods described herein with respect to FIGS. **68** through **71** have a number of advantages. By avoiding grazing angles of the illumination light **6837** and the image light **6835** at all the surfaces in the compact optical display assembly, scattering of light in the assembly is reduced and as a result the contrast of the image presented to the user's eye **6880** is higher with blacker blacks. In addition, the reflective image source **6810** can include a compensating retarder film **6815** as is known to those skilled in the art, to enable the reflective image source **6810** to provide a higher contrast image with more uniform contrast over the area of the displayed image. Further, by providing an optical display assembly that is largely comprised of air, the weight of the compact optical display assembly is substantially reduced. By using coincident optical axes for the illumination light **6837** and the image light **6835** and overlapping the illumination light **6837** and image light **6835** for a substantial portion of the optical display assembly, the overall size of the compact optical display assembly is reduced. Where the coincident optical axes are provided by passing the illumination light **6837** and the image light **6835** in opposite directions through the field lens **6820**. To maintain a uniform polarization state for the illumination light **6837**, the field lens **6820** is made from a low birefringence material such as glass or a plastic such as OKP4 as available from Osaka Gas Chemicals. By positioning the polarized light source **6850** and the associated illumination light **6837** below the field lens **6820**, and by

folding the optical path of both the illumination light **6837** at the reflective polarizer **6830** and the image light **6835** at the beam splitter **6870**, the overall height of the compact optical display assembly is greatly reduced. For example the overall height of the compact optical display assembly can be less than 24 mm as measured from the reflective image source **6810** to the bottom edge of the rotationally curved partial mirror **6860** for a display that provides a 30 degree diagonal field of view with a 6×10 mm eyebox.

(351) In a preferred case, the light control structure in the polarized light source **6850** includes a positive lens, such as for example a positive Fresnel lens, a positive diffractive lens or a positive refractive lens. Wherein a positive Fresnel lens or a positive diffractive lens is preferred because they can be very thin. The illumination light **6837** is thereby focused to form a smaller area or pupil at the reflective polarizer **6830** that has a direct relationship to the area of an eyebox **6882** at the other end of the optics wherein image light **6835** is provided to the user's eye **6880** as shown in FIG. **68**. Where the positive lens concentrates the illumination light **6837** from the polarized light source **6850** both in terms of intensity and angular distribution to match the etendue of the optical system and thereby fills the eyebox with image light **6835**. By using the positive lens to converge the light from the polarized light source **6850** as provided to the reflective polarizer **6830** and then using the field lens **6820** to expand the illumination light **6837** to illuminate the active area of the reflective image source **6810**, efficiency is improved since illumination light **6837** is substantially delivered only where needed to form image light **6835**. Further, illumination light **6837** outside the pupil can be controlled by the positive lens and clipped by masked edges of the positive lens. By focusing the illumination light **6837** and clipping light outside the pupil, illumination light **6837** is prevented from impinging adjacent surfaces at grazing angles in the compact optical display assembly to reduce scattering of light and thereby increase contrast in the image provided to the user's eye **6880** by providing blacker blacks.

(352) It should be noted that while FIGS. **68**, **69** and **70** show optical layouts wherein the illumination light **6837** is provided from behind the rotationally curved partial mirror **6860**, other optical layouts are possible within the invention. The location of the polarized light source **6850** can be changed for example to be at the side of the rotationally curved partial mirror **6860** wherein the reflective polarizer **6830** is oriented to receive the illumination light **6837** from the side. And reflect it toward the reflective image source **6810** (not shown).

(353) In a further embodiment, the portion of the image light **6835** that is reflected back toward the polarized light source **6850** is recycled in the polarized light source **6850** to increase the efficiency of the polarized light source **6850**. In this case, a diffuser and a reflective surface is provided behind the polarized light source **6850** so the polarization of the light is scrambled and reflected back toward the reflective polarizer **6830**.

(354) In yet another embodiment, another reflective polarizer is provided in the polarized light source **6850** and behind the linear polarizer previously disclosed. Wherein the respective transmission axes of the reflective polarizer and the linear polarizer are parallel to one another. The other reflective polarizer then reflects the light back into the backlight that has the polarization state that would not be transmitted by the linear polarizer. The light that is reflected back into the backlight passes through diffusers associated with the polarized light source **6850** where the polarization state is scrambled and reemitted thereby recycling the light and increasing efficiency.

(355) In another embodiment, the system according to the principles of the present invention includes an eye imaging system. FIG. **69** is an illustration of a compact optical display assembly, which includes an eye imaging camera **6992** that captures an image of the user's eye **6880** that is coaxial with the displayed image provided to the user so that a full image of the user's iris can be reliably captured. The eye imaging camera **6992** is reflected into the lower optics by a reflective polarizer **6930** that includes a notch mirror coating, facing the eye imaging camera **6992**, that reflects the wavelengths of light that are captured by the eye imaging camera **6992** (e.g. near infrared wavelengths) while transmitting wavelengths associated with the image light **6835** (e.g.

visible wavelengths). Eye light rays **6995** shown in FIG. **69** illustrate how the field of view associated with the eye imaging camera **6992** is a relatively narrow field of view because it is multiply reflected through the lower optics to capture an image of the user's eye **6880**. However, to enable the eye imaging camera **6992** to focus onto the user's eye **6880**, the eye imaging camera **6992** needs to have a very near focus distance (e.g. 35 mm). In addition, the field of view and focus distance of the eye imaging camera must take into account the reducing effect of the optical power provided by the rotationally curved partial mirror **6860**. To increase the efficiency of capturing the light reflected from the user's eye **6880** and thereby enable a brighter image of the eye, the rotationally curved partial mirror **6860** can be coated with a partial mirror coating that acts as a full mirror in the wavelengths being captured by the eye imaging camera **6992**, for example the coating can reflect 50% of visible light associated with the image light and 90% of near infrared light associated with the eye light **6995**. Where the reflections and associated changes in polarization state are similar to those associated with the image light **6835** but in the opposite order since the eye light rays **6995** are coming from the user's eye **6880**. LEDs or other miniature lights are provided adjacent to the user's eye **6880** to illuminate the user's eye **6880** wherein the wavelengths associated with the LED's or other miniature lights are different than the wavelengths associated with the image light **6835** such as for example near infrared wavelengths (e.g. 850 nm, 940 nm or 1050 nm). Alternatively, the image light **6835** is used to illuminate the user's eye **6880** and a reflective polarizer **6930** with a low extinction ratio in reflection (e.g. reflective extinction ratio <15) is used so that some of the eye light rays are reflected toward the eye imaging camera **6992**.

(356) In an alternative embodiment, the reflective and partially reflective surfaces can extend laterally to the sides of the areas used for displaying an image to the user. In this case, the eye imaging camera can be located adjacent to the field lens and pointed in a direction to image the user's eye after reflecting from the beam splitter and the rotationally curved partial mirror as shown in FIG. **70**. Where FIG. **70** is an illustration that shows an eye imaging camera **7092** positioned to the side of the field lens **6820** and reflective polarizer **6830**. The eye imaging camera **7092** is pointed such that the field of view captured by the eye imaging camera **7092** includes the user's eye **6880** as illustrated by the eye light rays **7095**. The quarter wave film **6890** is also extended laterally to change the polarization state of the eye light **7095** in the same way that the polarization state of the image light is changed so that the eye light passes through the beam splitter **6870** and quarter wave **6890**, is partially reflected by the rotationally curved partial mirror **6860** and is then reflected by the beam splitter **6870** and is then captured by the eye imaging camera **7092**. By positioning the eye imaging camera **7092** to the side of the field lens **6820** and reflective polarizer **6830**, the complexity of the optics associated with displaying an image to the user is reduced. In addition, the space available for the eye imaging camera **7092** is increased since interferences with the display optics are reduced. By positioning the eye imaging camera **7092** adjacent to the display optics, the eye image is captured nearly coaxially with the displayed image.

(357) In a yet another embodiment, the systems according to the principles of the present invention include a field lens with an internal reflective polarizer and one or more surfaces with optical power. FIG. **71** is an illustration of the upper optics including a field lens **7121** comprised of upper prism **7122** and lower prism **7123**. The upper prism **7122** and the lower prism **7123** can be molded to shape or grind and polished. A reflective polarizer **7124** is interposed on the flat surface between the upper prism **7122** and the lower prism **7123**. The reflective polarizer **7124** can be a wiregrid polarizer film or a multilayer dielectric polarizer as previously mentioned. The reflective polarizer **7124** can be bonded into place with a transparent UV curable adhesive that has the same refractive index as the upper prism **7122** or the lower prism **7123**. Typically the upper prism **7122** and the lower prism **7123** would have the same refractive index. Wherein upper prism **7122** includes an angled surface for illumination light **6837** to be provided to illuminate the reflective image source **6810**. The illumination light is provided by a light source that includes lights such as LEDs, a backlight **7151**, a diffuser **7152** and a polarizer **7153** as has been previously described. The lower

prism **7123** includes a curved surface on the exit surface for controlling the wavefront of the image light **6835** as supplied to the lower optics. The upper prism may also include a curved surface on the upper surface next to the reflective image source **6810** as shown in FIG. **71** for manipulating the chief ray angles of the light at the surface of the reflective image source **6810**. Illumination light **6837** is polarized by the polarizer **7153** prior to entering the upper prism **7122**. The transmission axes of the polarizer **7153** and the reflective polarizer **7124** are perpendicular to one another so that the illumination light **6837** is reflected by the reflective polarizer **7124** so that the illumination light is redirected toward the reflective image source **6810**. The polarization state of the illumination light **6837** is then changed by the reflective image source **6810** in correspondence with the image content to be displayed as previously described and the resulting image light **6835** then passes through the reflective polarizer **7124** to form the bright and dark areas associated with the image that is displayed to the user's eye **6880**.

(358) In another embodiment, the field lens **7121** of FIG. **71** comprises a polarizing beam splitter cube including two prisms, upper prism **7122** and lower prism **7123**. In this case, the reflective polarizer **7124** is replaced by a coating that is polarization sensitive so that light of one polarization state (typically S polarized light for example) is reflected and light of the other polarization state is transmitted. The illumination light **6837** is then provided with the polarization state that is reflected by the coating and the image light is provided with the polarization state that is transmitted by the coating. As shown in FIG. **71**, the beam splitter cube includes one or more curved surfaces in the upper prism **7122** or the lower prism **7123**. The beam splitter cube can also include one or more angled surfaces where the illumination light is supplied. The angled surface can include light control structures such as a microlens array to improve the uniformity of the illumination light **6837**, or a lenticular array to collimate the illumination light **6837**.

(359) In yet another embodiment, the curved surface(s) or the angled surface(s) illustrated in FIG. **71** can be molded onto a rectangularly shaped beam splitter cube by casting a UV curable material (e.g. UV curable acrylic) onto a flat surface of a beam splitter cube, placing a transparent mold with a cavity that has the desired curve onto the flat surface to force the UV curable material into the desired curve and applying UV light to cure the UV curable material. The beam splitter cube can be made of a material that has the same or different refractive index than the UV curable material.

(360) In a further embodiment, polarization sensitive reflective coatings such as dielectric partial mirror coatings, can be used in place of reflective polarizers or beam splitters as shown in FIG. **68**. In this case, the reflective films and plates that comprise the reflective polarizers **6830** and beam splitters **6870** include polarization sensitive coatings that substantially reflect light with one polarization state (e.g. S polarization) while substantially transmitting light with the other polarization state (e.g. P polarization). Since the illumination light source includes a polarizer **7153**, the illumination light **6837** is one polarization state and it is not important that the reflective polarizer **7124** be sensitive to the polarization state in reflection, the polarization state just needs to be maintained and presented uniformly over the surface of the reflective image source **6810**. However, it is important that the reflective polarizer **7124** be highly sensitive to polarization state in transmission (e.g. extinction ratio >200) to be an effective polarizer analyzer and to provide a high contrast image (e.g. contrast ratio >200) to the user's eye **6880**.

(361) In a further embodiment, the field lens **7121** shown in FIG. **71** can comprise a reflective polarizer **7124** with a curved surface (not shown) instead of a flat surface and wherein the reflective polarizer **7124** is not a film and instead is a polarization sensitive coating, a printed wiregrid polarizer or a molded wiregrid pattern that is then metallized. In this case, the upper prism **7122** and the lower prism **7123** are made as a matched pair with mating curved surfaces that together form the surface of the reflective polarizer. Wherein the polarization sensitive coating, the printed wiregrid or the molded wiregrid pattern are applied to the mating curved surface associated either the upper prism **7122** or the lower prism **7123** and a transparent adhesive is applied to the other mating surface to bond the upper prism **7122** and lower prism **7123** together to form the field lens

7121 with an internal curved reflective polarizer **7121**.

(362) Another aspect of the present invention relates to manufacturing and providing an optical element for use in a see-through computer display system. In embodiments, a lightweight low-cost and high optical quality optical element.

(363) In a head mounted display, a beam splitter can be used to direct illuminating light from a light source toward a reflective image source such as an LCOS or a DLP. Where it is desirable to have a low weight beam splitter with a flat partially reflective surface to provide good image quality. The flat partially reflective surface is particularly important when an eye camera is provided for eye imaging that utilizes the flat partially reflective surface for directing the field of view of the eye camera toward the user's eye.

(364) Systems and methods provide for a lightweight beam splitter comprised of molded plastic elements and an internal plate element to provide a flat partially reflective surface. Together the pieces form a triplet beam splitter optic including two molded elements and a plate element. By providing the plate element internal to the beam splitter, the matching surfaces of the molded elements do not have to be optically flat, instead the plate element provides the flat surface and an index matching material is used to join the plate element to the molded elements. All three elements can be plastic elements to reduce the weight and cost of the lightweight beam splitter. To provide a more uniform refractive effect, the molded elements and the plate element are preferentially made from plastic materials with similar refractive index and have low birefringence.

(365) FIG. 72 shows an illustration of the two molded elements **7210** and **7220**. These molded elements are molded with a relatively uniform thickness to provide uniform flow of the plastic material during molding (either injection molding, compression molding or casting) and thereby enable a low birefringence in the elements as molded. To further reduce birefringence in the molded elements as molded, materials with low viscosity and low stress optic coefficients are preferred including: OKP4 from Osaka Gas Company; Zeonex F52R, K26R or 350R from Zeon Chemical; PanLite SP3810 from Teijin.

(366) The molded elements **7210** and **7220** can include flat surfaces and surfaces with optical power, where the surfaces with optical power can include spherical or aspheric curved surfaces, diffractive surfaces or Fresnel surfaces. Flat surfaces, diffractive surfaces or Fresnel surfaces are preferred on the surfaces associated with light that illuminates the image source and flat surfaces, spherical surfaces or aspheric surfaces are preferred on the surfaces associated with image light. Molded element **7210** is shown with a spherical or aspheric surface **7215** and molded element **7220** is shown with a flat surface **7225**, however, any of the surfaces shown can be molded as flat surfaces or surfaces with optical power.

(367) After molding the molded elements **7210** and **7220** are machined to provide matching angled surfaces. Molded element **7210** is shown in FIG. 73 where a milling cutter **7328** is shown machining angled surface **7329**. FIG. 74 shows an illustration of molded elements **7210** and **7220** after they have been machined to respectively provide beam splitter elements **7430** and **7440** that are prisms. The angled surface of beam splitter elements **7430** and **7440** are machined to have matching angles. Alternatively, beam splitter elements **7430** and **7440** can be machined from sheet material or molded pucks. In either case of using machined angled surfaces or molded angled surface in the beam splitter elements, the surfaces will not be optically flat.

(368) FIG. 75 shows an illustration of the assembled triplet beam splitter optic, wherein the beam splitter elements **7430** and **7440** have been assembled with a partially reflecting plate element **7560** to form a beam splitter cube. Wherein the beam splitter elements **7430** and **7440** are made from either the same material or different materials that have a very similar refractive index (e.g. within 0.05 of each other). An index matching material is used at the interfaces between the beam splitter elements and the plate element. The index matching material can be a fluid, a light curing adhesive, a moisture curing adhesive or a thermally curing adhesive. The index matching material should have a refractive index that is very similar to that of the beam splitter elements (e.g. within 0.1).

(369) The partially reflective plate element **7560** can be a transparent plate with a partially reflective layer that is either a partially reflective coating or a laminated partially reflective film. The transparent plate is preferably a cast sheet such as cell cast acrylic that has low birefringence, or a molded plaque of a low birefringence material such as OKP4, Zeonex F52R, Zeonex K26R, Zeonex 350R or PanLite SP3810. In addition, the transparent plate should be optically flat (e.g. within 20 microns over the surface and with a surface finish of less than 15 nanometers), however optically flat surfaces are easily obtained in sheet stock. By using an index matching material at the interfaces between the beam splitter elements **7430** and **7440** and the partially reflective plate element **7560**, the lack of optical flatness of the surface of the beam splitter elements **7430** and **7440** can be filled by the index matching material so that the flatness of the reflective surface is determined by the flatness of the more easily obtained partially reflective plate element **7560** thereby providing a manufacturing advantage. The partially reflective layer can be a partial mirror, a reflective polarizer or a wiregrid polarizer where the reflective polarizer can be a coating or a film and the wiregrid polarizer can be a film or a molded structure that is partially coated with a conductive layer. Where a suitable reflective polarizer film can be obtained from 3M available under the trade name of DBEFQ and a wiregrid polarizer film can be obtained from Asahi-Kasei available under the trade name of WGF. In a preferred embodiment, the transparent plate of the partially reflective plate element **7560** has a refractive index that is very similar (e.g. within 0.1) to the refractive indices of the beam splitter elements **7430** and **7440**.

(370) FIG. **76** shows an illustration of an optical system for a head mounted display system. The system includes a reflective display as an image source **7667**, a light source **7665** that can be a white light source or a sequential color light source as appropriate for the image source **7665**. Wherein the light source **7665** provides illumination light **7674** that can be polarized light provided that a quarter wave layer is associated with the image source **7667** or the partially reflecting plate element **7560** so that the polarization of the illumination light **7674** is changed before becoming image light **7672**. The illumination light **7674** is reflected by a surface of the partially reflecting plate element **7560**, and then reflected by the image source **7667**, whereupon it passes through the partially reflective plate element **7560** thereby becoming image light **7672**. The image light **7672** is then reflected by a partially reflective combiner **7682** so that the image light is directed toward the user's eye **7680** to display an image to the user while simultaneously providing a see-through view of the environment. In the optical system, an index matching material can be used at the interface between the image source **7665** and the beam splitter element **7440** so that the surface of the beam splitter element **7440** does not have to be flat. It is contemplated by the current inventions that the optical system may include additional lenses and other optical structures that are not shown to improve the image quality or change the form factor of the optical system.

(371) In another embodiment, beam splitter elements **7430** and **7440** are molded directly to shape using injection molding or casting. The molded beam splitter elements are then assembled as shown in FIG. **75** as described previously herein.

(372) In further embodiments, surfaces of the beam splitter elements are molded or machined to have additional structures to provide further features. FIG. **77** shows an illustration of lightweight beam splitter **7750** that includes an extended partially reflective plate element **7760** and an extended beam splitter element **7740**, wherein the partially reflective surface is extended to provide additional area for the illumination light **7674** to be reflected toward the image source **7665**. Where having an extended partially reflective surface is particularly useful when the image source **7665** is a DLP and the illumination light **7665** must be delivered at an oblique angle. FIG. **78** shows a lightweight beam splitter **7850** that includes an entrance surface **7840** for the illumination light **7674** that is angled so the illumination light **7674** passes substantially perpendicularly through the entrance surface **7840**.

(373) In yet further embodiments, beam splitter element elements **7430** and **7440** are machined from a single molded element. Where the single molded element is designed to provide the desired

optical surfaces. For example, the molded element **7210** as shown in FIG. **72** has surfaces that could be used for both surface **7215** and **7225**. A series of molded elements **7210** could then be molded and some would be used to make machined beam splitter elements **7430** and some for beam splitter elements **7440**. A partially reflective plate element **7560** would then be bonded with the beam splitter element **7430** and **7440** using index-matched adhesive as previously described herein. Alternatively, the single molded element **7210** could be designed with extra thickness across the dimension where the partially reflective plate element **7560** will be added, so that a single molded element **7210** could be sawn, machined or laser cut into beam splitter elements **7430** and **7440**.

(374) In another embodiment, a first molded optical element is molded in a geometry that enables improved optical characteristics including: low birefringence; more accurate replication of the optical surfaces of the mold (reduced power and irregularity deviation). The first molded optical element is then cut to a different shape wherein the cutting process leaves an optically rough surface finish. A second optical element with an optically smooth surface is then bonded to the optically rough surface of the first molded optical element using an index-matched adhesive to provide a combined optical element. The index-matched adhesive fills in the optically rough surface on the first molded optical element so that the optically rough surface is no longer visible and an optically smooth surface is provided in the combined optical element by the second optical element. The optical characteristics of the combined optical element are improved as compared to a directly molded optical element that has the geometry of the combined optical element. The cut surface can be flat or curved, as long as the cut surface of the first molded optical element is substantially similar to the bonding surface of the second optical element. In addition, both the first molded optical element and the second optical element can provide optical surface with independent optical features such as optical power, wedge, diffraction, grating, dispersion, filtering and reflection. For example, optically flat surfaces can be difficult to mold on plastic lenses. A lens can be molded to provide a spherically curved surface and another surface that is subsequently milled off to provide a flat surface with a rough surface finish. An optically flat sheet can then be bonded to the milled surface using an index matched adhesive to provide a combined optical element with an optically flat surface.

(375) In yet further embodiments, surfaces of the beam splitter elements include molded or machined structures to collimate, converge, diverge, diffuse, partially absorb, redirect or polarize the illumination light **7674** or the image light **7672**. In this way, the number of parts in the lightweight beam splitter is reduced and the cost and manufacturing complexity is reduced.

(376) The multi-piece lightweight solid optic has been described in connection with certain embodiments; it should be understood that the multi-piece lightweight solid optic may be used in connection with other optical arrangements (e.g. other see-through head-worn display optical configuration described herein elsewhere).

(377) In embodiments, the invention provides methods for aligning images, along with methods and apparatus for controlling light within the optics of the display assembly associated with a HMD to prevent scattering and also to trap excess light to thereby improve the image quality provided to the user.

(378) FIG. **79a** is a schematic illustration of a cross section of a display assembly for a HMD. Wherein, the display assembly includes upper optics **795** and lower optics **797** that operate together to display an image to a user while simultaneously providing a see-through view of the environment. Aspects of the upper optics **795** will be discussed in more detail herein. The lower optics **797** can comprise optical elements to manipulate image light **7940** from the upper optics **795** and thereby present the image light **7940** to the user's eye **799**. Lower optics **797** can comprise one or more lenses **7950** and a combiner **793**. The combiner **793** presents the image light **7940** to the user's eye **799** while simultaneously allowing light from the environment **791** to pass through to the user's eye **799** so that the user sees the displayed image overlaid onto a view of the environment.

(379) FIG. 79 is a schematic drawing of a cross section of the upper optics 795 for a HMD. Included are a light source 7910, a partially reflective layer 7930, a reflective image source 7935 and a lens 7950. The light source 7910 provides illumination light 7920 to the HMD. The illumination light 7920 is redirected by the partially reflective layer 7930 to illuminate the reflective image source 7935. The illumination light 7920 is then reflected by the reflective image source 7935 in correspondence with the image content in the displayed image so that it passes through the partially reflective layer 7930 and thereby forms image light 7940. The image light 7940 is optically manipulated by the lens 7950 and other optical elements (not shown) in the lower optics 797 so that a displayed image is provided to a user's eye 799. Together, the light source 7910, the partially reflective layer 7930 and the reflective image source 7935 form a frontlighted image source. Where, the reflective image source 7935 can comprise a LCOS, a FLCOS, DLP or other reflective display. FIGS. 79, 80, 82 and 83 are shown with the illumination light 7920 provided so that it is incident on the reflective image source 7935 at an oblique angle as is required for a DLP. FIGS. 84c, 84d, 85, 86, 87, 88 and 89 are shown with the illumination light 7920 provided perpendicular to the reflective image source 8535 as is required for an LCOS or FLCOS. The principles of the invention described herein apply to any type of reflective image source where stray reduction is needed. The light source 7910 can include light sources such as LEDs, laser diodes or other light sources (e.g. as described herein) along with various light control elements including: diffusers, prismatic films, lenticular films, Fresnel lenses, refractive lenses and polarizers. Polarizers included in the light source 7910 polarize the light exiting the light source 7910 so that the illumination light 7920 is polarized. The partially reflective layer 7930 can be a partial mirror coating on a substrate or it can be a reflective polarizer film such as a wire grid film supplied by Asahi-Kasei under the name WGF or a multilayer polarizer film supplied by 3M under the name DBEF. When the partially reflective layer 7930 is a reflective polarizer, the illumination light 7920 is supplied as polarized light wherein the polarization axis of the illumination light 7920 is oriented relative to the polarization axis of the reflective polarizer so that the illumination light 7920 is substantially reflected. The reflective image source 7935 then includes a quarter wave retarder (e.g. a quarter wave film) so that the polarization state of the illumination light 7920 is reversed in the process of being reflected by the reflective image source 79345. This enables the reflected illumination light 7920 to then be substantially transmitted by the reflective polarizer. After passing through the partially reflective layer 7930, the light becomes image light 7940. The image light 7940 then passes into a lens 7950 which is part of the lower optics 797 or display optics which manipulates the light to provide a displayed image to the user's eye. While the partially reflective layer 7930 is illustrated as a flat surface, the inventors have contemplated that the surface may be curved, shaped, have simple or complex angles, etc. and such surface shapes are encompassed by the principles of the present invention.

(380) In HMDs that provide images to both eyes of the user, it is desirable to provide the images so that they are aligned to one another. This is particularly important when the images are viewed as stereo images where the perceived alignment of the images seen with each eye is critical to achieving the perception of depth. To provide an accurate alignment of the images, an active alignment of the optics can be performed after the optics have been assembled into a rigid frame of the HMD. Where active alignment includes aligning the images for each eye to one another by moving portions of the display assembly and affixing the portions into position relative to one another. To this end, FIG. 79 shows space 7952 that extends around the reflective image source 7935 so that the reflective image source 7935 can be moved laterally and rotationally. The light source 7910 and partially reflective layer 7930 are arranged to illuminate the area that includes the reflective image source 7935 and a portion of the adjacent space 7952. As a result, the reflective image source 7935 can be moved within the space 7952 during the active alignment process without losing illumination or degrading the brightness of the displayed image. Where movements of the reflective image source 7935 during the active alignment can include movements that

correspond to horizontal, vertical and rotational movements of the image provided to one eye relative to the image provided to the other eye of the user. The movements can be 0.5 mm in size for example when the reflective image source **7935** is approximately 5×8.5 mm in size (this equates to approximately 10% of the reflective image source dimension) and as such the space **7952** can be 0.5 mm wide or wider.

(381) However, by including the space **7952**, in the illuminated area, visible artifacts can occur due to light scattering or reflecting from the edges of the reflective image source **7935** or from structures adjacent to the space **7952**. Consequently, a mask **8055** is provided that extends from the edge of the active area of the reflective image source **7935** across the space **7952** to cover the edges of the reflective image source **7935** and structures adjacent to the space **7952** as shown in FIG. **80**. The mask **8055** is black and non-reflecting so that incident illumination light **7920** is absorbed. In addition the mask **8055** is designed to not interfere with the movements of the reflective image source **7935** that occur during active alignment. To this end, the mask **8055** can be stiff (e.g. a black plastic or a black coated metal) and designed to slide under the adjacent structures such as the light source **7910**, the edge of the partially reflective layer **7930** and the sides of the housing that contain the frontlight. Alternatively, the mask **8055** can be flexible (e.g. a black plastic film or a black rubber film or tape) so that it deforms when it contacts the adjacent structures. FIG. **81a** shows an illustration of the reflective image source **7935**, the light source **7910** and the space **7952** as viewed from above. As is typically found with image source of all kinds, there is a mask **8168** applied to the surface of the image source that surrounds the active area **8165**, however this mask **8168** does not cover the space **7952**. FIG. **81b** shows a further illustration of the system shown in FIG. **81a** wherein the mask **8055** is applied to the reflective image source **7935** so that it attaches within the mask **8168** in a way that covers the space **7952** and does not block the active area **8165**.

(382) In another embodiment, the image produced by the image source does not use all of the active display area of the image source so there is room to shift the image in an x and/or y perspective within the active display area for alignment of the content. For example, if a misalignment is observed (as indicated above) rather than physically moving the image source, or in addition to moving the image source, the image is digitally shifted in the x and/or y direction to create better combined content alignment. The originally inactive display area around the content may be referred to as a content shift buffer zone.

(383) In a further embodiment for aligning images in a HMD with see-through, a first image containing features is provided to one eye of the user using a display assembly similar to that shown in FIG. **79a** or FIG. **85**. A second image containing features in the same locations is provided to the other eye of the user. The position of at least one of the image sources is then moved within the space provided for adjustment to align the first image to the second image as seen by the user's eyes. This image alignment can also be done using cameras in place of the user's eyes.

(384) In the case where the first and second images are smaller in size than the active area of the reflective image source, thereby leaving a digital space adjacent to the images that can be used for digital shifting of the images for further alignment adjustment. This adjustment can be used in combination with physical movements of the reflective image sources to align the first image to the second image.

(385) FIG. **82** is an illustration of upper optics **825** that includes the elements of upper optics **795** with the addition of a trim polarizer **8260**. Where the polarization axis of the trim polarizer **8260** is oriented so the image light **7940** is transmitted to the lower optics (not shown). Light that has the opposite polarization state compared to the image light **7940** is absorbed by the trim polarizer **8260**. As such, light that is scattered from surfaces such as the walls of the housing **8262** that typically has a mixed polarization state will be partially absorbed by the trim polarizer **8260**. The trim polarizer **8260** can also absorb a portion of colored light caused by birefringence in the lens **7950** provided the trim polarizer **8260** is located after the lens **7950**. In this case, the trim polarizer

8260 absorbs the light that has the opposite polarization state caused by the birefringence and transmits the light that has the polarization state of the image light **7940** prior to the lens **7950**. In some cases, it is advantageous to change the polarization state of the image light **7940** to improve the reflection of the image light **7940** from the combiner **793** so that a half wave retarder is needed in addition to the trim polarizer **8260**. For proper operation, the half wave retarder is positioned with its fast axis oriented at 45 degrees to the transmission axis of the trim polarizer **8260**. In this case, it is advantageous to position the half wave retarder (not shown) below the trim polarizer **8260** so that the trim polarizer can absorb any elliptical polarization that may be present due to birefringence in the lens **7950** before the image light is acted upon by the half wave retarder. In this way, any variation in retardation with wavelength that may be present in the half wave retarder will not act to increase the elliptical polarization or act to increase color artifacts in the image light **7940** caused by birefringence in the lens **7950**. In an example, the trim polarizer can be a polarizer film that is laminated to a half wave retarder film and antireflection coatings can be applied to the outer surfaces.

(386) In FIG. **83**, the partially reflective layer **8330** is a laminated multiple polarizer film comprised of a reflective polarizer film **8332** laminated to an absorptive polarizer film **8331**. Where, the reflective polarizer film **8332** is only big enough to reflect the illumination light **7920** that illuminates the active area **8165** of the reflective image source **7935**. The absorptive polarizer film **8331** is larger than the reflective polarizer film **8332** and extends across the entire aperture between the reflective image source **7935** and the lens **7950**, so that no edges of the absorptive polarizer film **8331** are visible and all the light reflected from the reflective image source **7935** passes through the absorptive polarizer **8331**. For the case when the reflective image source **7935** is an LCOS, the absorptive polarizer **8331** acts as an analyzer polarizer to only allow the polarization state of the image light to be transmitted. As such, the reflective polarizer film **8332** only covers a portion of the absorptive polarizer film **8331**. The polarization axes of the reflective polarizer film **8332** and the absorptive polarizer film **8331** are aligned so that polarized light that is transmitted by the reflective polarizer film **8332** is also transmitted by the absorptive polarizer film **8331**. In contrast, polarized light that is reflected by the reflective polarizer film **8332** is absorbed by the absorptive polarizer film **8331**. Thereby, illumination light **7920** that is incident onto the reflective polarizer film **8332** is reflected toward the reflective image source **7935** where the polarization state is reversed so that it is transmitted by the reflective polarizer film **8332** and the absorptive polarizer film **8331** as it becomes image light **7940**. At the same time, illumination light **7920** that is incident onto the absorptive polarizer film **8331** in the area surrounding the reflective polarizer film **8332** is absorbed by the absorptive polarizer film **8331**. By absorbing this excess illumination light **7920**, that would not illuminate the active area **8165** of the reflective image source **7935**, stray light is reduced within the display assembly and the contrast in the image presented to the user's eye is increased as a result. By aligning the polarization axes of the reflective polarizer film **8332** and the absorptive polarizer film **8331**, the transmission is only reduced by approximately 12%, in the regions that include both reflective polarizer film **8332** and absorptive polarizer film **8331** compared to the regions that include just absorptive polarizer film **8331**. Given the location of the partially reflective layer **8330** in the optical system and the fact that it is remote from the reflective image source **7935**, having local differences in transmission on the partially reflective layer **8330** comprised of a laminated multiple polarizer will have a very small effect on the brightness uniformity in the image provided to the user's eye. In addition, the fact that the partially reflective layer **8330** is remote from the reflective image source **8330** makes the edges of the reflective polarizer film **8332** indistinct as seen by the user.

(387) FIGS. **84a** and **84b** show illustrations of examples of partially reflective layers **8330**, comprised of a reflective polarizer film **8430** and **8431** laminated to an absorptive polarizer film **8432**. The reflective polarizer films **8430** and **8431** are cut to a shape that covers only the area where illumination light **7920** will be reflected to illuminate the active area **8165** of the reflective

image source **7935**. The shape required for the reflective polarizer film will vary depending on the type of frontlight. For the frontlight shown in FIG. **83** where the partially reflective layer **8330** is located adjacent to the reflective image source **7935**, the shape of the reflective polarizer film **8431** will be rectangular or oval such as shown in FIG. **84b**. For the frontlight included in the display assembly shown in FIG. **85** where the lens **8550** is located between the partially reflective layer **8530** and the reflective image source **8535**, the influence of the illumination light **8520** passing through the lens **8550** changes the distribution of illumination light **8520** needed from the light source **8510**. As a result, the illumination light **8520** can cover only a portion of the partially reflective layer **8530** and the use of a laminated multiple polarizer is advantageous. In embodiments, the reflective polarizer film can cover less than 80% of the area of the absorptive polarizer film in the laminated partially reflective layer. In further embodiments, the reflective polarizer film can cover less than 50% of the area of the absorptive polarizer film in the laminated partially reflective layer. In this case, the partially reflective layer **8530** can include a reflective polarizer film **8430** with a shape similar to that shown in FIG. **84a**. In any case, the shape of the reflective polarizer film is selected in concert with the optical elements in the frontlight and display optics associated with the display assembly of the HMD.

(388) FIG. **84c** shows an example illustration of a frontlight for a display assembly similar to that shown in FIG. **85** wherein a laminated multiple polarizer film **8436** is shown with a complex curved shape that resembles an S with a central flat portion and curved ends. The laminated multiple polarizer **8436** includes a reflective polarizer film **8438** and an absorptive polarizer film **8437**. Illumination light **8520** includes rays **8522** that are incident on the reflective polarizer film **8438** and rays **8521** that are incident on the absorptive polarizer film **8437**. Due to the alignment of the polarization of the illumination light **8520** to the polarization axes of the reflective polarizer film **8438** and the absorptive polarizer film **8437** as previously described herein, rays **8522** are reflected by the reflective polarizer film **8438** and rays **8521** are absorbed by the absorptive polarizer film **8437**. In this way, rays **8521** are prevented from contributing to stray light. It is beneficial to absorb rays **8521** since they cannot contribute to image light **8540** because if they were reflected by the laminated multiple polarizer **8436** they would be incident on the reflective image source **8535** outside of the active area **8165**, and if they were transmitted by the laminated multiple polarizer **8436**, they would be incident on the housing sidewalls **8262**. Consequently, by absorbing rays **8521**, the laminated multiple polarizer **8436** reduces stray light and thereby increases the contrast in the image displayed to the user.

(389) FIG. **84d** shows a further example illustration of a frontlight for a display assembly similar to that shown in FIG. **79** wherein the partially reflective layer **7930** comprises a laminated multiple polarizer film with a curved surface. The laminated polarizer includes an absorptive polarizer film **8442** with a laminated reflective polarizer film **8441**. The reflective polarizer film **8441** is positioned in the central portion of the absorptive polarizer film **8442** where the illumination light **7920** is reflected toward the reflective image source **7935**. The polarization axes of the reflective polarizer film **8441** and the absorptive polarizer film **8442** are aligned in parallel to each other and perpendicular to the polarization axis of the illumination light **7920** as provided by the polarized light source **7910**. The rays **8421** of the illumination light **7920** that are incident on the partially reflective layer **7930** outside of the reflective polarizer film **8441** are absorbed by the absorptive polarizer film **8442**. The reflective light source **8535** includes a quarter wave layer **8443** so that the polarization axis of the illuminating light **7920** is changed during the process of being reflected from the reflective image source **8535**. As a result, the reflected illumination light **7920** is transmitted by the reflective polarizer film **8441** and the absorptive polarizer film **8442**, thereby becoming image light **7940**. By absorbing the rays **8421**, before they are incident on external surfaces such housing walls or other optical surfaces, stray light is reduced and as a result the contrast in the image provided to the user's eye is increased. It should be noted that while FIGS. **84c** and **84d** show the reflective polarizer film being positioned to reduce stray light from the left

and right sides as shown in the figure, the reflective polarizer can similarly be positioned to reduce stray light in the direction in and out of the paper as shown in the figure. FIGS. **84a** and **84b** show reflective polarizer films **8430** and **8431** positioned in a center portion of the absorptive polarizer **8432** so that stray light can be reduced in all directions. An important aspect of the invention is that this stray light reduction is obtained without a reduction in the brightness of the image provided to the user's eye since the reflective polarizer films **8430** and **8431** reflect illumination light over the entire area that is needed to fully illuminate the reflective image source.

(390) FIG. **85** shows a schematic illustration of a display assembly for a HMD wherein the optical elements of the frontlight are overlapped with the display optics, as the lens **8550** is located between the partially reflective layer **8530** and the reflective image source **8535**. The display assembly is then comprised of upper optics and lower optics. The upper optics include a reflective image source **8535**, a lens **8550**, a partially reflective layer **8530** and a light source **8510**. The upper optics convert illumination light **8520** into image light **8540**. As shown, the lower optics comprise a beam splitter plate **8580**, a quarter wave film **8575** and a rotationally curved partial mirror **8570** (lower optics similar to those shown in FIG. **79a** are also possible). The lower optics deliver the image light **8540** to a user's eye **8582**. As previously stated herein, the display assembly provides the user with image light **8540** that conveys a displayed image along with scene light **8583** that provides a see-through view of the environment so that the user sees the displayed image overlaid onto a view of the environment.

(391) FIG. **85** shows a display assembly wherein the partially reflective layer **8530** is a single flat film. However, it can be advantageous to use a segmented partially reflective layer **8630** such as is shown in FIG. **86**. In this way, the angle of the central portion **8631** of the partially reflective layer **8630** can be selected to position the light source **8610** differently to reduce the clipping of illumination light **8620** by the lens **8550** or other portions of the supporting structure associated with the display assembly and thereby improve brightness uniformity in the displayed image seen by the user's eye **8582**. To this end, a comparison of FIG. **85** to FIG. **86** shows that by changing the angle of the central portion of the partially reflective film, the position of the light source **8610** is moved downward and the clearance of the illumination light **8620** is increased relative to the lens **8550**.

(392) Segmented partially reflective layers can be used which a variety of geometries and makeups. FIG. **86** shows a segmented partially reflective layer **8630** that includes a folded Z shape with three flat sections. FIG. **87** shows a segmented partially reflective layer that includes an S shape with a central flat section **8731** and ends that are curved similar to that shown in FIG. **84c**. The segmented partially reflective layer can comprise a single partially reflective layer such as a reflective polarizer film or a partial mirror film. In addition, illumination light **8620** can be reflected from just the central flat section or it can be reflected from the central flat section plus one or more of the other segments of the segmented partially reflective layer. Alternatively, the partially reflective layer **8630** can comprise a multiple polarizer film to selectively provide a partially reflective layer over just the portions of the partially reflective layer that are actually needed to reflect illumination light to uniformly illuminate the reflective image source **7935** as previously described herein. FIG. **88** shows a display assembly wherein the partially reflective layer **8830** is comprised of a laminated multiple polarizer film with a central portion **8831** that includes a reflective polarizer film and the remainder of which is an absorptive polarizer as previously described herein. Where the segmented shape of the partially reflective layer **8830** is similar to that shown in FIG. **86**. FIG. **89** shows a display assembly wherein the partially reflective layer **8930** is comprised of a laminated multiple polarizer film with a central portion **8931** that includes a reflective polarizer film and the remainder of which is an absorptive polarizer as previously described herein. Where the segmented shape of the partially reflective layer **8930** is similar to that shown in FIG. **87**. While FIGS. **88** and **89** show the reflective polarizer film as just occupying the flat central segment of the segmented partially reflective layers **8830** and **8930** respectively, the reflective polarizer can extend into the adjacent

segments as needed to reflect the illumination light **8620** in the pattern needed to uniformly illuminate the reflective image source **8535**. Alternatively the segments associated with the segmented partially reflective layers **8830** and **8930** can have three dimensional shapes when the reflective polarizer portion is shaped like that shown in FIG. **84a** to keep the reflective polarizer **8430** portion flat.

(393) In a further embodiment, the reflective polarizer film is laminated to a flexible transparent carrier film to increase the flexibility and the absorptive polarizer film is a separate layer. FIG. **90** shows a partially reflective layer **9030** comprised of a reflective polarizer film **8441** laminated to a flexible transparent carrier film **9043**. Where the flexible transparent carrier film **9043** does not reflect the illumination light **7920** or change polarization state of the illumination light **7920** and as a result rays **8421** pass through the flexible transparent carrier film **9043**. The purpose of the flexible transparent carrier film is to support the reflective polarizer film **8441** while allowing the partially reflective layer **9030** to be substantially as flexible as the reflective polarizer film **8441** alone. Absorptive polarizer film **9042** is then provided as a separate layer positioned adjacent to the partially reflective layer **9030**. While the absorptive polarizer film **9042** can be flat or curved as needed to fit within the available space, in a preferred embodiment, the absorptive polarizer film **9042** is curved to be better positioned to absorb rays **8421** that are incident on the partially reflective layer **9030** outside of the reflective polarizer film **8441** as shown in FIG. **90**.

(394) In yet another embodiment, the reflective polarizer film is modified to make the portions transparent and non-reflective where illumination light is incident that is not needed to illuminate the active area of the reflective image source and a separate absorptive polarizer is provided to absorb light that is transmitted through the non-reflective portions. FIG. **91** is an illustration of a partially reflective layer **9130** comprised of a reflective polarizer film wherein portions **9143** are modified to be transparent and non-reflective while the portion **9141** is a reflective polarizer. As such, polarized illumination light **7920** is reflected by the reflective polarizer portion **9141** and is transmitted by the modified portions **9143**. An absorptive polarizer **9042** is provided as a separate layer adjacent to the partially reflective layer **9130** so that rays **8421** of the illumination light **7920** are transmitted by the modified portions **9143** and absorbed by absorptive polarizer. Wherein the transmission axis of the reflective polarizer portion **9141** is parallel aligned to the transmission axis of the absorptive polarizer **9042**. The modification of the reflective polarizer film can be accomplished by etching the reflective polarizer film, when the reflective polarizer film is a wiregrid polarizer, and thereby removing the metal wires of the wiregrid in the modified portions. Alternatively the wiregrid polarizer can be masked during the metal deposition step to provide shaped portions of wiregrid polarizer during manufacturing. An advantage provided by modifying the reflective polarizer film is that the flexibility of the partially reflective layer **9130** is substantially unchanged by the modification and as a result the partially reflective layer **9130** remains uniformly flexible in both the modified portions **9143** and the reflective polarizer portion **9141**. Another advantage provided by using a modified reflective polarizer film is that the transition from the modified portion **9143** to the reflective polarizer portion **9141** does not include a sharp edge that can cause visible artifacts in the image provided to the user's eye due to scattering by the edge or a change in optical density from a thickness change. This embodiment can also be applied to other types of display assemblies such as for example that shown in FIG. **85**.

(395) In a yet further embodiment, the partially reflective layer comprises a reflective polarizer film laminated to an absorptive polarizer and the partially reflective layer includes a flat portion and a curved portion. FIG. **92** is an illustration of a frontlight for a display assembly similar to that shown in FIG. **79a** with the addition of a laminated partially reflective layer **9230** that has a portion that is a reflective polarizer laminated to an absorptive polarizer **9230**. Where the partially reflective layer **9230** is segmented with a flat segment and a curved segment. By including a flat segment in the portion of the partially reflective layer **9230** that is a reflective polarizer **9241**, the uniformity of illumination light **7920** that is reflected onto the reflective image source **7935** is

improved because a larger portion of the light source **7910** is mapped to the image as can be seen in FIG. **92**. Wherein when using a small scale light source and associated light control films such as diffusers, it is important to map a large portion of the light source area to avoid darker or brighter lines across the image produced by a dark or bright spot on the light source. Including a flat segment in the partially reflective layer **9230** also reduces local distortions in the image provided to the user's eye that are caused by local changes in optical path length or localized refraction due to changes in the surface angles that the light is exposed to. This embodiment can also be applied to other types of display assemblies such as for example that shown in FIG. **85**.

(396) In head mounted displays that provide a displayed image overlaid onto a see-through view of the environment, it is advantageous to have high see-through transmission both so the user can better interact with the environment and so that people in the environment can see the user's eyes so they feel more engaged with the user. It is also advantageous to have a thin optics module with low height to make the head mounted display more compact and thereby more attractive.

(397) FIG. **93** shows an illustration of an optics module that provides the user with a displayed image while simultaneously providing high see-thru transmission. In this way, the user is provided with a displayed image overlaid onto a clear view of the environment. The optics module includes a combiner **9320** that can have a partial mirror coating that transmits a majority (greater than 50% transmission of visible light) of the available light from the environment, with transmission higher than 70% preferred. For example, the combiner **9320** can have a broadband partial mirror that reflects less than 30% and transmits over 70% of the entire visible wavelength band. Alternatively, the combiner **9320** can have a notch mirror coating where the reflectivity band of the notch mirror coating is matched to the wavelength bands provided by the light source **9340**, where the light source **9340** can include one or more LEDs, QLEDs, diode lasers or other light source, each with narrow wavelength bands (e.g. 50 nm wide bands or less, full width half max). The notch mirror coating can provide for example, greater than 20% reflectivity (e.g. 50% reflectivity) in the wavelengths bands provided by the light source **9340** while providing greater than 80% transmission in the remaining wavelength bands in the visible. For full color images to be provided by the optics module, at least three LEDs with complimentary colors are required such as red, green and blue light or, cyan, magenta and yellow light. In a preferred embodiment, the combiner **9320** has a tristimulus notch mirror that reflects over 50% of the light within the wavelength bands provided by the light source **9340** and transmits an average of over 80% across the entire visible wavelength band. In this way, the tristimulus notch mirror coating provides improved efficiency compared to the partial mirror coating previously described. In an example, if the combiner is to provide 75% transmission of visible light from the environment **9362**, the partial mirror coating will reflect only 25% of image light **9360** so that 75% of the image light will be transmitted through the combiner and will not contribute to the brightness of the image provided to the user's eye **9310**. In contrast, a tristimulus notch mirror coating can be used to reflect over 50% of the image light **9360** over the wavelengths of light provided by the LEDs in the light source **9340** while transmitting over 90% of the remaining wavelengths of visible light that are not provided by the LEDs so that the average transmission over the entire range of visible light is over 75%. Consequently, the tristimulus notch mirror is twice as efficient as the partial mirror in terms of the ability to reflect image light **9360** toward the user's eye **9310**.

(398) To enable the optics module to operate with a combiner **9320** as shown in FIG. **93**, image light **9360** is provided to a lens **9330** which focuses the image light **9360** at the user's eye **9310**. Wherein lens **9330** is shown as a single lens element for simplicity, but multiple lens elements are also possible. The image light **9360** is provided from illumination light **9364** that comes from the light source **9340**. Where, the illumination light **9364** is reflected by a beam splitter **9352** toward a reflective image source **9350**. The image source **9350** can be a liquid crystal on silicon display (LCOS), a ferroelectric liquid crystal display (FLCOS) or other such reflective display. A polarizer **9342** can be associated with the light source **9340** to provide polarized illumination light **9364**. The

beam splitter **9352** can then be a reflective polarizer that is oriented to substantially reflect the polarized illumination light **9364**. The image source **9350** changes the polarization state of the illumination light **9364** when the light is reflected by the image source **9350** to form image light **9360** that has a polarization state that is opposite to that of the illumination light **9364**. By changing the polarization state of the illumination light **9364** to the polarization state of the image light **9360**, the image light **9360** can then be transmitted by the reflective polarizer of the beam splitter **9352**. It is important to note that the image light **9360** is polarized to enable a folded illumination system and not because polarized light is required by the combiner **9320**. In fact, to provide a transmission of light from the environment **9362** that is greater than 50%, the combiner **9320** cannot include a polarizer.

(399) FIG. **94** is an illustration of an optics module that includes multiply folded optics to reduce the overall height of the optics module. In this case, illumination light **9464** is transmitted by the beam splitter **9452** so that it passes directly toward the image source **9450** wherein the beam splitter **9452** is a reflective polarizer and the light source **9340** includes a polarizer **9342** that is oriented so the transmission axis of the polarizer **9342** is parallel to the transmission axis of the beam splitter **9452**. The illumination light **9464** is then reflected and changed in polarization state by the image source **9450** so that the image light **9360** with its changed polarization state is reflected by beam splitter **9452** toward the lens **9330**. As can be seen by comparing FIG. **93** to FIG. **94**, the overall height of the optics module shown in FIG. **94** is substantially reduced.

(400) However, the orientation of the additional fold in the optical path of the image light **9360** in the optics module of FIG. **94** increases the thickness of the optics module, where thickness is defined as the distance from the closest back surface of the optics module that is nearest to the user's eye to the farthest front surface of the optics module that is farthest from the user's eye. FIGS. **95** and **96** show illustrations of an optical module where the added fold in the optical path of the image light **9360** is oriented perpendicular to the fold shown in FIG. **94**. In this case, the optics module in FIGS. **95** and **96** is wider but thinner than that shown in FIG. **94**. FIG. **95** shows the optics module from the side and FIG. **96** shows the optics module from the position of the user's eye **9310**. As such, in the multiply folded optics shown in FIGS. **95** and **96**, optical axis **935** associated with the illumination light **9464** is perpendicular to both the optical axis **934** associated with the image light **9360** as it passes through the lens **9330** and the optical axis **933** associated with the image light **9360** as it proceeds toward the user's eye **9310** in the eyepiece. In the case of a head mounted display, it can be very important to have a thin optics module because a thick optics module can cause the head mounted display to stick outward from the user's forehead, which can be uncomfortable and unattractive. Thus, the multiply folded optics module shown in FIGS. **95** and **96** are shorter and thinner than the optics module shown in FIG. **93**. The optics module shown in FIGS. **95** and **96** is wider than the optics module shown in FIG. **93**, but in a glasses configuration of the head mounted display, wider optics modules can be better fit into the glasses frames than taller or thicker optics modules.

(401) A further advantage that is provided by an optics module that includes multiply folded optics is that twists can be introduced at the fold surfaces to modify the orientation of different portions of the optics module relative to each other. This can be important when the optics module needs to fit into a thin curved glasses frame, a visor or a helmet where the increased width associated with the upper portion of the multiply folded optics module can make it more difficult to fit into structures that are not parallel to the combiner. In this case, the upper portion including for example (based on FIG. **96**), the light source **9340**, the polarizer **9342**, the beam splitter **9452** and the image source **9450**, can be twisted relative to the lower portion including the lens **9330** and the combiner **9320**. Where to avoid distortion of the image due to the compound angles between the fold surfaces, a twist of the upper portion about the axis **934** must be combined with a corresponding twist of the lower portion about the axis **933**. In this way, the effects of the increased width of the upper portion of the multiply folded optics can be reduced when fitting the optics module into a curved structure

such as glasses frames, a visor frame or a helmet structure.

(402) FIG. **99** shows a further embodiment wherein the lens **9930** includes a diffractive surface **9931** to enable a more compact and shorter optical design with reduced chromatic aberration. Where the diffractive surface **9931** can be comprised of a series of small annular sections of a refractive lens curve such as for example in a Fresnel lens. The diffractive surface **9931** can be flat as shown in FIG. **99** or it can have a base curve to provide additional optical power. The diffractive surface **9931** can be a single order diffractive or a multiple order diffractive. To reduce scattering of wide angle illumination light **9964** that could be incident on the diffractive surface **9931**, an absorptive polarizer **9932** is provided and is oriented with its transmission axis perpendicular to the transmission axis of the reflective polarizer of the beam splitter **9452**. In this way, illumination light **9964** that is transmitted by the beam splitter **9452** in the direction that would cause it to be incident on the diffractive surface **9931** is absorbed by the absorptive polarizer **9932** before it can be scattered by the diffractive surface **9931**. At the same time, image light **9360** has a polarization state that is opposite to that of the illumination light **9964** so that it is reflected by the beam splitter **9452** and transmitted by the absorptive polarizer **9932** as it passes into the lens **9930**.

(403) FIG. **100** shows an illustration of an optics module that includes a reduced angle between the beam splitter **9452** and the lens **9930** to reduce the overall height of the optics module. The fold angle of the image light **9360** (the deflection angle between 934 and 1005) is then more than 90 degrees and as a result, the upper edge of the beam splitter is closer to the lens **9330** thereby providing a reduced overall height of the optics module.

(404) FIG. **100** also shows a compact planar light source **10040** comprised of a thin edge-lit backlight similar to what is provided in displays used in displays for mobile devices like cellphones. The compact planar light source **10040** is positioned directly behind the beam splitter **9452** to reduce the overall size of the optics module. The compact planar light source can include a light guide film or light guide plate with an edge lit light such as one or more LEDs and a reflector on the side opposite the beam splitter **9452**. The compact planar light source can include a polarizer so the illumination light **10064** is polarized as previously described herein. To direct the illumination light **10064** toward the image source **9450** for improved efficiency, a turning film **10043** is positioned between the compact planar light source **10040** and the beam splitter **9452**. A 20 degree prismatic turning film can be obtained for example, from Luminit **103C** (Torrance, CA) under the name DTF. To obtain greater degrees of turning, such as 40 degrees, multiple layers of turning film **10043** can be stacked together provided they are oriented such that the turning effect is additive. A diffuser layer (not shown) can be used in addition to the turning film **10043** to reduce artifacts such as linear shadows that can be associated with prismatic structures that are typically associated with turning films **10043**. FIG. **101** shows an illustration of an optics module as seen from the position of the user's eye, which is similar to that shown in FIG. **100** but with a perpendicular orientation of the added fold in the image light **10164** to reduce the thickness of the optics module as previously described herein. As in the optics module shown in FIGS. **95** and **96**, the multiply folded optics shown in FIG. **101** have an optical axis **1005** associated with the illumination light **10164** that is perpendicular to both the optical axis **934** associated with the image light **9360** as it passes through the lens **9330** and the optical axis **933** associated with the image light **9360** as it proceeds toward the user's eye **9310** in the eyebox. As a result, the optics module of FIG. **101** is thinner and shorter than the optics module of FIG. **93**. FIG. **101** also includes a field lens **10130** to improve the optical performance of the optics module. The addition of this second lens element is possible because of the change in fold orientation so that the field lens **10130** does not increase the thickness of the optics module, instead the added length of the optical path from the field lens **10130** occurs in the width of the optics module where space is more readily available in the head mounted display.

(405) FIG. **102** shows an illustration of an optics module similar to that shown in FIG. **99** but with a different orientation of the upper portion of the optics module relative to the combiner so that the

combiner **10220** can be more vertical. This rearrangement of the elements within the optics module can be important to achieve a good fit of the head mounted display onto the user's face. By making the combiner **10220** more vertical, the optics module can be made to have less interference with the user's cheekbones.

(406) FIGS. **103**, **103a** and **103b** show illustrations of optics modules as seen from the position of the user's eye, that include multiply folded optics and digital light projector (DLP) image sources **10350**. In this case, the illumination light **10364** is provided at an oblique angle to the image source **10350** as required by the micromirrors in the DLP, to reflect image light **9360** along the optical axis **934** of the lens **9930**. Where, in the case of a DLP image source **10350**, image light **9360** is comprised of on-state light reflected by on-state micromirrors in the DLP image source **10350** along optical axis **934**, in correspondence to the brightness of pixels in the image to be displayed to the user's eye **9310** in the eyebox. The micromirrors in the DLP image source **10350** also reflect off-state light **10371** to the side of the optics module in correspondence to the dark image content and as a result, a light trap **10372** is provided in the optics module to absorb light **10371**. The light trap **10372** can be a black absorptive surface or a textured black surface. The purpose of the light trap **10372** is to absorb incident light **10371** and thereby reduce stray light and subsequently improve the contrast of the image displayed to the user's eye **9310**. As previously described in other embodiments herein, the light source **10340** is provided to the side of the optics module with a multiply folded optical path to reduce the overall thickness and height of the optics module. FIG. **103** provides the DLP image source **10350** at the top of the optics module so that the image light **9360** proceeds straight along the optical axis **934**, through the lens **9930** and down to the combiner **9320** where the image light is reflected toward the user's eye **9310** located in the eyebox. A polarizer **10341** is provided with the light source **10340** so that polarized illumination light **10364** is reflected by the beam splitter **9452** to illuminate the DLP image source **10350**. Where, the beam splitter **9452** in this case, is a reflective polarizer that is aligned with the polarizer **10341** so that the polarized illumination light **10364** is reflected by the beam splitter **9452** and image light **9360** is transmitted by the beam splitter **9452**. A quarter wave film **10351** is located adjacent to the surface of the DLP image source **10350** so that the polarization state of the image light **9360** is opposite to that of the illumination light **10364** after being reflected by the DLP image source **10350**. The light source **10340** and the reflective polarizer **9452** are angularly arranged so that the illumination light **10364** is incident onto the DLP image source **10350** at the oblique angle required so that the image light **9360** when reflected by the on-state pixels in the DLP image source **10350** proceeds along the optical axis **934** of the lens **9930**. A field lens (similar to **10130** as shown in FIG. **101**) or other lens elements may be included in the optics of FIG. **103** but is not shown, in which case, the illumination light **10364** and the image light **9360** may pass thru the field lens or other lens elements in opposite directions.

(407) FIG. **103a** is an illustration of another optics module with a multiply folded optical path that includes a DLP image source **10350** and is shown from the position of the user's eye. The light source **10340** is again provided to the side of the optics module to reduce the thickness of the optics module. In this case, the light source **10340** is provided on the same side of the lens **9930** and combiner **9320**, as the DLP image source **10350**. Lens **9930** can optionally include one or more diffractive surfaces **9931**. The light source **10340** directly illuminates the DLP image source **10350** where the illumination light **10364** is incident on the DLP image source **10350** at an oblique angle so that the image light **9360**, after being reflected by the on-state micromirrors in the DLP image source **10350**, proceeds along the folded optical axis **934**. At least one light trap **10372** is also provided to absorb light **10371** that is reflected from off-state micromirrors in the DLP and thereby improve the contrast of the displayed image as seen by the user. A field lens **10332** is provided between the DLP image source **10350** and the fold mirror **10352**. The illumination light **10364** in this case can be unpolarized light whereupon the fold mirror **10352** can be comprised of a full mirror coating (e.g. a coating that reflects the entire visible light spectrum) on a substrate. The field lens

10332 can be a single lens element as shown in FIG. **103a** or it can include multiple lens elements as needed. The field lens **10332** is designed to provide a large air gap between the field lens **10352** and the DLP image source **10350**, so that the illumination light **10364** can be introduced to the optics module to directly illuminate the active area associated with the DLP image source **10350**. By using unpolarized illumination light **10364**, the optics module shown in FIG. **103a** has improved efficiency over the optics module with DLP image sources **10350** shown in FIGS. **103** and **103b**.

(408) FIG. **103b** is an illustration of another optics module with multiply folded optical path that includes a DLP image source **10350** and is shown from the position of the user's eye **9310** in the eyebox. As in the optics modules shown in FIGS. **103** and **103a**, the optics module of FIG. **103b** has the light source **10340** positioned at the side of the optics module to reduce the height and thickness of the optics module. The DLP image source **10350** is positioned opposite the light source **10340** however in this embodiment they do not share an optical axis. The illumination light **10364** passes through the beam splitter **10352**, which in this case can be a first reflective polarizer. A second reflective polarizer **10332** is positioned adjacent to the lens **9930** so that the illumination light **10364** is reflected toward the DLP image source **10350**. To reflect the illumination light **10364**, the first reflective polarizer (beam splitter **10352**) and the second reflective polarizer **10332** are oriented with perpendicular transmission axes. A quarter wave film **10351** (or quarter wave coating on the DLP cover glass) is provided adjacent to the DLP image source **10350** so that the polarization state of the illumination light **10364** is changed upon reflection from the DLP image source **10350** as it becomes image light **9360**. As a result, the polarization of the illumination light **10364** is opposite to that of the image light **9360**. Consequently, the illumination light **10364** is transmitted by the beam splitter **10352** and reflected by the second reflective polarizer **10332**, while the image light **9360** is reflected by the beam splitter **10352** and transmitted by the second reflective polarizer **10332**. The light source **10340** is oriented relative to the second reflective polarizer **10332** so that it is reflected at an oblique angle relative to the DLP image source **10350** as required to provide image light **9360** reflected from on-state micromirrors in the DLP image source **10350** along the folded optical axis **934**. The second reflective polarizer **10332** can be extended beyond the lens **9930** to provide the required oblique angle to fully illuminate the DLP image source **10350** as shown in FIG. **103b**. Because the light source **10340** is located behind the beam splitter **10352**, which is a reflective polarizer, the light source **10340** does not affect the image light **9360** and as a result, the light source **10340** can be a different size and orientation than the beam splitter **10352**. One or more light traps **10372** are provided to absorb light **10371** that is reflected from off-state micromirrors in the DLP image source **10350** and thereby improve the contrast of the displayed image. In this case, the light trap **10372** can be positioned under the second reflective polarizer **10332** because the polarization state of the light **10371** is such that it is reflected by the beam splitter **10352** and transmitted by the second reflective polarizer **10332**. The combined orientation of the light source **10340**, the beam splitter **10352** and the DLP image source **10350** provides an optics module that is relatively thin and relatively short compared to optics modules where the image source or the light source are positioned above the fold mirror or beam splitter (e.g. such as the optics module shown in FIG. **103**).

(409) FIGS. **97** and **98** show illustrations of optics modules similar to those shown in FIG. **94** but with the addition of an eye imaging camera **979** for capturing images of the user's eye **9310** during use. In these cases, the light source **9340** and image source **9450** are positioned opposite one another so that the eye imaging camera **979** can be positioned directly above the lens **9340** so that the optical axis **934** is shared between the optics module and the eye imaging camera **979**. By sharing a common optical axis, the eye imaging camera **979** can capture an image of the user's eye **9310** that has a perspective from directly in front of the user's eye **9310**. Image light **9360** can then be used to illuminate the user's eye **9310** during image capture. A portion of the light reflected from the user's eye **9310**, which can be unpolarized, passes through the beam splitter **9452** before being

captured by the eye imaging camera **979**. Because the eye imaging camera **979** is located above the beam splitter **9452**, if the beam splitter **9452** is a reflective polarizer, the polarization state of the image light **9360** will be opposite to that of the light **978** captured by the eye imaging camera **979**. The eye imaging camera **979** can be used to capture still images or video. Where video images can be used to track movements of the user's eye when looking at displayed images or when looking at a see-through view of the environment. Still images can be used to capture images of the user's eye **9310** for the purpose of identifying the user based on patterns on the iris. Given the small size of available camera modules, an eye imaging camera **979** can be added to the optics module with little impact on the overall size of the optics module. Additional lighting can be provided adjacent to the combiner **9320** to illuminate the user's eye. The additional lighting can be infrared, so the user can simultaneously view images displayed with visible light. If the additional lighting is infrared, the eye camera **979** must be capable of capturing images at matching infrared wavelengths. By capturing images of the user's eye from the perspective of directly in front of the user's eye, undistorted images of the user's eye can be obtained over a wide range of eye movement.

(410) FIG. **120** shows an illustration of another embodiment of an eye imaging camera associated with the optics module shown in FIG. **101**, however the eye imaging camera can be similarly included in optics modules such as those shown in FIGS. **99**, **100**, **103**, **103b**. These optics modules include absorptive polarizers **9932** to reduce stray light as previously disclosed herein. These optics modules can also include a diffractive surface **9931**, but the diffractive surface **9931** is not required for the operation of the eye imaging camera **979**. In this embodiment, the polarization state of the image light **9360** is the same as that of the light that is reflected by the user's eye and captured by the eye imaging camera **979** since they both pass through the absorptive polarizer **9932**. In this embodiment, the eye imaging camera **979** is positioned adjacent to the beam splitter **9452** and the compact planar light source **10040** and between the beam splitter and the field lens **10130**. The optical axis **12034** of the light reflected by the eye is then angled somewhat relative to the optical axis **934** of the image light **9360**, so that the center of the user's eye **9310** and the associated eyebox are within the field of view of the eye imaging camera **979**. In this way, the eye imaging camera **979** captures images of the user's eye from nearly directly in front and only slightly to the side of the user's eye **9310** as shown in FIG. **120**. While FIG. **120** shows the eye imaging camera **979** positioned adjacent to an end of the beam splitter **9452**, it is also possible to position the eye imaging camera **979** adjacent to a side of the beam splitter **9452**. The advantage of this embodiment is that the eye imaging camera **979** is provided with a simple optical path so that high image quality is possible in the captured images of the user's eye **9310**. It should be noted that the optics associated with the eye imaging camera must take into account the effect of the lens **9930** since the light reflected by the user's eye **9310** that is captured by the eye imaging camera passes through the lens **9930**. Also, the addition of the eye imaging camera **979** does not substantially increase the volume of the optics module as can be seen by comparing FIG. **120** to FIG. **101**.

(411) FIG. **121** shows an illustration of a further embodiment of an optics module that includes an eye imaging camera **979**. Similar to the embodiment shown in FIG. **120**, this optics module also includes an absorptive polarizer **9932** to reduce stray light and a diffractive surface **9931** may be included, but is not required. In this embodiment, the eye imaging camera **979** is positioned between the beam splitter **9452** and the field lens **10130** and pointed towards the beam splitter **9452**. In this way, light reflected by the user's eye **9310** is reflected upwards by the combiner **9320**, passes through the lens **9930** and the absorptive polarizer **9932** and then is reflected laterally toward the eye imaging camera **979** by the beam splitter **9452**. The light captured by the eye imaging camera **979** is thereby the same polarization state as the image light **9360**, so that it is reflected by the beam splitter **9452** and transmitted by the absorptive polarizer **9932**. The light reflected by the user's eye **9310** can be unpolarized as initially reflected by the user's eye **9310**, however, after passing through the absorptive polarizer **9932**, the light becomes polarized with the same polarization state as the image light **9360**. An advantage of this embodiment is that it is even

more compact than the embodiment shown in FIG. **120**. This arrangement of the eye imaging camera **979** is also possible in the optics modules shown in FIGS. **99**, **100**, **103**, **103a** and **103b**. (412) In the embodiments shown in FIGS. **120** and **121**, the user's eye **9310** and the associated eyebox can be illuminated by image light **9360** or an additional light source can be provided for example, by an LED positioned adjacent to the combiner **9320**. Where the LED can provide visible light or infrared light, provided the eye imaging camera can capture at least a portion of the wavelengths of light provided by the LED.

(413) In an alternative embodiment for the optics module shown in FIG. **103a**, the light source **10340** provides polarized illumination light **10364** and the fold mirror **10352** is a reflective polarizer plate so that an eye camera (not shown) can be positioned above the fold mirror **10352** and along the optical axis **934** for capturing images of the user's eye **9310** similar to that shown in FIGS. **97** and **98**. The eye camera and the optics module then share a common optical axis **934** so that images of the user's eye **9310** are captured from directly in front of the eye. In this arrangement, the polarization state of the image light **9360** is opposite to that of the light captured by the eye camera because the image light **9360** is reflected by the fold mirror **10352** and the light captured by the eye camera is transmitted by the fold mirror **10352**.

(414) FIG. **104** shows an illustration of the optics module of FIG. **95** with the additional element of a controllable light blocking element to improve contrast in portions of the displayed image and also to improve the appearance of opacity in displayed objects such as augmented reality objects. Where the controllable light blocking element can operate by absorbing the incident light or scattering the incident light as provided, for example, by an electrochromic element, a polymer stabilized liquid crystal or a ferroelectric liquid crystal. Examples of suitable light blocking elements includes: 3G Switchable Film from Scienstry (Richardson, TX); Switchable Mirror or Switchable Glass from Kent Optronics (Hopewell Junction, NY). The controllable light blocking element **10420** is shown in FIG. **104** as being attached to the lower surface of the combiner **9320** so that it doesn't interfere with the displayed image while blocking see-thru light from the environment **9362**. Provided the combiner **9320** is flat, the addition of controllable light blocking elements **10420** adjacent to the combiner **9320** is easily done either by attaching directly to the combiner or attaching to the sidewalls of the optics module housing. The controllable light blocking element **10420** can have a single area that can be used to block a selectable portion of the see-through light from the environment over the entire combiner **9320** area thereby enabling a selectable optical density. Alternatively the controllable light blocking element **10420** can provide an array of areas **10520**, as shown in FIG. **105**, that can be separately selectably controlled to block portions of the combiner **9320** area that correspond to areas in the displayed image where high contrast areas of the image are located. FIG. **105** shows an illustration of an array of separately controllable light blocking elements **10520**. FIGS. **106a**, **106b** and **106c** are illustrations of how the array of separately controllable light blocking elements **10520** can be used. FIG. **106a** shows how the array of separately controllable light blocking elements **10520** can be put into blocking modes in areas **10622** and non-blocking modes in areas **10623**. Where the blocking mode areas **10622** correspond to areas where information or objects are to be displayed such as is shown in the corresponding areas in the illustration of FIG. **106b**. FIG. **106c** shows what the user sees when the image of FIG. **106b** is displayed with the array of controllable light blocking elements **10520** used in light blocking modes **10622** and non-blocking modes **10623**. The user then sees the displayed information or objects overlaid onto a see-through view of the environment, but in the areas where information or objects are displayed, the see-through view is blocked to improve the contrast of the displayed information or object and provide a sense of solidness to the displayed information or objects.

(415) In addition, FIG. **104** shows a rear optical element **10490** that can be a protective plate or a corrective optic. The protective plate can be connected to sidewalls and other structural elements to stiffen the positioning of the combiner **9320** and to prevent dust and dirt from getting onto the inner

surface of the combiner **9320**. The corrective optics can include a prescriptive optic, which includes the ophthalmic prescription (optical power and astigmatism for example) of the user to improve the viewing experience.

(416) Head mounted displays provide the user with freedom to move their head while watching displayed information. See-through head mounted displays also provide the user with a see-through view of the environment whereupon the displayed information is overlaid. While head mounted displays can include various types of image sources, image sources that provide sequential color display typically provide higher perceived resolution relative to the number of pixels in the displayed images because each pixel provides image content for each of the colors and the image perceived by the user as a displayed full color image frame is actually the sum of a series of rapidly displayed sequential color subframes. For example, the image source can sequentially provide subframe images comprised of a red image, a green image and then a blue image that are all derived from a single full color frame image. In this case, full color images are displayed at an image frame rate that includes a series of at least three sequentially colored subframes that are displayed at a subframe rate which is at least $3\times$ the image frame rate. Sequential color images sources include reflective image sources such as LCOS and DLP.

(417) The color breakup that occurs with a sequential color display occurs because the different color subframe images that together provide the user with a full color frame image are displayed at different times. The inventors realized that with sequential color display in a head mounted display, when there is movement of the head-mounted display or movement of the user's eyes, such that the user's eyes do not move in synch with the displayed image that under such movement conditions the perceived locations of each of the sequential color image subframes are different within the user's field of view. This can happen when the user moves his head and the user's eyes do not follow the same trajectory as the head mounted display, which can be due to the user's eyes moving in a jerky trajectory as the eyes pause to look at an object in the see-through view of the environment. Another way this can happen is if an object passes through the see-through view of the environment and the user's eyes follow the movement of the object. Due to this difference in perceived locations within the user's field of view, the user sees the sequential color images slightly separated at the edges of objects. This separation of colors at the edge of objects is referred to as color breakup. Color breakup may be easily perceived during certain movements because the sequential colors are vividly colored in areas where they do not overlap one another. The faster the user moves their head or the faster the user's eyes move across the display field of view, the more noticeable the color breakup becomes, because the different color subframe images are separated by a greater distance within the field of view. Color breakup is particularly noticeable with see-through head mounted displays, because the user can see the environment and the user's eyes tend to linger on objects seen in the environment as the user turns his head. So even though the user may turn his head at a steady rotational rate, the user's eye movement tends to be jerky and this creates the conditions where color breakup is observed. As such there are two different conditions that tend to be associated with color breakup: rapid head movement and rapid eye movement.

(418) It is important to note that when the user is not moving his head and the head mounted display is not moving on the user's head, color breakup will not be observed because the subframe images are provided at the same positions within the field of view of the user's eyes. Also, if the user were to move his head and the user moves his eyes in synch with the head movement, color breakup will not be observed. So movement of the head mounted display is indicative of conditions that can lead to color breakup and is also indicative of the degree of color breakup that can occur if the user moves his eyes relative to the movement of the head mounted display. Color breakup is less of an issue with head mounted displays that do not have see-through to the environment, because only the displayed image content is visible to the user and it moves in synch with the movement of the head mounted display. Color breakup is also not an issue if a monochrome image is displayed with a monochrome light source (i.e. there are no sequential color subframes, instead

there are only single color frames) since all the displayed images are comprised of the same color. Thus, color breakup is an issue that is most noticeable with head mounted displays that provide a see-through view of the environment.

(419) Systems and methods according to the principles of the present invention reduce color breakup and thereby improve the viewing experience provided by a head-mounted display with see-through when the user is moving through the environment.

(420) In embodiments, systems and methods are provided where the head-mounted display detects the speed of movement of the head-mounted display and in response, the resolution of the image is reduced or the bit depth of the image is reduced, while the image frame rate at which the image is displayed and the associated subframe rate are correspondingly increased. In this way, the bandwidth associated with the display of the image can be maintained constant, in spite of the frame rate being increased. Where, by increasing the frame rate associated with the display of images, the time between the display of each sequential color subframe image is reduced and as a result the visually perceived separation between the sequential color images is reduced. Similarly the image frame rate can be reduced while the subframe rate is increased by increasing the number of subframes displayed for each image frame.

(421) In further embodiments, systems and methods are provided where the sequential color subframe images are shifted laterally or vertically relative to one another by a number of pixels that corresponds to the detected movement of the head mounted display. In this way, the color sequential subframe images are displayed to the user such that they are visually overlaid on top of each other within the displayed field of view. This compensates for separation between subframes and thereby reduces color breakup.

(422) In yet another embodiment, systems and methods are provided where an eye-imaging camera in the head-mounted display is used to track the movement of the user's eyes. The movement of the head-mounted display may be simultaneously measured. An accommodation in the presentation may then be made to reduce color breakup. For example, the resolution of the images and the frame rate may be changed or the image frame rate can be reduced while increasing the subframe rate, in correspondence to the difference in movement of the user's eyes and the movement of the head mounted display. As another example, the subframes may be shifted to align the subframes in correspondence to the determined difference in movement between the user's eyes and the head mounted display. As a further example, the color saturation of the content may be reduced to reduce the perception of color breakup due to the fact that the colors, while positionally separated as perceived by the user, are not as separated in color space. In yet a further example, the content could be converted to monochrome imagery which is displayed as a single color image (e.g. white) during the detected movement so that color breakup is not visible.

(423) FIG. 107 shows an example of a full color image 10700 that includes an array of pixels, including portions of red, green and blue pixels. For sequential color display, three subframe images are created that are each comprised of only one color, such as only red or only green or only blue. Those skilled in the art will recognize that sequential color images that together provide a perceived full color image can also be comprised of subframes of cyan, magenta and yellow. These subframe images are rapidly displayed in sequence to the user on the head-mounted display so that the user perceives a full color image that combines all three colors. With a reflective display such as an LCOS or a DLP, the subframe images are displayed by changing the reflective display to provide the respective image content associated with the particular subframe image and then illuminating the reflective display with the associated color light, so the light is reflected to provide the subframe image to the optics of the head-mounted display and from there to the user's eye.

(424) If the subframe images are accurately aligned with each other, then the full color image perceived by the user will be full color out to the edges of the image and there will be no color breakup. This is what is typically seen by the user of a head-mounted display when the head-mounted display is stationary on the user's head and the user is not moving his eyes. However, if

the user moves his head or the head-mounted display moves on the user's head (such as due to vibration) and the user's eyes are not moved in unison with the displayed image, the user will perceive the subframe images to be laterally (or vertically) offset relative to one another as shown by illustrations **10802** and **10804** in FIGS. **108A** and **108B**. The perceived amount of lateral offset between the displayed subframe images is related to the speed of movement of the head-mounted display and the time between the display of the sequential subframe images, which is also known as subframe time or 1/subframe rate. The lateral shifting between subframe images, that is perceived by the user, is the color breakup and color breakup is perceived as fringes of color at the edges of objects. When the user moves his head (or eyes) quickly and the subframe rate is slow, color breakup can be substantial as illustrated in FIG. **108A**. If the user moves his head slowly or the subframe rate is higher, the color breakup is less as illustrated in FIG. **108B**. If the color breakup is less than one pixel, in digital imaging, in lateral shifting, the user will perceive there to be no color breakup.

(425) Display frame rate in a head-mounted display is typically limited by either the bandwidth of the processor and associated electronics or by the power required to drive the processor and associated electronics, which translates into battery life. The bandwidth required to display images at a given frame rate is related to the number of frames displayed in a period of time and the number of pixels in each frame image. As such, simply increasing the frame rate to reduce color breakup is not always a good solution as it requires a higher bandwidth which the processor or associated electronics may not be able to support and power usage will be increased thereby reducing battery life. Instead, systems and methods in accordance with the principles of the present invention provide a method of display wherein the number of pixels in each subframe image is reduced thereby reducing the bandwidth required to display each subframe image while simultaneously increasing the subframe rate by a corresponding amount to maintain bandwidth while reducing color breakup. This embodiment is suitable for situations wherein subframe images can be provided with different numbers of pixels and different frame rates. For example, it would be suitable in camera and display systems where the capture conditions can be changed to provide images with a lower resolution that can then be displayed with a faster subframe rate. Static images such as text or illustrations can be displayed with a lower frame rate and a faster subframe rate to reduce color breakup since the image content doesn't change quickly. Alternatively, images can be modified to be displayed at lower resolution (fewer pixels) with a faster frame rate or subframe rate to reduce color breakup.

(426) FIG. **109** shows an illustration of the timing of a sequential color image comprised of sequential display of a red subframe image **10902** followed by a green subframe image **10904** followed by a blue subframe image **10908** in a repeating process. As long as the subframes together are displayed at a full color image frame rate that is greater than approximately 24 frames/sec, such that the sequential color subframes are displayed at a subframe rate of greater than 72 subframes/sec, the human eye will perceive full color moving images without flicker. This condition is suitable for displaying a video image without color breakup when the head-mounted display is stationary or moving relatively slowly. However, if the user moves his head such that the head-mounted display moves rapidly, color breakup will occur. This color breakup occurs because rapid head movements are typically a reaction of the user to something occurring in the environment (e.g. a loud noise) so that the user's eyes are searching the environment during the rapid head movement, which leads to jerky eye movements and substantial color breakup.

(427) Movement of the head-mounted display can be detected by an inertial measurement unit, which can include accelerometers, gyro sensors, magnetometers, tilt sensors, vibration sensors, etc. Where only the movements within the plane of the display field of view (e.g. x and y movements and not z movement) are important for detecting conditions where color breakup may occur. If the head-mounted display is detected to be moving above a predetermined threshold where color breakup is predicted to occur (e.g. greater than 9 degrees/sec), in embodiments, the resolution of

the images may be reduced (thereby reducing the number of pixels in the images and effectively making each pixel larger within the display field of view) and the subframe rate may be correspondingly increased. Note that the subframe rate can be increased without changing the image frame rate by increasing the number of subframes that are displayed sequentially, for example six subframes could be displayed for each image frame wherein the sequential color subframe images are each displayed twice. By increasing the number of subframes displayed for each image frame, the subframe rate can be increased without having to increase the image frame rate, which can be more difficult to change because the image frame rate is typically provided by the source of the image content such as in a movie. FIG. 110 shows an illustration of a faster subframe rate, wherein the display time for each subframe, **11002**, **11004**, and **11008** is reduced and the time between display of each sequential subframe is also reduced. FIG. 110 shows a subframe rate that is approximately twice as fast as that shown in FIG. 109. The associated image frame rate can be twice as fast in FIG. 110 as compared to FIG. 109, where both the image frame rate and the subframe rate are doubled. Alternatively, as previously described, the image frame rate can be unchanged between FIGS. 109 and 110, where only the subframe rate is doubled to reduce color breakup. To enable the bandwidth associated with the display of the images shown in FIG. 110 to be approximately the same as the bandwidth associated with the display of subframe images shown in FIG. 109, the resolution (number of pixels in each subframe image) is reduced by approximately a factor of two.

(428) While reducing the resolution of the displayed subframe images in correspondence to an increase in the subframe rate may seem to degrade the image quality perceived by the user, the human eye is not capable of perceiving high resolution when there is substantial movement. As such, color breakup is more visible than a reduction in the resolution of the image when the eye is moving. Consequently, the systems and methods of the present invention trade reduced image resolution for increased image frame rate to reduce color breakup without a perceptible loss in resolution, and bandwidth is thereby maintained. This technique can be used, for example, to reduce color breakup by a factor of up to 16, where the resolution of the displayed image is reduced to 1/16th the original resolution and the frame rate of the displayed image is increased by 16 $\times$.

(429) In another embodiment of the invention, when movement of the head-mounted display is detected, the subframe images associated with a full color frame image are digitally shifted relative to one another in a direction counter to the detected direction of movement and with an amount that corresponds to the detected speed of movement. This effectively compensates for the perceived offset between the displayed subframe images that causes color breakup. The digital shifting is applied only to the subframes that together comprise a full color frame image. This is different from typical digital image stabilization wherein full color frame images are digitally shifted relative to one another to compensate for movement as described, for example, in United States patent publication 2008/0165280. By applying the digital shifting to the subframes that constitute a single full color frame image, the amount of digital shifting required to reduce color breakup is typically only a few pixels even when the detected movement speed is high, this is in contrast to typical digital image stabilization where fast movements result in accumulating shifts of the frame image so that the image effectively moves outside of the display field of view or the amount of digital stabilization that can be applied is limited. FIGS. 111a and 111b illustrate this embodiment. FIG. 111a shows how sequentially displayed subframe images, **11102**, **11104**, and **11108** would be perceived by the user when there is substantial movement, wherein the different colors associated with the subframes are separately visible along the edges of objects, evenly spaced across the field of view in the direction of movement. In contrast, FIG. 111b shows how the visibility of the subframes is changed when the subframes are digitally shifted to compensate for the detected movement and thereby reduce the separation between the subframes across the field of view, and as a result the user perceives a series of full color frame images **11120** with reduced color breakup. As shown in FIG. 111b, the full color frame images are not image stabilized or digitally shifted in

response to the detected movement.

(430) In embodiments, movement direction and speed of the head-mounted display is detected by the IMU sensor immediately prior to the display of each full color frame image. If the movement speed is above a predetermined threshold, the sequentially displayed color subframes associated with each full color frame are digitally shifted relative to one another so that they are displayed in an aligned position within the display field of view. The magnitude of the shift corresponds to the speed of the detected movement and the direction of the shift is counter to the detected direction of movement.

(431) In an example, the movement of the head-mounted display is detected immediately prior to display of a first subframe associated with a full color frame image. The first subframe associated with the full color frame image can then be displayed without a shift. The second subframe can be shifted by an amount and direction that compensates for the movement that occurs between the display of the first and second subframes and then is displayed. The third subframe can be shifted by an amount and direction that compensates for the movement that occurs between the display of the first subframe and the third subframe and is then displayed. The movement of the head-mounted display is then detected again to determine the shifts to be applied to the subframes associated with the next full color frame image. Alternatively, the subframes can be shifted by an amount that compensates for a portion of the movement that occurs between the subframes.

(432) In a further example, the direction and speed of movement of the head-mounted display is detected immediately prior to the display of a reference subframe. Subsequent subframes are then shifted to compensate for movement that occurs between the time the reference subframe is displayed and the time that the subsequent subframe is displayed. Wherein the time that the reference subframe is displayed and the time that the subsequent subframe is displayed may be up to 5 frame times.

(433) An advantage of this embodiment is illustrated by examining the effective frame rates associated with the color breakup and the blur of the image. If the full color image is displayed with an image frame rate of 60 frames/sec, the subframes would typically be displayed at a subframe rate of 180 frames/sec to provide three subframes for each image frame. The described system and method effectively shifts the subframes so that they are positioned on top of one another, so the color breakup is reduced to an amount that corresponds to 180 frames/sec. At the same time, the blur perceived by the user between image frames corresponds to 60 frames/sec since each of the subframes is derived from the same full color frame image.

(434) In further embodiments, the digital shifting of the subframes that is based on detected movement immediately prior to the display of each full color frame image can be combined with digital image stabilization that is applied between the full color frame images.

(435) In yet further embodiments, the method of digital shifting of subframes is combined with the method of increasing frame rate with a simultaneous reduction in image resolution. These two methods of reducing color breakup operate on different aspects of the image processing associated with displaying an image in a head mounted display, as such they can be independently applied in either order in the image processing system associated with the processor.

(436) In yet another embodiment, the head mounted display includes a camera for detecting the eye movements of the user (e.g. as described herein) relative to the movement of the head mounted display. The eye camera can be used to measure the speed of eye movement and the direction of eye movement. In embodiments, the resolution of eye cameras can be relatively low (e.g. QVGA or VGA) so that the frame rate can be relatively high (e.g. 120 frames/sec) without introducing bandwidth limitations. The detected eye movements relative to the head-mounted display can be used to determine when to apply methods to reduce color breakup including, for example, increasing the frame rate and digitally shifting the subframes as has been previously described herein. For example, if the detected eye movement is above a predetermined angular speed, the resolution of the displayed images can be reduced and the subframe rate can be increased. In

another example, the detected eye movement can be used to determine the amount and direction of digital shifting applied to subframes within an image frame prior to display of the subframes. In yet another example, measured eye movements can be used in combination with detected movements of the head-mounted display to determine the amount and direction of digital shifting applied to subframes within an image frame prior to display of the subframes. The amount and direction of digital shifting applied to the subframes can be in correspondence to the difference between the detected movements of the head mounted display and the detected eye movements of the user. Where the detection of a condition where the user's eye is moving one direction and the head mounted display is moving in an opposing direction represents a situation where particularly bad color breakup can occur. In this case, combined methods for reducing color breakup are advantageous.

(437) In another yet further embodiment, when movement of the head-mounted display or eye movement is detected above a predetermined threshold, the images are changed from color sequentially displayed full color images to monochrome images. The monochrome images can be comprised of combined image content from each of the color sequential subframes associated with each full color image frame. Where the monochrome images can be grey scale or luma images wherein the luma code values (Y) for each pixel can be calculated for example as given in Equation 1 below as taken from <http://en.wikipedia.org/wiki/Grayscale> and as referenced to the CIE 1931 standard for digital photography:

$$Y=0.2126R+0.7152G+0.0722B \quad \text{Equation 1}$$

(438) where R is the red code value for the pixel, G is the green code value for the pixel and B is the blue code value for the pixel. Alternatively, monochrome images can be comprised of single color images such as the green subframe image, and this image can be displayed either with a single color or preferably with simultaneous application of all the sequential colors (e.g. red, green and blue) so that the applied illumination onto the reflective image source is white light and as a result, the displayed image appears as a grey scale image.

(439) Several more specific examples are provided below.

Example 1

(440) For a 26 deg display field of view and a 1280 pixel horizontally wide image, a pixel occupies 0.020 deg within the display field of view. If the frame rate of the full color images is 60 Hz, with three color sequential subframes images, the subframe time is 0.006 sec. The rotational speed of the head mounted display needed to produce one pixel of color breakup is then 3.6 deg/sec. If the number of horizontal pixels in the display field of view is reduced to 640 pixels and simultaneously the frame rate of the full color images is increased to 120 Hz, with three color sequential subframes images, the subframe time is reduced to 0.003, the size of a pixel is increased to 0.041 deg and the rotational speed to produce one pixel of color breakup is 14.6 deg/sec.

Example 2

(441) For a 26 deg display field of view and a 1280 pixel horizontally wide image, a pixel is 0.020 deg within the display field of view. If the smallest size that the user can detect for color breakup is one pixel wide, then a rotational speed of over 3.6 deg/sec is required if the subframe rate is 180 Hz, before color breakup is detected by the user. Even though the color breakup is an analog effect, the user's eye does not have the resolution to detect the color fringes that are present during movement below this speed. So below this rotational speed, color breakup management is not required.

Example 3

(442) For a 26 deg display field of view and a 1280 pixel horizontally wide image, a pixel is 0.020 deg within the display field of view. If the user can detect color breakup as small as one pixel wide, then a rotational speed of 3.6 deg/sec will require a shift of the subframes relative to each other of one pixel if the subframe rate is 180 Hz, to align the subframes so that color breakup is not visible to the user. If the user rotates their head at 15 deg/sec, then the subframes will require a shift of 4

pixels relative to one another to align the subframes so that color breakup is not visible. If the image frame begins with the display of the red subframe image, then no digital shifting is required for the red subframe image. A 4 pixel shift is required for the green subframe image. And, an 8 pixel shift is required for the blue subframe image. The next red subframe associated with the next image frame would then be effectively shifted 12 pixels relative to the previous red subframe within the field of view.

(443) Each of the color breakup reduction technologies described herein may be used in combination with each of the other color breakup reduction technologies.

(444) The inventors appreciated that fitting see-through computer displays into certain head-worn form factors is a challenge, even when reduced in size as described herein. A further advantage that is provided by an optics module that includes multiply folded optics is that twists can be introduced at the fold surfaces to modify the orientation of different portions of the optics module relative to each other. This can be important when the optics module needs to fit into a thin curved glasses frame, a visor or a helmet where the increased width associated with the upper portion of the multiply folded optics module can make it more difficult to fit into structures that are not parallel to the combiner. As such, another aspect of the present invention relates to twisting certain optical components within the see-through computer display such that the optical components better fit certain form factors (e.g. glasses) yet continue to perform as high quality image displays. In embodiments, optics systems with dual mirror systems to fold the optical path (e.g. optical systems described herein with respect to FIGS. 6, and 93 through 106) are provided such that the image production module (e.g. upper module), which includes a first image light reflective surface, is turned about a first optical axis leading from the upper module to the lower module and in a direction to fit the upper module more compactly into a frame of a head-worn computer. At the same time, to avoid distorting the image provided to the eye of the user, the image delivery optics (e.g. lower module), which includes a second image light reflective surface, is turned about a second optical axis that leads to the user's eye and in the opposite direction relative to the image, thereby introducing a compound angle between the first image light reflective surface and the second image light reflective surface. Provided that the first and second optical axes are perpendicular to one another in the non-twisted state, the distortion in the image associated with the twist about the first axis is compensated by a twist of the same angular magnitude about the second axis so that the image presented to the eye of the user is undistorted by the twisting.

(445) FIG. 112 illustrates a head-worn computer with see-through displays in accordance with the principles of the present invention. The head-worn computer has a frame 11202 that houses/holds the optics modules in position in front of the users eyes. As illustrated in FIG. 112, the frame 11202 holds two sets of optical modules 11204 and 11208 each of which have upper and lower optics modules. Optics module 11204 is non-twisted and is presented to illustrate the difficulty in fitting the non-twisted version into the frame. One will note that the dotted box, which represents the outer bounds of the optics module 11204 doesn't fit within the bounds of the frame 11202. Fitting optics module 11204 into the frame 11202 would normally require that the frame 11202 become thicker, from front to back, which would lead to more offset of the glasses form factor from the face of the user, which is less desirable and is less compact. In contrast, optics module 11208 is a twisted optics module, where the upper module is twisted (or rotated) to better fit into the confines of the frame 11202 as shown in FIG. 112.

(446) FIG. 113 shows a more detailed illustration of the twists imparted within multiply folded optics in optics module 11208. Upper module 11214 is twisted relative to the lower module 11218 along optical axis 934 to better fit into the frame 11202, It is this twist which enables optics module 11208 to better fit within the frame 11202 as shown in FIG. 112 and as a result frame 11202 can be thinner and more compact than if non-twisted optics modules were used. To avoid distorting the image provided to the user, a second twist is required to introduce a compound angle between the first reflecting surface 11225 in the upper module 11214 and second reflecting surface 11226 in the

lower optics module **11218**. The second twist is imparted to the second reflecting surface about the optical axis **933** and in an opposite direction relative to the image from the twist in the upper module **11214**. In this way, the effects of the increased width of the upper portion of the multiply folded optics can be reduced when fitting the optics module into a curved structure such as glasses frames, a visor frame or a helmet structure. Where it is preferred, but not required that the optical axis **934** be perpendicular to the optical axis **933** so that the magnitude of the angular twist imparted to the first reflecting surface **11225** can be the same as the twist imparted to the second reflecting surface **11226** to provide an image to the user's eye that is not distorted due to the twisting.

(447) Another aspect of the present invention relates to the configuration of the optics and electronics in a head-worn frame such that the frame maintains a minimal form factor to resemble standard glasses. In embodiments, a see through optical display with multiply folded optics to provide a reduced thickness (e.g. as described herein) may be mounted in the frame. In embodiments, the multiply folded optical configuration may be twisted at the fold surfaces (e.g. as described herein) to better fit the optics into the frame. In embodiments, the electronics that operate the displays, processor, memory, sensors, etc. are positioned between, above, below, on a side, etc. of the optical modules and oriented to provide a reduced thickness in the frame to match the thickness of the optics. Orienting the board can be particularly important when the board includes large components that limit the width of the board, such as for example the processor chip. For example, an electronics board or components on the electronics board may be mounted in a vertical orientation between and/or above the optical modules to reduce the thickness of the electronics board as mounted into the frame. In another configuration the board may be mounted between the optical modules at a height near the top of the optical modules to minimize the height of the glasses frame. In yet another configuration the board may be mounted such that it extends over the optical modules to minimize the thickness of the frame. In further embodiments, the board may be mounted in an angled configuration to enable the thickness and height of the frame to be reduced simultaneously. In embodiments, the electronics may be divided between multiple boards. For example, a longer board over a shorter board where the space between the optical modules is used for the lower board. This configuration uses some of the space between the eyes for some of the electronics.

(448) FIG. **114** illustrates a top view and front view of a configuration including optical modules **11208**, electronics board **11402** and a heat sink **11404**. The board **11402** is mounted in a vertical orientation to maintain a thin frame portion that sits across the user's brow. As illustrated, the optical modules **11208** include upper modules **11214** and a second reflecting surface **11226** in front of the user's eye. The upper module may have a flat reflecting surface and the upper **11214** may be turned or twisted with respect to the second reflecting surface **11226** as described herein. The second reflecting surface **11226** may be a partial mirror, notch filter, holographic filter, etc. to reflect at least a portion of the image light to the eye of the user while allowing scene light to transmit through to the eye.

(449) FIG. **115** illustrates a front view of a configuration that includes optics illustrated in FIG. **114**; however, the electronics board **11402** is mounted in the space between the optical modules at a height that is similar to the height of the optical modules. This configuration reduces the overall height of the frame.

(450) FIG. **116** illustrates a front view of a configuration that includes optics illustrated in FIGS. **114** and **115**. The electronics layout in this configuration is done with multiple boards, **11402**, **11602** and **11604**. The multiple board configuration allows the boards to be thinner from front to back thereby enabling the brow section of the frame to be thinner. A heat sink **11404** (not shown in FIG. **116**) may be mounted on the front face between the optical modules. This configuration also causes the heat to be drawn in a direction away from the user's head. In embodiments, the processor, which is a main heat generator in the electronics, is mounted vertically (e.g. on board

11604) and the heat sink **11404** may be mounted in front such that it contacts the processor. In this configuration, the heat sink **11404** causes heat to spread to the front of the device, away from the user's head. In other embodiments, the processor is mounted horizontally (e.g. on board **11602** or **11402**). In embodiments, the board(s) maybe tilted (e.g. 20 degrees) from front to back to create an even thinner brow section.

(451) Another aspect of the present invention relates to concealing the optical modules such that a person viewing the user does not clearly see the optical modules, electronics or boards. For example, in configurations described herein, the optical modules include lenses that hang below the top of the brow section of the head-worn device frame and the electronics board(s) hang down as well so that the see-through view is partially blocked. To conceal these features and thereby provide the head worn computer with the appearance of conventional glasses, an outer lens may be included in the glasses frame so that it covers a portion of the frame that contain the optical modules or electronics, and the outer lens may include a progressive tint from top to bottom. In embodiments, the tint may have less transmission at the top for concealment of a portion of the frame that includes the optical modules or electronics board while having higher transmission below the concealment point such that a high see-through transmission is maintained.

(452) Aspects of the present invention provide multiply folded optics to reduce the thickness of the optics modules along with vertically oriented or angled electronics to reduce the mounted thickness of the electronics and progressively tinted outer lenses to conceal a portion of the optics or electronics. In this way, a head worn computer is provided with a thinner form factor and an appearance of conventional glasses.

(453) Another aspect of the present invention relates to an intuitive user interface mounted on the HWC **102** where the user interface includes tactile feedback to the user to provide the user an indication of engagement and change. In embodiments, the user interface is a rotating element on a temple section of a glasses form factor of the HWC **102**. The rotating element may include segments such that it positively engages at certain predetermined angles. This facilitates a tactile feedback to the user. As the user turns the rotating element it 'clicks' through it's predetermined steps or angles and each step causes a displayed user interface content to be changed. For example, the user may cycle through a set of menu items or selectable applications. In embodiments, the rotating element also includes a selection element, such as a pressure-induced section where the user can push to make a selection.

(454) FIG. **117** illustrates a human head wearing a head-worn computer in a glasses form factor. The glasses have a temple section **11702** and a rotating user interface element **11704**. The user can rotate the rotating element **11704** to cycle through options presented as content in the see-through display of the glasses. FIG. **118** illustrates several examples of different rotating user interface elements **11704a**, **11704b** and **11704c**. Rotating element **11704a** is mounted at the front end of the temple and has significant side and top exposure for user interaction. Rotating element **11704b** is mounted further back and also has significant exposure (e.g. 270 degrees of touch). Rotating element **11704c** has less exposure and is exposed for interaction on the top of the temple. Other embodiments may have a side or bottom exposure.

(455) As discussed above, a specially designed lens may be used to conceal portions of the optics modules and/or electronics modules. FIG. **119** illustrates an embodiment of one such lens **11902**. Two lenses, **11902** are illustrated with Base 6 and 1.3 mm thickness but other geometries with, for example, different curvatures and thicknesses can be used. The lenses **11902** are shaped to look like conventional glasses lenses with features including magnetic mounting attachment and special tinting in portions of the lenses **11902** where opaque structures such as electronics are located behind the lenses.

(456) The lenses **11902** includes blind holes **11904** for the mounting of a magnetic attachment system (not shown). The magnetic attachment system may include magnets, magnetic material, dual magnets, opposite polarization magnets, etc. such that the lenses **11902** can be removed and

remounted to the head-worn computer (e.g. HWC **102**). In the magnetic attachment system, the lenses **11902** are held by magnetic force into the frame of the HWC. The magnets can be inserted into the blind holes **11904** or inserted into the frame of the HWC in corresponding matching positions. As long as either the lens **11902** or the matching position on the frame of the HWC includes a magnet and the other position has a similar sized piece of magnetic material or another magnet oriented to attract the lens **11902** and hold it in the frame of the HWC. To this end, the frame of the HWC can provide guidance features to position the lens **11902** in front of the optics modules in the HWC. Where the guidance features can be a ridge or flange that the lens is seated in so the lens **11902** cannot move laterally when held in place by the magnetic attachment system. In this way, the function of the magnetic attachment system is simply to hold the lenses **11902** in place, while the guidance features position the lenses **11902**. The guidance features can be robustly made to hold the lenses **11902** in place when dropped or subjected to impact even when the force provided by the magnetic attachment system is relatively low, so that the lenses **11902** can be easily removed by the user for cleaning or replacement. Where easy replacement enables a variety of lenses with different optical features (e.g. polarized, photochromic, different optical density) or different appearance (e.g. colors, level of tinting, mirror coating) to be changed out by the user as desired.

(457) FIG. **119** also illustrates an example of how the lens **11902** may be tinted to conceal or at least partially conceal certain optical components (e.g. the non-see-through components or opaque components) such as, electronics, electronics boards, auxiliary sensors such as an infrared camera and/or other components. As illustrated, the blind holes **11904** may also be concealed or at least partially concealed by the tinting. As illustrated in FIG. **119**, a top portion **11908**, approximately 15 mm as illustrated, may be more heavily tinted (e.g. 0 to 30% transmission) or mirrored to better conceal the non-see through portions of the optics and other components. Below the top portion **11908**, the lens **11902** may have a gradient zone **11909** where the tinting level gradually changes from top to bottom and leads into the lower zone **11910**. The lower zone **11910** includes the area where the user primarily views the see-through surrounding and this zone may be tinted to suit the viewing application. For example, if the application requires a high see through, the lower zone **11910** may be tinted, between 90% and 100% transmissive. If the application requires some see-through tint, then the lower area may be more heavily tinted or mirrored (e.g. 20% to 90%). In embodiments, the lower area **11910** may be a photochromic layer, an electrochromic layer, a controllable mirror or other variable transmission layer. In embodiments, the entire lens or portions thereof may have a variable transmission layer such as a photochromic layer, electrochromic layer, controllable mirror, etc. In embodiments, any of the areas or whole lens **11902** may include polarization.

(458) Another aspect of the present invention relates to cooling the internal component through the use of micro-holes sized such they are large enough to allow gas to escape but small enough to not allow water to pass through (e.g. 25 μm , 0.2 mm, 0.3 mm, etc.). The micro-holes may be included in a heat sink, for example. The heat sink, or other area, may be populated with hundreds or thousands of such micro-holes. The micro-holes may be laser cut or CNC holes, for example, that are small enough to keep large droplets of water out of the device but allow air to exchange through the heat sink. Besides increasing surface area of the heat sink, they also have matching holes on the underside of the frame to enable convective cooling where cool air is pulled in from the bottom as the heat raises from the top, like a chimney and as such, the heat sink with the micro-holes is preferably located on the top or side of the frame of the HWC. In embodiments, the micro-holes are aligned in the troughs formed by the fins on the top of the heat sink. This causes the exiting air to flow through the troughs thereby increasing the heat transfer from the fins. In embodiments, the micro-holes may be angled such that the length of the hole in the heat sink material is increased and the air flow can be directed away from the head of the user. In addition, the micro-holes may be of a size to cause turbulence in the air flow as it passes through the micro-holes. Where, turbulence

substantially increases the heat transfer rate associated with the air flow through the heat sink. In embodiments, the heat management system of the HWC **102** is passive, including no active cooling systems such as fans or other energized mechanical cooling systems to force air flow through the micro-holes. In other embodiments, the heat management system includes energized mechanical cooling, such as a fan or multiple fans or other systems to force air movement through the HWC and the micro-holes.

(459) Another aspect of the present invention relates to finding items in the surrounding environment based on similarity to items identified. Augmented reality is often rigidly defined in terms of what is included and how it is used, it would be advantageous to provide a more flexible interface so people can use augmented reality to do whatever they want it to do. An example is to use the HWC camera, image analysis and display to designate items to be found. FIG. **122** shows an illustration of an image **12210** of a scene containing an object that the user would like the HWC to assist in looking for the object as the user moves through the environment. In this example, the user has circled the object **12220** that is being looked for, where in this case the object is a cat. The HWC then analyses the circled region of the image for shapes, patterns and colors to identify the target to be searched for. The HWC then uses the camera to capture images of the scene as the user moves about. The HWC analyses the images and compares the shapes, patterns and colors in the captured images of the scene and compares them to the shapes, patterns and colors of the target. When there is a match, the HWC alerts the user to a potential find. The alert can be a vibration, a sound or a visual cue in an image displayed in the HWC such as a pointer, a flash or a circle that corresponds to the location of the potential find in the scene. This method provides a versatile and flexible augmented reality system wherein an item is described visually and a command of “find something like this” is given to the HWC. Examples of ways to identify an object to be searched for include: circle an item in a previously captured image that is stored on the HWC (as shown in FIG. **122**); point to an item in a physical image held in front of the camera in the HWC; point to an item in the live image provided by the camera in the HWC and viewed in the see-through display of the HWC, etc. Alternately, text can be input to the HWC with a command of “find wording like this”, e.g. a street sign or an item in a store and the HWC can then search for the text as the user moves through the environment. In another example, the user can indicate a color with a command of “find a color like this”. The camera used to search for the item can even be a hyperspectral camera in HWC to search for the item using infrared or ultraviolet light to thereby augment the visual search that the user is conducting. This method can be extended to any pattern that the user can identify for the HWC such as sounds, vibrations, movements, etc. and the HWC can then use any of the sensors included in the HWC to search for the identified pattern as the target. As such the finding system provided by the invention is very flexible and can react to any pattern that can be identified by the sensors in the HWC, all the user has to do is provide an example of the pattern to look for as a target. In this way the finding system assists the user and the user can do other things while the HWC looks for the target. The finding system can be provided as an operating mode in the HWC where the user selects the mode and then inputs the pattern to be used as the search target by the HWC. Examples of items that can be searched for include: household objects, animals, plants, street signs, weather activity (e.g. cloud formations), people, voices, songs, bird calls, specific sounds, spoken words, temperatures, wind direction shifts as identified by wind sound relative to the compass heading, vibrations, objects to be purchased, brand names in stores, labels on items in a warehouse, bar codes or numbers on objects and colors of objects to be matched. In a further embodiment, the rate of searching (e.g. how often an analysis is conducted) can be selected by the user or the rate can be automatically selected by the HWC in response to the rate of change of the conditions related to the target. In a yet further embodiment, the sensors in the HWC include a rangefinder or a camera capable of generating a depth map to measure the distance to an object in an image captured by the camera. The HWC can then analyze the image along with the distance to determine the size of the object. The user can then input the size of the object to the

finding system as a characteristic of the target pattern to enable the HWC to more accurately identify potential finds.

(460) Another aspect of the present invention relates to assisting a person in reading text that is presented in a physical form, such as a book, magazine, on a computer screen or phone screen, etc. In embodiments, the camera on the HWC can image the page and the processor in the HWC can recognize the words on the page. Lines, boxes, or other indicators may be presented in the HWC to indicate which words are being captured and recognized. The user would then be viewing the page of words through the see-through display with an indication of which words have been recognized. The recognized words can then be translated or converted from text that is then presented to the user in the see-through display. Alternately, the recognized words can be converted from text to speech, which is then presented to the user through the head worn speakers, headphones, visual displays, etc. This gives the user a better understanding of the accuracy associated with the text recognition relative to the translated text or converted speech.

(461) In a further aspect of the invention, a magnetic attachment structure is provided for the combiner to enable the combiner to be removable. In the optics associated with a HWC **102** such as for example the optics shown in FIG. **6**, it is important that the combiner **602** be accurately positioned and rigidly held below the frame of the HWC and the upper optical module **202** located inside the frame. At the same time, the combiner **602** can become damaged so that it needs to be replaced, or it may need to be cleaned periodically so that it is advantageous for the combiner to be removable. FIG. **123** shows an illustration of a cross section of a single combiner **12360** with the magnetic attachment structure as shown from the side to show the angle of the combiner **12360**. FIG. **124** shows an illustration of two combiners **12360** with magnetic attachment structures attaching the combiners **12360** to the frame of the HWC **12350** as shown from the front of the HWC. The combiner **12360** has two or more pins **12365** that are attached to the combiner **12360** such that the pins have parallel axes. The pins **12365** are shown as being inserted into holes drilled through the combiner **12365** and attached in place with adhesive such as UV cured adhesive. The pins **12365** are made of a magnetic material such as for example 420 stainless steel. The pins **12365** extend into parallel bores in the frame of the HWC **12350** so that the combiner **12360** is fixedly held in place relative to the frame of the HWC **12350**. The attachment and bend of the pins **12365** establish the angle between the combiner **12360** and the optics in the frame of the HWC **12350**. A magnet **12370** is bonded into the frame of the HWC **12350** such that the pin **12365** attracted by the magnet **12370** and thereby the pin **12365** and the attached combiner **12360** are held in place relative to the frame of the HWC **12350**. The magnet **12370** is selected so that the force exerted by the magnet **12370** onto the pin **12365** is strong enough to hold the combiner **12360** in place during normal use, but weak enough that removal of the combiner **12350** is possible by the user. By having the pins **12365** and associated bores parallel, the combiner **12350** can be easily removed for cleaning, or replaced if damaged. To provide a more rigid and repeatable connection between the combiner **12360** and the frame of the HWC **12350**, the pins can fit into an extended tight bore in the frame of the HWC **12350**. In addition, the pins **12365** can include a flange as shown that seats onto an associated flat surface of the frame of the frame **12350** or a flat surface of the magnet **12370** to further establish the angle of the combiner **12360** and the vertical position of the combiner **12360**. In a preferred embodiment, the magnet **12370** is a ring magnet and the pin **12365** extends through the center of the ring magnet. The magnet **12370** can also be included in an insert (not shown) that further includes a precision bore to precisely align and guide the pin **12365**. The insert can be made of a hardened material such as a ceramic to provide a bore for the pin **12365** that is resistant to wear during repeated removal and reinstallation of the combiner **12360**. The pins can be accurately positioned within the combiner through the use of a jig that holds the pins and the combiner. The holes for the pins in the combiner are then made larger than the pins so there is a clearance to allow the combiner and pins to be fully positioned by the jig. An adhesive such as a UV curing adhesive is then introduced to the holes and cured in place to fasten the pins to the

combiner in a position that is established by the jig. In a further embodiment, the combined structure of the pins **12365** and the combiner **12350** are designed to break if subjected to a high impact force, to thereby protect the user from injury. Where the pin **12365** or the combiner are designed to break at a previously selected impact force that is less than the impact force required to break the frame of the HWC **12350** so that the combiner **12350** with the attached pins **12365** can be simply replaced when damaged. In yet a further embodiment, by providing a method for easily replacing the combiners **12360**, different types of combiners can also be provided to the user such as: polarized combiners, combiners with different tints, combiners with different spectral properties, combiners with different levels of physical properties, combiners with different shapes or sizes, combiners that are partial mirrors or combiners that are notch mirrors, combiners with features to block faceglow as previously described herein.

(462) Although embodiments of HWC have been described in language specific to features, systems, computer processes and/or methods, the appended claims are not necessarily limited to the specific features, systems, computer processes and/or methods described. Rather, the specific features, systems, computer processes and/or methods are disclosed as non-limited example implementations of HWC. All documents referenced herein are hereby incorporated by reference.

Claims

1. A wearable head device comprising: an optics module comprising: a polarized light source configured to provide an illumination light; a reflective image source configured to form an image light based on the illumination light; and a partial reflector configured to transmit the illumination light toward the reflective image source, receive the image light from the reflective image source, and transmit the image light toward a first lens configured to converge the image light; a combiner configured to transmit the converged image light, along with an environment light from a surrounding environment, toward a user's eye concurrently with the user wearing the wearable head device; and a frame configured to hold a transparent element, wherein: the transparent element comprises a first region having a first transmissivity configured to occlude viewing of the optics module through the first region of the transparent element, the transparent element further comprises a second region having a second transmissivity configured to transmit the environment light toward the user's eye concurrently with the user wearing the wearable head device, and the second transmissivity is greater than the first transmissivity.
2. The wearable head device of claim 1, wherein the transparent element comprises a corrective lens.
3. The wearable head device of claim 1, wherein the first transmissivity is less than 30 percent.
4. The wearable head device of claim 1, wherein the second transmissivity is greater than 90 percent.
5. The wearable head device of claim 1, wherein the transparent element further comprises a gradient zone, the gradient zone comprising a tint gradient from the first transmissivity to the second transmissivity.
6. The wearable head device of claim 1, wherein the frame is configured to hold the transparent element via a magnetic attachment system.
7. The wearable head device of claim 1, wherein the first region corresponds to an upper region of the transparent element and the second region corresponds to a lower region of the transparent element.
8. The wearable head device of claim 1, wherein: the transparent element further comprises an electrochromic element; the wearable head device further comprises one or more processors in communication with the electrochromic element, the one or more processors configured to vary the second transmissivity via the electrochromic element.
9. The wearable head device of claim 8, wherein varying the second transmissivity via the

electrochromic element comprises varying the second transmissivity based on a signal received via a software application executed by the one or more processors.

10. The wearable head device of claim 1, further comprising an eye imaging camera positioned adjacent to the partial reflector and configured to capture light reflected by the user's eye.

11. A method comprising: providing, via a polarized light source of an optics module of a wearable head device, an illumination light; transmitting, via a partial reflector of the optics module, the illumination light toward a reflective image source of the optics module; receiving an image light from the reflective image source, the image light based on the illumination light; transmitting, via the partial reflector, the image light toward a first lens of the wearable head device; converging the image light via the first lens; and concurrently transmitting, toward a user's eye via a combiner of the wearable head device, the converged image light and an environment light from a surrounding environment, wherein: the wearable head device comprises a frame configured to hold a transparent element, the transparent element comprises a first region having a first transmissivity configured to occlude viewing of the optics module through the first region of the transparent element, the transparent element further comprises a second region having a second transmissivity configured to transmit the environment light toward the user's eye, and the second transmissivity is greater than the first transmissivity.

12. The method of claim 11, wherein the transparent element comprises a corrective lens.

13. The method of claim 11, wherein the first transmissivity is less than 30 percent.

14. The method of claim 11, wherein the second transmissivity is greater than 90 percent.

15. The method of claim 11, wherein the transparent element further comprises a gradient zone, the gradient zone comprising a tint gradient from the first transmissivity to the second transmissivity.

16. The method of claim 11, wherein the frame is configured to hold the transparent element via a magnetic attachment system.

17. The method of claim 11, wherein the first region corresponds to an upper region of the transparent element and the second region corresponds to a lower region of the transparent element.

18. The method of claim 11, wherein: the transparent element further comprises an electrochromic element; the method further comprises varying the second transmissivity via the electrochromic element.

19. The method of claim 18, wherein varying the second transmissivity via the electrochromic element comprises varying the second transmissivity based on a signal received via a software application executed by one or more processors in communication with the wearable head device.

20. The method of claim 11, wherein: the wearable head device further comprises an eye imaging camera positioned adjacent to the partial reflector, and the method further comprises capturing, via the eye imaging camera, light reflected by the user's eye.
